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Result Year 2017 - No. 025 2017. 1. 20.

Fourth Issue 2015시감2118

κ Case Avoidance of abuse of dominant position by Myungqualcom

Incorporated, etc. P In 1 . Qualcomm Incorporated

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3. Qualcomm CDMA Technologies Asia-Pacific PTE Limited

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Closing date of the hearing Dec. 21, 2016.

主旨 statement

1. In entering into a modem chipset patent license agreement with a modem chipset manufacturer that expresses an intention to enter into a license agreement for the right to manufacture, sell, rent, use, repair, and otherwise dispose of modem chipsets (hereinafter referred to as a "modem chipset patent license") with respect to the CDMA, WCDMA, and LTE related mobile communication standard essential patents held by the Defendants, the Defendants shall not interfere with the business activities of the modem chipset manufacturer or other operators by imposing the following unreasonable terms and conditions.

a. Terms that limit some of the rights that are subject to the modem chipset patent license, such as the modem chipset manufacturer's sale or use of the modem chipset

b. A condition requiring the modem chipset manufacturer to provide sales information related to the modem chipset, such as the name of the customer, the number of sales per customer, and the product model, against the will of the modem chipset manufacturer. However, if it is necessary for the accurate calculation of the implementation fee, the

information may be requested within the scope agreed upon by both parties.

c. Require Respondents to license patents held by the modem chipset manufacturer against the will of the modem chipset manufacturer without payment of fair consideration or to refrain from asserting the patents against Respondents and third parties.

2. The defendants claim that a modem chipset manufacturer wishes to enter into a modem chipset patent license agreement with a

Follow the procedures in a. through d. below, but in good faith and in accordance with customary industry practice and good faith

Negotiate to enter into a modem chipset patent license agreement.

a. Within 60 days of receiving a written request from a modem chipset manufacturer to enter into a modem chipset patent license agreement (or such longer period as the Parties may agree), Respondents shall send to the modem chipset manufacturer a proposed modem chipset patent license agreement in compliance with the remedial order in 1. above. The proposed agreement shall specify or attach a list of the patents to be licensed, the claims, a breakdown of the key claims and their relevance to the standard, and the methodology for calculating the royalty.

b. If a modem chipset manufacturer offers an alternative to the proposed modem chipset patent license agreement in 2. a. above, Defendants shall negotiate in good faith and in accordance with customary industry practice.

c. If the Respondents and the modem chipset manufacturer are unable to reach final agreement on the terms of a modem chipset patent license agreement and either party gives written notice of termination of negotiations, the Respondents shall, within sixty (60) days of receipt of such notice, request a determination by an independent third party, such as the International Chamber of Commerce, the World Intellectual Property Organization, or a court, as agreed to by the modem chipset manufacturer, and accept the terms of the agreement as determined by the independent third party.

d. The Respondents shall not file any injunctive claim against the modem chipset manufacturer in any court or competent governmental

authority for the manufacture, importation, sale, etc. of the modem chipsets while the proceedings in paragraphs 2. a. through c. above are pending. However, this shall not apply to any investigation of a modem chipset manufacturer that refuses to comply with the procedures in paragraphs 2. a. through c. above.

3. Defendants have agreed to license their CDMA, WCDMA, and LTE related mobile communications standards essential patents as a condition of purchasing and using Defendants' modem chipsets by manufacturers of portable mobile communications devices ("mobile phones"), including cell phones and tablets, as follows

Carry by requiring or enforcing a license agreement first.

Not to interfere with the business activities of other operators such as phone manufacturers. However, this shall not apply if a mobile phone manufacturer has failed to negotiate a patent license agreement with the Respondents in good faith and has received a final injunction from a court or other authority for infringement of the essential patents of CDMA, WCDMA, and LTE related mobile communication standards held by the Respondents in products manufactured or sold by such mobile phone manufacturer.

a. Refusing, delaying, or restricting the supply of modem chipsets because the patent license agreement with the Respondents has not been concluded or has expired, or because the Respondents are not properly performing the patent license agreement.

B. Restricting the manufacture and sale of mobile phones equipped with modem chipsets before entering into patent license agreements with the defendants

4. If a mobile phone manufacturer expresses an intention to modify an existing modem chipset supply agreement with the Respondents, the Respondents shall modify and delete all provisions in the modem chipset supply agreement with the mobile phone manufacturer that require the mobile phone manufacturer to first enter into or execute a patent license agreement as a condition for the mobile phone manufacturer to purchase and use the modem chipset. The order to modify or delete such contractual provisions is not limited to the provisions of the modem chipset supply agreement between Respondents and the handset manufacturer, but also includes relevant provisions in other agreements that have substantially the same effect.

5. In entering into patent license agreements with mobile phone manufacturers for their CDMA, WCDMA, and LTE-related telecommunications standard essential patents, Defendants shall not interfere with the business activities of other operators, including mobile phone manufacturers, by forcing them to engage in the following unfair practices

A. Failure to distinguish between essential patents for mobile communication standards and other patents against the will of mobile phone manufacturers or to distinguish between patents for each mobile communication standard such as CDMA, WCDMA, LTE, etc.

Licensing with a blanket license

B. Unilaterally determining the royalty rate for a patent subject to a license agreement against the will of a mobile phone manufacturer without going through a fair consideration calculation procedure, such as failing to provide the list of patents, claims, analysis of key claims, relevance to standards, and the method of calculating the royalty rate necessary to evaluate the value of the patent.

c. Requiring the defendants to license patents held by the mobile phone manufacturer against the mobile phone manufacturer's will without a fair consideration calculation process, or requiring the defendants and third parties not to assert the patents.

6. In the event that a handset manufacturer expresses an interest in amending an existing patent license agreement with Defendants, Defendants shall follow the procedures in a. through d. below, but shall negotiate in good faith to enter into a patent license agreement in accordance with customary industry practice and in good faith.

a. Within sixty (60) days (or such shorter period as the parties may agree) of receiving a written request from a handset manufacturer to enter into an amendment to an existing patent license agreement, Defendants shall send the handset manufacturer a proposed patent license agreement that complies with the remedial order in paragraph 5. above.

b. Defendants shall negotiate in good faith and in accordance with customary industry practice and good faith if a handset manufacturer offers an alternative to the proposed patent license agreement in 6. a. above.

c. If the Respondents and the handset manufacturer are unable to reach final

agreement on the terms of a patent license agreement and either party gives written notice of termination of negotiations, the Respondents shall, within 60 days of receiving such notice, request a determination by an independent third party, such as the International Chamber of Commerce, the World Intellectual Property Organization, or a court, as agreed to by the handset manufacturer, and accept the terms of the agreement as determined by the independent third party.

d. While the proceedings in paragraphs 6. a. through c. above are pending, Defendants have filed a lawsuit against the cell phone manufacturer, the

Do not file a claim for an injunction against the manufacture, importation, sale, etc. before a court or competent governmental authority. However, this shall not apply to a mobile phone manufacturer that refuses to comply with the procedures in 6. a. through c. above.

7. Within 30 days from the date of receipt of this Remedial Order, the Respondents shall notify all modem chipset manufacturers and handset manufacturers that they have been notified by the KFTC of the receipt of this Remedial Order and the full contents of this Remedial Order, and shall submit the results to the KFTC within 14 days.

8. If Respondents enter into any new or amended modem chipset patent license agreements, patent license agreements, or modem chipset supply agreements with modem chipset manufacturers or handset manufacturers pursuant to this remedial order, they shall submit the new or amended agreements and related materials to the FTC within 14 days.

9. For purposes of this Order, modem chipset manufacturers and handset manufacturers are those modem chipset manufacturers and handset manufacturers that

A. Modem chipset manufacturer

1) A modem chipset manufacturer headquartered in South Korea and its domestic and international affiliates ("Affiliates")

2) Businesses and their affiliates that supply modem chipsets to mobile phone manufacturers in b. 1) through 3) of the following 9.

B. Phone manufacturer

1) Mobile phone manufacturers based in South Korea and their affiliates

2) Operators and their affiliates that sell mobile phones in or to South Korea

3) 9. b. above. Businesses and their affiliated companies that supply mobile phones to businesses in 2) above and their affiliated companies

c. Modem chipset manufacturers in a. above and cell phone manufacturers in b. above include any entity that manufactures, supplies, or sells modem chipsets or cell phones after the date of this Order.

10. Respondents may request reconsideration of this Order from the FTC if a binding final judgment or action or order of a foreign court or competition authority that is finalized after the date of this Order conflicts with this Order and simultaneous compliance is not feasible.

11. The Respondents shall pay the penalty to the National

Treasury as follows A. Amount of penalty: KRW

1,031,145,000,000 in total

1) Qualcomm Incorporated :
428,605,000,000,000 KRW
2) Qualcomm CDMA Technology Asia-Pacific PTI Limited : KRW

602,540,000,000 B. Due Date: Within the due date (60 days) specified in the Penalty

Payment Notice

C. Payment location: Bank of Korea Treasury Acceptance Agent or Post Office

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1. Basic facts

A. Status and general status of the subjects

1 Defendants Qualcomm Incorporated ("QI")[1], Qualcomm Technologies Incorporated ("QTI")[2], and Qualcomm CMDA Technologies Asia-Pacific PTE Limited ("QCTAP")[3] are corporations organized under the laws of the United States of America ("U.S.") or the Republic of Singapore ("Singapore"), In the United States and Singapore, as well as in the territory of the Republic of Korea, CMDA Technologies Asia-Pacific PTE Limited ("CMDA") and QCTAP are entities incorporated under the applicable laws of the United States of America ("U.S.") or the Republic of Singapore ("Singapore") that license their telecommunications patents, including Code Division Multiple Access ("CDMA"), Wideband Code Division Multiple Access ("WCDMA"), and Long Term Evolution ("LTE"), to mobile phone manufacturers[4] and collect royalties in exchange for the sale of related software, including modem chipsets[5] ("modem chipsets") for mobile phones.

2 Defendants[6] conduct their patent licensing business through Qualcomm Technology Licensing ("QTL") and their modem chipset business through Qualcomm CMDA Technologies ("QCT"), and the revenues from the licensing business and the revenues from the modem chipset business are accounted for as revenues of QTL and QCT, respectively.

3 With respect to the patent licensing business, QTL, which is responsible for the licensing business, exists within QI, which owns substantially all of Respondents' patents and is the party to the patent license agreements. With respect to the modem chipset business, on October 1, 2012, Respondents spun off QCT, which conducts the modem chipset business, from QI and incorporated it into QTI, a wholly owned subsidiary of QI.

1) Hereafter referred to as "QI".
2) Hereinafter referred to as "QTI".
3) Hereafter referred to as 'QCTAP'.
4) "Mobile phone" means a portable mobile communication device, including a cell phone,

tablet, etc. equipped with a modem chipset. "Cell phone manufacturer" also includes a business that manufactures cell phones, outsources the manufacture of cell phones to other businesses, or sells used cell phones.

5) The term "modem chipset" means a wireless baseband chipset or communication processor that is installed in a mobile phone for wireless communication. A "modem chipset manufacturer" also includes an entity that manufactures modem chipsets, outsources the manufacture of modem chipsets to another entity, or sells modem chipsets.

6) "Defendants" means QI with respect to the Patent Licensing Business, QI and QCTAP with respect to the Modem Chipset Business prior to the 2012 Spin-Off, QTI and QCTAP after the 2012 Spin-Off, and QI, QTI and QCTAP together with respect to both the Patent Licensing Business and the Modem Chipset Business.

Defendants' modem chipset business is operated by QTI through QCT, which exists within QTI.
On the other hand, most of the contracts for the supply of modem chipsets to mobile phone manufacturers are concluded by QCTAP8), which has been concluding contracts for the supply of modem chipsets to mobile phone manufacturers even before the Defendants' modem chipset business was transferred from QI to QTI. As a result, the Defendants have been operating the modem chipset business and licensing business through QTI and QCTAP, which are wholly-owned subsidiaries of QI, centered on QI as the top parent company.

<Table 1 General Status

(as of September 30, 2015, in millions of dollars)

Subject	Representative	Year Founded	Total assets	Sales	Net income
Qualcomm	Steve Molenkoff	1985	50,796	25,281	5,268
Qualcomm Technologies	Steve Molenkoff	2011	2,923	17,154	2,465
Qualcomm Technologies Asia-Pacific	Wei Xiong Li Qi Xian Chang, Rowen Chen	1999	*** B	*** B	*** B

* Source: Respondents' submissions.

4 Since its establishment in 1985, the Defendants led the introduction of CDMA technology and conducted business in the telecommunications industry, including base stations, mobile phones, and components, until the 1990s. In the late 1990s, the base station and mobile phone businesses were discontinued and sold, and since 2000, the Defendants have mainly focused on the patent license business and modem chipset manufacturing and sales business. The sales of patent samples and modem chipsets amounted to 25 billion USD per year as of 2015, of which royalties accounted for approximately 32%.

<Table 2 Patent royalties and modem chipset sales as a percentage of revenue

(Unit: million $)

分类	2013	2014	In 2015
Patent Implementation Fees	7,554	7,569	7,947
Modem chipset revenue	16,715	18,665	17,154

* Source: Respondents' 2015 Annual Report (10-K)

7) Defendants' filings and Defendants' 2012 Annual Report (10-K)
8) QCTAP was formed on December 11, 1999, and became a wholly owned subsidiary of QTI upon the formation of QTI and the transfer of QCT, Respondents' modem chipset division, from QI to QTI.

I. Jurisdiction over Defendants

6 The Respondents' activities related to the manufacture and sale of modem chipsets, software sales, patent licenses, and the collection of implementation fees are carried out both domestically and abroad in accordance with a consistent business policy, and the Respondents' activities affect the domestic market by affecting the business activities of domestic operators or ultimately affecting domestic consumers, and are therefore subject to regulation under Article 2(2) of the Act on Monopoly Control and Fair Trade, and the KFTC has jurisdiction over the Respondents.[9][10]

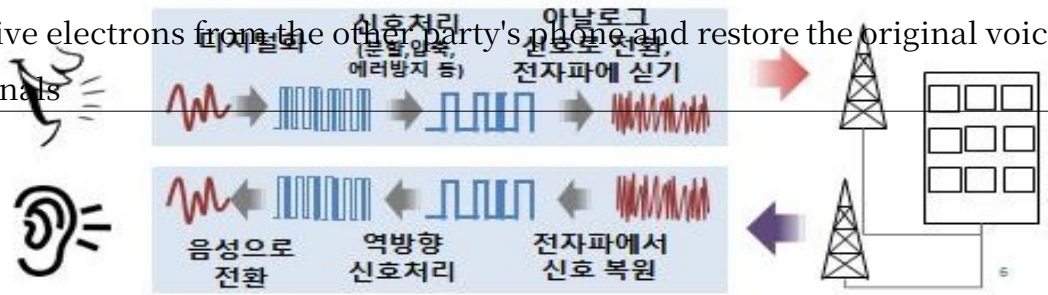
C. Telecom market overview

1) Overview of Telecommunications

7 Wireless Communication refers to a method of transmitting voice and data signals using radio waves without a wire connection, and includes radio, Wi-Fi, and mobile communication. Among them, mobile communication is a type of wireless communication method that enables wireless communication on the go rather than in a fixed location.

<Table 3 How cellular communication works

Your phone processes voice and data signals according to certain rules and transmits them to the nearest base station.
The cell tower receives it and transmits it back to the cell tower near the other end of the call.
③ Receive electrons from the other party's phone and restore the original voice and data signals



9) Joint price determination o f graphite electrode rods by Showa Denko and six other companies (Supreme Court, March 24, 2006, Decision 2004DU11275, Seoul High Court, August 19, 2004, Decision 2002NU6110)

10) Hartford Fire Ins. Co. v. California, 509 U.S. 764 (1993), etc.

a) Changing the way we communicate

8 In order for different phones to make calls or send data back and forth, they need "standards" to ensure that they follow the same promised rules. As the number of mobile phone users and the amount of data continues to grow, mobile technology has evolved and mobile standards have evolved with it to better utilize limited frequency bands and process information faster.

<Table 4 Key standards by cellular method generation

Generation	Key Standards
1st generation	AMPS, TACS, etc.
Generation 2 (1990-2012)	gsm, cdma, tdma, d-amps
3rd Generation (2003-2011)	CDMA2000, UMTS (WCDMA)[11], HSDPA, HSUPA, HSPA+, Wibro
Generation 4 (2012 and later)	LTE(Advanced)[12], Wibro-evolution

9 However, the evolution of cellular technology from generation to generation does not mean that the standards for each generation are temporarily switched and the old standards disappear from the market. Since there are still cell phone subscribers on the market who have cell phones equipped with the old standard, the old standard service must be maintained for some time in order for the new standard to communicate with the old standard, a process known as backward compatibility. Therefore, not only LTE, the latest mobile communication standard, but also CDMA and WCDMA, the older standards, have an important place in mobile communications.

10 Mobile communication requires a device that processes voice or data information generated by one mobile phone according to mobile communication standards and restores the original information to another

mobile phone, and the core component that performs this function in a mobile phone is the modem chipset. Since modem chipsets need to be backward compatible to support both old and new standards, multi-mode chipsets are now commonly used.

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11) 'UMTS (WCDMA)' is written as 'WCDMA'.

12) "LTE (advanced)" is written as "LTE".

b) Characteristics and evolution of CDMA, WCDMA, and LTE cellular technologies

(1) CDMA and WCDMA cellular technologies

11 CDMA is a representative digital mobile communication system developed by the Defendants that assigns a unique code to each subscriber (terminal) and allows multiple users to transmit and receive codes over the same radio resource. The Telecommunication Industry Association (TIA) selected CDMA technology as a standard in the United States in 1993. The initial CDMA method could only transmit voice signals, but later, CDMA2000 method was developed to transmit data in addition to voice signals.

12 WCDMA is a standard developed by European mobile operators who formed a standardization organization called 3GPP (3rd Generation Partnership Project) to submit wireless communication technology that evolved from the second generation GSM standard.[13] WCDMA is similar to CDMA2000 in that it allows data communication, video communication, etc. in addition to voice communication, and multiple communications can be performed simultaneously using codes, but there are differences in the way codes are assigned.[14]

13 WCDMA has evolved with each release of the standard, including Release 99, which improved voice transmission speeds, HSDPA (Release 5), which increased downlink data rates, and HSPA+ (Release 7), which increased data rates on both the downlink and uplink.

<Table 5 Cellular technology method terminology

	Abbreviations	Technical Method	
CDMA15)	CDMAone	IS-95A	CDMAone IS-95A
		IS-95B	CDMAone IS-95B

	CDMA2000	1x	CDMA2000 1x

13) 3GPP submitted WCDMA technology as IMT-DS to the ITU-R's call for third generation mobile telecommunications standards (IMT-2000).
14) CDMA2000 uses one code as if it were another, with all base stations using the same code but with different code start points based on GPS absolute time, while WCDMA divides the world's base stations into 512 codes, with different starting points, so that each base station uses a different code.

		EVDO	CDMA2000 1xEVDO
		EVDV	CDMA2000 1xEVDV
WCDMA		Release 99	WCDMA
		Release 5	HSDPA
		Release 7	HSPA

(2) LTE cellular technology

14 In 2004, the Institute of Electrical and Electronics Engineers (IEEE) developed the mobile WiMax (802.16e) standard, which uses orthogonal frequency division multiplexing (OFDM) technology to increase data transmission speed by moving away from CDMA technology and combining it with multiple input multiple output (MIMO) antenna technology. The LTE standard offers significant performance improvements over previous methods in terms of data transfer rates and suitability for high-speed communication.

C) Development of domestic telecommunications

15 In 1984, Korea Mobile Communication Service Corporation, a subsidiary of Korea Telecommunications Corporation, was established, and in May of the same year, the AMPS cellular system was introduced to provide car phone service. From 1988 onward, the spread of mobile phones increased rapidly, and the concept of mobile phones began to change from car phones to 'handy phones', which are actually mobile phones.

16 In 1993, the then Ministry of Information and Communication selected CDMA, which was selected as a standard in the United States, as the next generation mobile communication method.[16] Since then, the world's first commercially successful CDMA-based mobile communication service was launched in 1996.

15) 'CDMA' includes 'CDMAone' and 'CDMA2000'.

16) It was considered that CDMA is superior to other methods in terms of subscriber capacity, and since it has not yet been commercialized, it can be self-sustaining if successful in Korea.

It has been operating telecommunication services. Korea Telecommunication Technology Association (Telecommunication
Technology Association ("TTA") established the CDMA mobile communication standard based on seven patents filed by the Respondents in 1996, and in March 2005, the third-generation standard mobile communication service based on WCDMA was launched, and in July 2011, LTE service was launched. Currently, the second and third generation CDMA communication methods17), WCDMA communication methods, and the fourth generation LTE communication methods coexist in Korea.

<Table 6 Number and share of domestic subscribers by telecom method

(Unit: people)

	2012	2013	2014	2015
CDMA	10,753,379	7,741,958	6,331,643	6,331,643
	19.7	13.9	10.9	10.9
WCDMA	27,059,688	18,489,445	14,874,490	14,870,808
	49.5	33.2	25.6	25.6
LTE	15,811,360	28,449,437	36,001,824	36,087,905
	28.9	51.1	62.0	62.1
WiBro	1,049,788	983,387	868,481	868,481
	1.9	1.8	1.5	1.5
Subtotal	54,674,215	55,664,227	58,076,438	58,158,837
	100.0	100.0	100.0	100.0

* Source: Korea Telecommunications Commission "Status of Wireless Communication Service Subscribers in November 2016"

2) Telecommunications and standardization

a) Standardization and Essential Patents

(1) Concepts of standard technology and standard essential patents

17 A "standard technology" is a technology that has been selected as a standard in a certain technology field by a government, standardization

organization, business group, or a group of companies with the same technology.[18] In other words, a standard setting organization (SSO) usually prevents duplication of investment in a certain technology field.

17) However, KT ended its 2G service in 2011, so o n l y SKT and LGT currently offer 2G services.
18) Korea Fair Trade Commission' s Guidelines for Examination of Unfair Exercise of Intellectual Property Rights (revised March 23, 2016.
No. 247) Ⅰ. 3. a. (5)

and selected as a standard to promote technology development in related fields, Standardization bodies, which are typically collaborative organizations formed by stakeholders from related industries to set arbitrarily specific standards, select and specify the technologies that make up a standard and agree to use only those technologies.

18 A "Standard Essential Patent (SEP)" is a patent that must be licensed in order to produce goods or provide services embodying the standard technology, and a patent that requires a voluntary commitment from the licensee to license the patent on fair, reasonable and non-discriminatory (FRAND) terms.) In other words, a standard-essential patent is a patent that is conceptually and essentially impossible for technical reasons to make, sell, lease, otherwise dispose of, repair, use, or operate a product or method without infringing the patented technology, considering the normal technical practice at the time of standardization or the state of the art generally available, and therefore must be licensed by the patentee in order to avoid becoming an infringer.

19 On the other hand, patents that are not directly related to standards are sometimes referred to as non-standard-essential patents (Non-SEPs) or other patents to distinguish them from standard-essential patents. A non-standard-essential patent is a patent that is not strictly necessary to implement a given technical standard or that can perform the same function through a circumvention or avoidance design. Therefore, unlike standard-essential patents, non-standard-essential patents are not subject to FRAND terms in the patent licensing process. FRAND terms are discussed in more detail below.

(2) Telecommunications-related standardization organizations

20 Standardization organizations in the telecommunications sector include

the International Telecommunication Union (ITU), the Institute of Electrical and Electronics Engineers (IEEE), the European Telecommunications Standard Institute (ETSI), the Telecommunications Industry Association (TIA), and the Korea Telecommunications Technology Association (TTA).

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19) Korea Fair Trade Commission' s Guidelines for Examination of Unfair Exercise of Intellectual Property Rights (revised March 23, 2016.
No. 247) Ⅰ. 3. a. (6)

There are also collaborative organizations, 3GPP (for WCDMA and LTE) and 3GPP2 (for CDMA).

(3) Standards development and selection process

21 The standards development process varies slightly depending on the standardization body and the nature of the standard in question, but is basically similar. Standardization organizations are generally membership-based, meaning that they receive proposals from members, deliberate them in technical committees, and finally approve them by vote. Therefore, the standard adoption process is a collective decision-making process among the participants of the standardization organization.

22 On the other hand, when a patentee declares and discloses to a standardization body that its technology is a standard-essential patent, the standardization body generally does not have a procedure to verify whether the declared standard-essential patent is in fact a valid patent or essential to the standard.[20] And it is known that in the course of related litigation, many of the patents declared to be standard-essential by the patentee are often determined to be invalid or not essential to the standard.[21] Thus, in general, when a patentee seeks to collect royalties for a patent that it has declared to be a standard-essential patent, the patentee must prove which patents are being infringed and in what manner, and the patentee must verify and contest the claims.

(4) Positive and negative effects of standardization

23 The positive effects of standardization include, first, network externality due to increased interoperability. When standardization eliminates uncertainty about market changes and ensures product compatibility, the utility of a product increases proportionally with the number of other users of that product. Second, standardization provides producers with the ability to

standardize production processes

20) ETSI's IPR Guide 3.2.1 states that ETSI does not perform any verification of the validity or essentiality of standard-essential patents ["ETSI does not perform any check on the status and validity of any Essential IPRs notified to ETSI" (ETSI Guide on IPRs, September 19, 2013)].

21) Fairfield Resources International, a specialized patent research firm, has published standards papers on GSM, WCDMA, and LTE standards.
Of the patents declared as essential, 50-73% were found not to be essential ("Analysis of Patent Declared as Essential to GSM as of June 6, 2007," "Review of Patents Declared as Essential to WCDMA Through December, 2008," "Review of Patents Declared as Essential to LTE and SAE (4G Wireless Standards) Through June 30, 2009").

enabling economies of scale through innovation and market expansion to reduce costs.

It can have a sensitization effect. Third, it can promote competition in the downstream market using standardized technologies and reduce consumers' product switching costs and transaction costs, thereby reducing prices due to competition, promoting the development of new technologies, and improving the quality of products.

24 On the other hand, standardization, on the other hand, firstly, the adoption of a particular technology as a standard-essential patent artificially eliminates competition between technologies that were actually or potentially substitutes before the selection of the standard, and creates a kind of barrier to entry in the market for that technology. Second, companies with dominant standard technologies may strengthen their market power as standard-essential patent holders through standardization and opportunistically use it to refuse licenses for standard-essential patents or impose unreasonable conditions on the implementation of the standard, thus increasing the risk of patent hold-up. Third, if a standardization organization selects a specific technology as a standard, the market will be locked into the standard, increasing the risk that the dominance of companies holding standard-essential patents will be abused by increasing the sunk costs of participants in the industry, such as component manufacturers and network operators.

b) Standard Essential Patents and FRAND Commitments

(1) The significance of FRAND commitments

25 In order to maximize the effectiveness of standardization by promoting the diffusion of standard technologies and further technological innovation, and to prevent the risk of patent abuse, most standardization organizations have an IPR policy that includes disclosure of patent information and negotiation of terms and conditions, such as requiring patent holders to

disclose their patent information in advance of the selection of a standard technology and to voluntarily declare an irrevocable "FRAND commitment" to license on fair, reasonable, and non-discriminatory terms prior to the selection of the standard, and excluding the patent from the standard if the patent holder refuses.

26 In light of this, a FRAND commitment is an indication of acceptance of the restriction of granting a license to practice on FRAND terms, and is therefore not a standard essential feature.

When a licensee obtains a FRAND commitment from a standardization organization for a standard-essential patent, Table
The holder of a quasi-essential patent would be obligated to grant the right to practice on FRAND terms and negotiate in good faith with those seeking to practice the standard-essential patent.[22)]

(2) Major standardization organizations' policies on FRAND commitments

27 In November 2004, the major international standardization organizations, the ITU, the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC), compared and harmonized their respective patent policies and published the Common Patent Policy for ITU-T/ITU-R/ISO/IEC in March 2007. The main content of the policy is that standardization organizations will require patent holders of standard-essential patents to submit a commitment to "provide a license on FRAND terms to a willing licensee willing to pay a fair and reasonable implementation fee."23) If the patent holder refuses to do so, the patent will not be included in the standard.

<Table 7 ITU-T/ITU-R/ISO/IEC Common Patent Policy

Participants in standards work must inform these standardization bodies from the outset of the work of any known or pending patents owned by them or by other entities.
The patent cannot be included in the standard if the standard has been developed and the information in ① has been disclosed and the patentee is unwilling to negotiate a free or paid license on a nondiscriminatory basis under reasonable terms.
(3) The patentee shall submit to the standardization organization a patent description and licensing declaration in accordance with the conditions of paragraph (2)

22) In a lawsuit between Samsung Electronics and Apple, the Seoul Central District Court (Aug. 24, 2012, Decision 2011-39552) ruled that "the plaintiff declared the grant of a license on fair, reasonable and non-discriminatory terms. Decision 2011-39552) stated, "The plaintiff declared that the FRAND declaration irrevocably grants a license on fair, reasonable and

non-discriminatory terms, but considering the wording of the declaration and the following circumstances, which are admitted by the aforementioned facts and evidence, it cannot be said that the patentee automatically grants the right to practice the patent to an unspecified third party immediately upon the FRAND declaration, or that it is a binding irrevocable grant of the right to practice, that a person using or intending to use the patent does not automatically acquire a right to practice by virtue of the FRAND declaration, but rather declares a general principle that imposes an obligation on the patentee to negotiate in good faith on FRAND terms in connection with the entry into a license agreement."

23) ETSI's Intellectual Property Policy Article 6.1 defines the scope of a "license" on FRAND terms as "including, at a minimum, the right to manufacture and the right to sell, rent, dispose of, repair, use, operate and use methods of the manufactured device", and the term "license" is hereinafter used to mean the right to manufacture and the right to sell, rent, dispose of, repair, use, operate and use methods of the manufactured device.

28 ETSI, another standardization organization, has a similar IPR policy. ETSI's IPR policy requires patent holders to submit a written commitment that they will provide an irrevocable license on FRAND terms to anyone who wants to license their patent for a fair consideration in order for it to be adopted as a standard.[24)]

29 In addition, in February 2015, IEEE, which has selected major communications standards such as Wi-Fi, revised its intellectual property policy to further clarify the meaning of FRAND commitments. IEEE's Board of Directors approved a policy that requires IEEE to (1) license an unrestricted number of users on a worldwide basis who request a license at no cost or on RAND terms, and (2) limit injunctions against licensees who actively seek to enforce,
③ Revised the IPR policy to include a provision that RAND obligations will be inherited when standard essential patents are transferred to a third party.

<Table 8 IPR policies of major standardization organizations

	ISO/IEC/ITU	ETSI	IEEE
Publication rules	・Scope: Essential patents and patents Obligations to disclose application information When: Start of standardization	・Scope: Essential patents and patents Filing information disclosure obligations Timing: Drive early disclosure in a timely manner	・Scope: Essential patents and patents Application information can be disclosed, but not required When: Provide an opportunity to reveal to attendees at the start of the meeting
Patent Search Obligations		State that it is not mandatory	・If you don't do a patent search Explicit

Licensin g rules	、RAND 條款 Licensing Or licensing with no implementation fee	Licensing on FRAND terms 、A third party standard essential patents	、Real world Without Licensing or RAND conditional licensing

24) "the Director-General of ETSI shall immediately request the owner to give within three months an irrevocable undertaking in writing that it is prepared to grant irrevocable licenses on fair, reasonable and non-discriminatory ("FRAND") terms and conditions under such IPR" (ETSI Intellectual Property Right Policy 6.1)

		If you have one, ETSI recommends using the You can also license your declaration to third parties on FRAND terms. Can be required to	
If you decline licensing	ㆍPatentee's 해당 patent Notification of numbers and claims and related standard descriptions (Recommended by ISO/IEC, mandatory by ITU) Exclusion of patented technologies from relevant committees (standard revision or revocation)	ㆍExcluded from the standard Related IPRs circumventing the relevant IPR Deliberate delays in patent disclosure by members can be sanctioned by the ETSI General Assembly.	ㆍExcluded from the standard
Negotiation rules	Understanding Parties between ISO/IEC/ITU Solving outside	ㆍSpecific within ETSI No commercial discussions or negotiations, such as secondary sourcing terms	ㆍPatent Rights scope or price Prohibit discussions of collusion ㆍPossible relative comparison for optimal technology selection when promoting standardization

(3) Effects of breaching FRAND commitments

(a) Competition Law Effects of Breaching FRAND Commitments

30 The FRAND commitments required of patent holders by standardization organizations can be viewed from a competition law perspective as the only means of controlling abuses of dominance by

standard-essential patent holders, as they function as a substitute for the existence of competing technologies that would have existed in the absence of the standard selection process. Thus, a breach of the FRAND covenant for a standard-essential patent, given the absence of any other substitute technology within the standard, per se removes the only means of preventing anticompetitive behavior. Specifically, the competition law effects of a breach of FRAND include

(i) the European Commission has found that it has anticompetitive effects, including distortion of the standardization process and the imposition of unilaterally disadvantageous licensing agreements on licensees;[25] (ii) the U.S. Federal Trade Commission has found that it has distorted the standardization process, including reducing the completeness and efficiency of standards adoption, and

25) EU Commission, Case AT.39985 - Motorola - Enforcement of GPRS Standard essential patents, Apr. 29, 2014

and passing it on to consumers, resulting in anticompetitive effects.[26)]

31 The anticompetitive effects of such breaches of FRAND commitments may also be assessed differently depending on the existence of alternative standards that could be substituted for the standard in question and the competitive situation in the downstream product markets in which the standard is implemented.

32 First, the abuse of dominance by a standard-essential patent holder can be adequately controlled if there are other standards that can replace the standard, even if no other technology exists in the technology licensing market that replaces the standard-essential patent. In addition, even if no substitute standard exists, the influence of a standard-essential patent may be reduced if there is strong competition from other products that do not use the standard in the submarket for which the standard is applied, and the monopolistic position of the standard-essential patent holder may be checked by competition in the submarket.[27)] Even if the standard-essential patent holder's breach of the FRAND covenant raises production costs in the product market, consumers have access to substitute products, so the standard-essential patent holder's abuse of dominance is limited by the competitive structure of the submarket.

33 However, if the above two conditions are not met, i.e., no other standard exists as a substitute for a particular standard and there is no way to control the abuse of monopoly power through the competitive structure of the downstream product market to which the standard is applied, then a FRAND commitment in a standardization body is the only means of preventing the patentee from abusing its monopoly power.

34 However, when it comes to mobile communication standards, the WCDMA and LTE standards have been finalized by the International

Organization for Standardization (ISO) through the selection of globally uniform mobile communication standards, and there is virtually no substitute for them, and the products to which they are applied, modem chipsets and mobile phones, are not

26) United States of America before the Federal Trade Commission, Docket No. C-4410

27) For example, there is one standard for producing and using Blu-ray media, one of the removable storage devices used in computers, but there are other substitutable goods, such as USB, which is a removable storage medium, and there are other standards for producing and using USB. In this case, there is no substitutability between the standards applied to each storage medium, but substitutability can be recognized between Blu-ray and USB, which are products produced using each standard. Therefore, if there is substitutability between products in the submarket using the standard, the exercise of the monopoly power of the standard-essential patent required to utilize the standard of each storage medium may be limited through competition between substitutes in the submarket.

There is also a virtually non-existent adjacent product market for cell phones, so the cellular table

In the case of Jun, violation of FRAND commitments directly increases the manufacturing costs of products covered by the standard, such as modem chipsets and cell phones, thereby impeding the diffusion of the standard, restricting related technological innovation, and ultimately leading to a reduction in consumer welfare. In addition, violations of FRAND commitments made during the standards selection process distort the standards selection process, which is a process of competition among standards, and prevent competition among technologies from functioning properly.

(b) the effect of a breach of FRAND commitments by a dominant vertically integrated operator

35 FRAND commitments are particularly important for vertically integrated monopolies that dominate markets by participating directly in downstream markets. As we have seen, the purpose of selecting a telecommunications standard is not to artificially grant a monopoly position in the technology licensing market, but rather to foster innovation and promote competition in the sub-markets that use the technology.

36 However, if the standard-essential patent holder simultaneously participates in the sub-markets, such as manufacturing and selling products covered by the standard-essential patent, the standard-essential patent holder will have the status of a vertically integrated business in both the standard-essential patent license market and the sub-component market. If the standard-essential patent holder, as a vertically integrated entity, uses its monopoly power in the technology market to restrict competition in the downstream product market, the standard-setting benefits are lost and the harms of monopolization remain. In other words, the vertically integrated entity's breach of the FRAND commitment directly results in the exclusion

of its competitors in the downstream component market. For example, if a vertically integrated entity refuses to license its standard-essential patents to a competing product manufacturer in a subcomponent market in which it participates, in violation of its FRAND commitments, the competing component manufacturer will be forced to manufacture and sell components without a license to the standard-essential patents and will be at a disadvantage in competing with the vertically integrated entity because it will be selling defective products that may infringe the patents.

37 This effect is more pronounced when the vertically integrated entity is a monopolist or market power player in the subcomponent market. Monopoly or market power in subcomponent markets

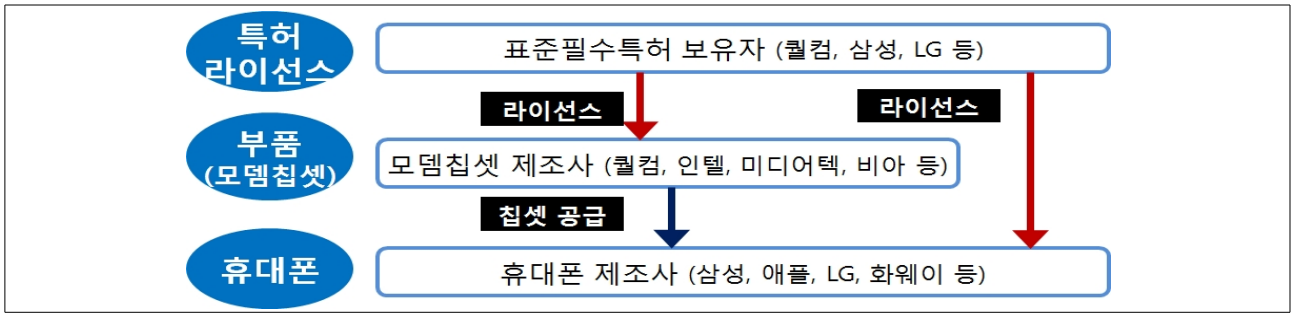
Vertically integrated operators that are not adversaries may license standard essential patents in subcomponent markets.
By refusing to license, the vertically integrated entity may be able to exclude competitors from the submarket and consolidate its dominance in the submarket, but it will also reduce the royalty income that it would have received if it licensed to competitors, and it will be more difficult to use its lack of dominance in the subcomponent market to require the purchaser of the subcomponent to enter into a royalty agreement. However, the situation is different if the vertically integrated entity is a monopolist or dominant player in the subcomponent market, and by refusing to license the standard-essential patent in the subcomponent market, it can exclude competitors from the subcomponent market and further consolidate its dominance in the subcomponent market, while using its dominance in the subcomponent market to maximize its profits by entering into license agreements with purchasers of the subcomponent and collecting royalties.

38 Therefore, if the standard-essential patentee is also a monopolist or market dominant player in the subcomponent market, ensuring that it licenses to competitors in compliance with FRAND commitments is even more important to control anticompetitive effects in the patent licensing market as well as in the subcomponent market.

3) Structure of the telecom market

39 The telecommunications market consists of three main parts: the patent license market, the component market such as modem chipsets that make up mobile phones, and the mobile phone market.

<Table 9 Overview of market structure in the telecommunications industry


특허
라이선스
표준필수특허 보유자 (퀄컴, 삼성, LG 등)
라이선스
라이선스
부품
(모뎀칩셋)
모뎀칩셋 제조사 (퀄컴, 인텔, 미디어텍, 비아 등)
칩셋 공급
휴대폰
휴대폰 제조사 (삼성, 애플, LG, 화웨이 등)

40 In the mobile communications patent licensing market, the original technology source that developed the communications method Yuja becomes a supplier. The majority of the CDMA standard essential patents are held by the accused, and the WCDMA and LTE standard essential patents are also held by the accused, but operators such as Samsung Electronics, InterDigital (IDC), LG Electronics, Nokia, and Ericsson also hold significant patents. Original technology owners license their patented technologies to component manufacturers such as modem chipsets, mobile phone manufacturers, and telecommunications equipment manufacturers, and receive royalties in return.

41 Component manufacturers, such as modem chipsets, license the original technology owners to make the various components that go into cell phones and sell them to cell phone manufacturers, and in some cases, such as Defendants, the original technology owners manufacture the cell phone components themselves.[28)]

42 Cell phone manufacturers build cell phones with components and software and pay an implementation fee for each patented technology implemented in a cell phone. Cell phone manufacturers produce cell phones and sell them to carriers or consumers, who in turn provide wireless communications services to consumers.

a) Patent License Market by Telecom Standard

(1) Percentage of licensed mobile phone market by telecom standard

43 For the second generation of standards, 19% of the global mobile phone market adopted CDMA, while the remaining 81% adopted the GSM standard. For the third generation of standards, which began in the early 2000s, the WCDMA market accounts for 85% of the total, CDMA2000 accounts for 13%, and TD-SCDMA, which was adopted in China, accounts for 2%. For the fourth generation of standards,

which began around 2012, the global mobile phone market follows a single standard, LTE.

44 As a result, for the second generation standard, only 19% of the total cell phone market was covered by the defendants' standard essential patent licenses, whereas after the third generation standard, virtually all of the world's cell phone

28) As described above, these entities are referred to as "vertically integrated entities," and the Respondents are vertically integrated entities that operate in both the patent license market and the modem chipset market, which are upstream levels of the overall market structure, and are also dominant vertically integrated entities that hold dominant positions in both markets.

The patent licensing market and the modem chipset market because the entire market is subject to licensing by the accused.
In all cases, the conduct of the Respondents as vertically integrated operators has taken on greater significance in relation to the competitive situation in the relevant markets.

<Table 10 Market share by cellular standard29)

		Mobile phone	market share
Generation 2 (1990-2012)	GSM	Voice/Text	81
	CDMA		19
3rd Generation (2003-2011)	WCDMA	Voice/Video, Data Communications	85
	CDMA2000		13
	TD-SCDMA		2
Generation 4 (2012 and later)	LTE	Superfast data communications	100

(2) Status of standard essential patents of patent holders by communication standard

45 In the telecommunications patent licensing market, the share of standard-essential patents held by the Respondents was over 90% of the total second-generation standard-essential patents, as the second-generation CDMA standard was virtually independently developed by the Respondents. However, the share of standard-essential patents held by the Respondents fell to 27% and 16% for the third-generation WCDMA and fourth-generation LTE standards, respectively, as many patent holders other than the Respondents contributed to the development of the technology. In this regard, even a single standard-essential patent is necessary for the manufacture and sale of products that implement the

technology, and given the inherent nature of such standard-essential patents, the decrease in the share of standard-essential patents held by the Respondents does not reduce the need to enter into a license agreement with the Respondents. However, the change in the proportion of standard essential patents may be a factor that can be taken into account in the calculation of the license fee in some cases.

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29) Based on Strategy Analytics, "VENDOR SHARE Global Handset Market by Technology," where the second generation share is based on cumulative sales from 1990 to 2002 and the third generation share is based on cumulative sales from 2003 to 2015. The original data does not distinguish between CDMA as second generation (CDMA One) and third generation (CDMA2000), so we categorized CDMA as second generation until 2002 and third generation after 2003, based on the year 2003, when the third generation became widely available.

<Table 11> WCDMA standard essential p a t e n t s by company

(per ETSI homepage, unit: cases)

	Company name	Public	Registration	Total
1	Subjects	786	1042	1,828 (27.3%)
2	Id.	293	670	963 (14.4%)
3	Nokia	214	745	959 (14.3%)
4	Ericsson	175	431	606 (9.1%)
5	Elgie	78	426	504 (7.5%)
6	Huawei	224	183	407 (6.0%)
7	Blackberry	141	213	354 (5.3%)
8	Apple	97	185	282 (4.2%)
9	Motorola	48	219	267 (4.0%)
10	nc	128	137	265 (4.0%)
11	Samsung Electronics	59	201	260 (3.9%)
Total number of reports		2,243	4,452	6,695

<Table 12 LTE standard essential patent holdings by company

(per ETSI homepage, unit: cases)

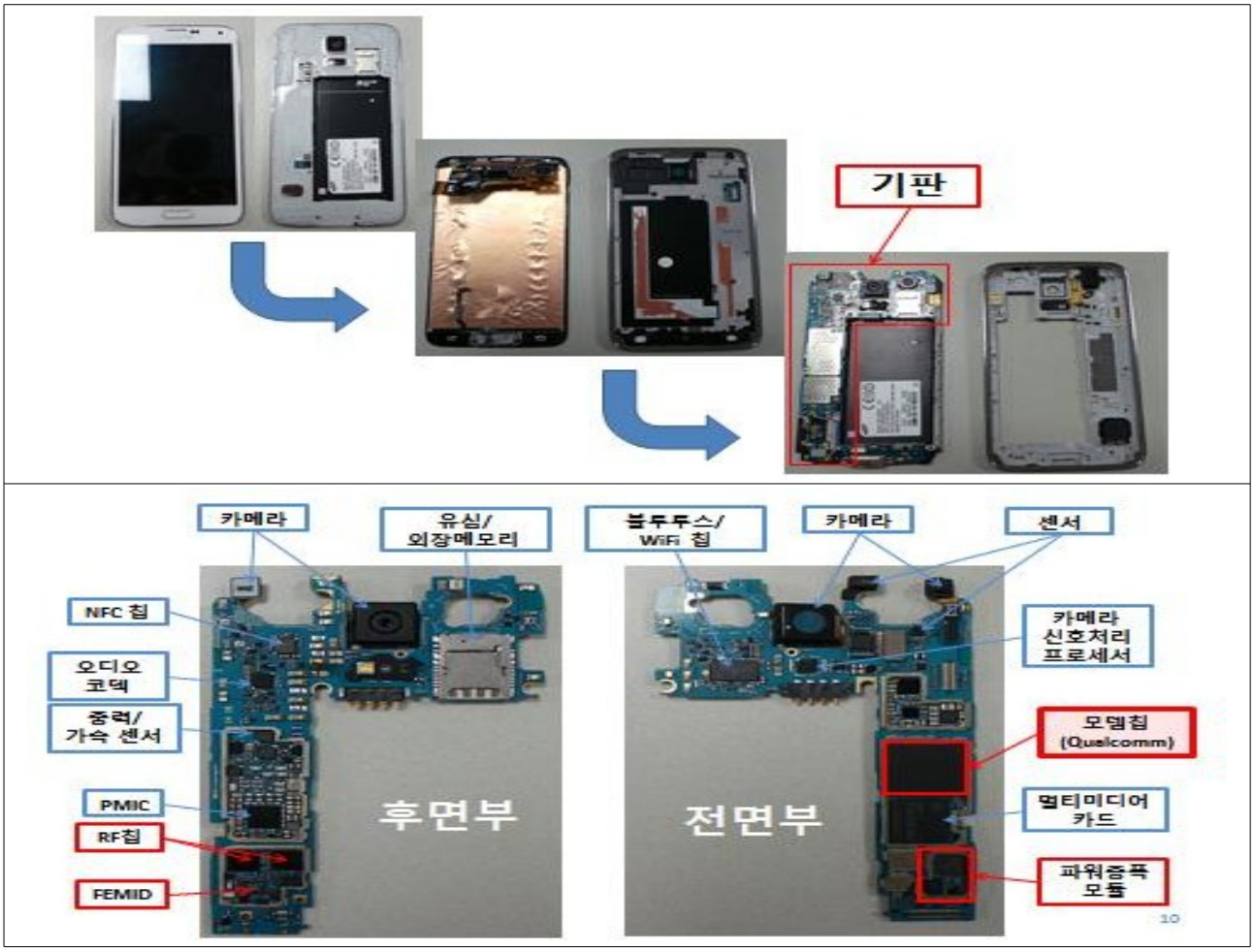
	Company name	Public	Registration	Total
1	Subjects	808	839	1,647 (16.0%)
2	Samsung Electronics	503	536	1,039 (10.0%)
3	Id.	346	560	906 (8.8%)
4	Elgie	398	481	879 (8.5%)
5	Nokia	196	521	717 (6.9%)
6	Ericsson	315	391	706 (6.8%)
7	Huawei	255	176	431 (4.2%)
8	Panasonic	137	260	397 (3.8%)
9	Motorola	110	284	394 (3.8%)
10	nc	170	167	337 (3.3%)
Total number of reports		4620	5708	10,328

b) Parts Market

(1) The parts that make up your phone

46 In the early days of cell phones, they were virtually all mobile phones for voice communication, and the core functions of mobile communication were concentrated in the modem chipset. In contrast, smartphones, which have recently become popular, are considered to be comprehensive IT devices that go beyond the functions of mobile phones. As a result, recent smartphones incorporate convenient operating systems, fast program processing speeds, high-quality cameras, touchscreens, and displays, and various components such as displays, memory, wireless modules, camera modules, application processors (APs), and power management PM chips are used to achieve this. The internal structure and major parts of a cell phone are shown below, and the major parts in red are the mobile communication-related parts.

<Table 13 Your phone's internal structure and major components



47 With the recent transformation of cell phones into feature-rich, all-in-one IT devices, the cost of today's smartphones has risen from $350 to $700, with expensive components such as high-performance cameras and high-resolution displays. As a result, the modem chipset has become an average of $20 to $40, or about 4% of the total cost of the phone.

<Table 14 Phone major components and schematic pricing structure

Part name	Price	Non-weighted
Modem chipsets (communications chipsets)	*** B	*** B
Memory	*** B	*** B
Application processor (AP)	*** B	*** B
High-resolution displays	*** B	*** B
High-performance cameras	*** B	*** B
WiFi, Bluetooth, GPS	*** B	*** B
Battery	*** B	*** B
Other parts	*** B	*** B
(Partial Subtotal)	*** B	*** B
Marketing costs, R&D costs, labor, expenses, profit, etc.	*** B	*** B

* Source: Phone manufacturer submission

(2) Modem Chipset

Market (a)

Modem Chipset

Functions

48 Among cell phone components, a modem chipset is a chipset that is built to perform modulation and demodulation functions on a single chip30). In the case of cellular modems, the chipset must have a high

30) The term modem is a combination of the terms modulation and demodulation. Modulation is the conversion of digital information into symbols or signals suitable for the characteristics of the transmission channel in order to transmit it, while demodulation is the recovery of digital information from symbols or signals that have been converted to be suitable for the channel.

Since the signal is transmitted over a frequency transmission channel, the modulated and output signal from the modem chipset is the new The arc is converted back to a higher frequency and sent to the antenna. Conversely, the high-frequency signal received by the antenna is converted to a lower frequency and input to the modem chipset. For this reason, a modem chipset is also called a baseband chipset or baseband chipset.

49 In recent years, with the development of semiconductor integration technology, AP-modem integrated chipsets have emerged, which integrate AP31) and modem32), which were previously produced separately. AP-modem integrated chipsets have the advantage of lowering the manufacturing cost of mobile phones and making mobile phones smaller than if the modem chip and AP were implemented as separate chips, but they also have the disadvantage of increasing the chipset development period and slowing down the response to technological advancements.

50 On the other hand, although the number of mobile phones that follow the LTE standard has recently increased, the "backward compatibility" of mobile communication standards requires that it is also possible to communicate with mobile phone users that use the older standards of the second generation CDMA and third generation WCDMA, so the modem chipsets currently in use are multi-mode baseband chipsets that support both the fourth generation LTE and the older CDMA and WCDMA mobile communication standards.

(b) Modem chipset market status

51 The modem chipset market has grown significantly since 2008, reaching a total revenue of
This is more than double the 2008 level of USD 21,264 million. During this growth in the overall modem chipset market, the Respondents

have continued to maintain their position as the number one player in the market, with their share increasing significantly from 36.8% in 2008 to 59.4% in 2015. During the same period, most of the other modem chipset manufacturers either lost share or exited the market, while new entrants to the modem chipset market gained meaningful share.

31) An application processor (AP) is a processor in a mobile phone that runs the operating system and application programs and controls many peripherals and interfaces, such as user memory, camera, keypad, display, etc.

32) The subjects' share of the global smartphone AP market (including AP-modem integrated chipsets) was 34.8% in 2013, and in 2014
It continued to rise to 36.6% in Q1 and 41.4% in Q2 2014.

33) The term "LTE modem chipset" is used to include not only modem chipsets that support the LTE standard alone, but also multimode chipsets that support backward standards such as CDMA or WCDMA simultaneously, as is the case with "WCDMA modem chipset".

There was no chipset manufacturer.[34)]

<Table 15 Global Modem Chipset Market Status[35)]

(Unit: million $)

	2015							
Subjects	4,092 (36.8%)	4,417 (40.0%)	5,248 (39.9%)	6,764 (45.0%)	9,174 (52.6%)	12,251 (63.1%)	14,660 (66.1%)	12,626 (59.4%)
MediaTek	1,291 (11.6%)	1,890 (17.1%)	1,911 (14.5%)	1,736 (11.5%)	2,132 (12.2%)	2,486 (12.8%)	3,737 (16.9%)	4,133 (19.4%)
Spreadtrum	98 (0.9%)	78 (0.7%)	295 (2.2%)	530 (3.5%)	647 (3.7%)	967 (5.0%)	1,125 (5.1%)	1,452 (6.8%)
Samsung Electronics	-.	-.	13 (0.1%)	72 (0.5%)	103 (0.6%)	139 (0.7%)	231 (1.0%)	1,249 (5.9%)
Intel	643 (5.8%)	872 (7.9%)	1,590 (12.1%)	2,315 (15.4%)	2,227 (12.8%)	1,470 (7.6%)	561 (2.5%)	346 (1.6%)
Mabel	125 (1.1%)	154 (1.4%)	250 (1.9%)	316 (2.1%)	345 (2.0%)	508 (2.6%)	674 (3.0%)	353 (1.7%)
VIA	45 (0.4%)	62 (0.6%)	96 (0.7%)	152 (1.0%)	156 (0.9%)	108 (0.6%)	72 (0.3%)	35 (0.2%)
Broadcom	78 (0.7%)	240 (2.2%)	483 (3.7%)	813 (5.4%)	889 (5.1%)	699 (3.6%)	365 (1.6%)	-.
Nvidia	14 (0.1%)	32 (0.3%)	68 (0.5%)	89 (0.6%)	48 (0.3%)	5 (0.0%)	12 (0.1%)	-.
Ericsson	1,417 (12.7%)	1,134 (10.3%)	1,045 (7.9%)	770 (5.1%)	896 (5.1%)	343 (1.8%)	16 (0.1%)	-.
Renesas	157 (1.4%)	143 (1.3%)	175 (1.3%)	142 (0.9%)	44 (0.3%)	5 (0.0%)	-.	-.
TI	2,538 (22.8%)	1,725 (15.6%)	1,715 (13.0%)	1,026 (6.8%)	288 (1.7%)	26 (0.1%)	-.	-.
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
Total	11,120	11,042	13,140	15,034	17,437	19,415	22,165	21,264

* Source: Strategy Analytics 'Baseband Market Share Tracker'

34) Samsung manufactures only modem chipsets for self-consumption under its patent license agreements with the Defendants.

35) The share is based on revenue a n d is as follows

52 This trend is also true for the respective modem chipset markets for CDMA, WCDMA, and LTE standards.

In the CDMA modem chipset market, the incumbents have consistently maintained a share of more than 90% until recently, and in the LTE modem chipset market, where their share of the total modem chipset market has increased dramatically in recent years, they remain the dominant operator, with a share of 96% in 2013 and a share of 70% until recently.

<Table 16 Worldwide CDMA Modem Chipset Market Share

Business name	2008	2009	2010	2011	2012	2013	2014	In 2015
Subjects	98.4	97.6	96.4	94.3	92.4	93.1	91.6	83.1
TAI	0.0	-.	-.	-.	-.	-.	-.	-.
VIA	1.6	2.4	3.6	5.7	7.6	6.9	8.4	16.9
Total	100	100	100	100	100	100	100	100

* Source: Strategy Analytics 'Baseband Market Share Tracker'

<Table 17 Global LTE modem chipset market share

	2015							
Subjects	-.	-.	34.2	58.8	94.5	96.0	84.8	69.4
MediaTek	-.	-.	-.	-.	-.	-.	4.1	13.7
Samsung Electronics	-.	-.	65.8	26.9	3.1	2.0	1.9	7.9
Spreadtrum	-.	-.	-.	-.	0.0	0.0	0.3	0.9
High silicon	-.	-.	-.	0.1	0.0	0.3	3.4	3.7
Intel	-.	-.	-.	-.	-.	0.7	1.2	1.0
Sequences	-.	-.	-.	0.0	0.0	0.0	0.1	0.2
Mabel	-.	-.	-.	-.	-.	0.2	3.3	2.0
Altair	-.	-.	-.	0.1	0.0	0.2	0.4	0.3
Leadcore	-.	-.	-.	-.	-.	-.	0.0	0.9
GCC	-.	-.	-.	4.4	0.6	0.4	0.1	0.0
Broadcom	-.	-.	-.	-.	-.	0.0	0.1	-.
Nvidia	-.	-.	-.	-.	0.0	0.0	0.1	0.0
Ericsson	-.	-.	0.0	0.0	0.0	0.0	0.0	-.
Renesas	-.	-.	-.	-.	-.	0.0	-.	-.
Other	-.	-.	-.	9.7	1.7	0.2	0.0	0.0
Total	-.	-.	100	100	100	100	100	100

* Source: Strategy Analytics 'Baseband Market Share Tracker'

53 In the WCDMA modem chipset market, the subjects believe that by 2005, the market share will be 20%.

However, since then, the market share of WCDMA modem chipsets has been rising steadily, maintaining a share of more than 50% from 2011 to 2013 and reaching 53.9% in 2013. However, although the Respondents' share of the WCDMA modem chipset market has declined in recent years, the rapid realignment of the modem chipset market toward the LTE standard since 2014 has reduced the size of the WCDMA modem chipset market to between one-third and one-fourth of the LTE modem chipset market in 2015, and has become more pronounced in recent years, with the WCDMA modem chipset market being less than one-tenth of the LTE modem chipset market in the third quarter of 2016. The Defendants have not introduced any new models of WCDMA modem chipsets since 2013.

<Table 18 Worldwide WCDMA Modem Chipset Market Share

Business name	2008	2009	2010	2011	2012	2013	2014	In 2015
Subjects	38.8	47.4	45.7	55.0	50.4	53.9	48.8	32.3
MediaTek	-.	-.	0.7	2.3	11.1	15.5	31.2	35.9
Spreadtrum	-.	-.	-.	-.	0.0	0.9	7.4	23.9
High silicon	0.4	0.7	0.5	0.7	0.6	0.2	0.5	0.5
Intel	4.7	9.5	14.4	16.8	15.6	11.8	3.8	2.9
AldiA	-.	-.	-.	-.	-.	-.	0.5	2.6
Mabel	0.9	1.8	3.0	3.0	1.4	4.3	2.9	0.8
Rockchip	-.	-.	-.	-.	-.	-.	-.	0.5
Freescale	2.2	1.0	0.5	0.3	0.1	0.0	-.	-.
Broadcom	2.1	1.3	1.3	3.9	9.7	9.3	4.7	0.5
Nvidia	0.4	0.8	1.2	1.2	0.6	0.1	0.0	0.0
Ericsson	15.1	11.7	6.8	3.6	7.1	3.4	0.1	-.

Renesas	4.6	3.6	3.0	1.9	0.5	0.1	-.	-.
TAI	30.7	22.3	22.9	11.3	3.0	0.3	-.	-.
Other	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0
Total	100	100	100	100	100	100	100	100

* Source: Strategy Analytics 'Baseband Market Share Tracker'

<Table 19 Global WCDMA and LTE modem chipset revenue as a percentage of total revenue

(Unit: Million$)

	2012	2013	2014	In 2015	In 2016		
					Q1	Q2	Q3
LTE	3,296	6,884	12,051	15,864	4,137	4,879	5,302
WCDMA	8,216	7,453	7,406	4,471	632	558	477

* Source: Strategy Analytics 'Baseband Market Share Tracker'

(C) Participants in the modem chipset market

Subjects

54 The Defendants first produced modem chipsets in 1999, and later began producing AP-modem integrated chipsets, and in 2008 began selling integrated chipsets supporting GSM, WCDMA, and CDMA2000 standards. In 2012, Defendants introduced AP-modem integrated chipsets that support GSM, WCDMA, CDMA2000 standards as well as LTE standards. As of 2015, the Defendants held a 59.4% share of the global modem chipset market.

Modem chipset manufacturers other than the Respondents

55 (i) MediaTek: Headquartered in Taiwan, MediaTek entered the WCDMA modem chipset market around 2010. It mainly supplies chipsets used in mid- to low-end smartphone products in Asian markets such as China, and since 2014, it has also been selling LTE modem chipsets. As of 2015, the company held a 19.4% share of the global modem chipset market.

56 (ii) Spreadtrum: Spreadtrum, a Chinese company, has been developing modem chips since 2004 and currently offers

modem chipsets that support GSM, W C D M A , LTE, e t c .

As of 2015, the company had a 6.8% share of the global modem chipset market.
is in the

57 (iii) Samsung Electronics: Samsung Electronics has been developing modem chipsets since 1999, and has recently produced and used modem chips and AP-modem integrated chips that support GSM, WCDMA, LTE, etc. in its smartphone models released in some regions, including Korea. As of 2015, Samsung Electronics held a 5.9% share of the global modem chipset market. However, Samsung Electronics is prohibited from selling its modem chipsets to other mobile phone manufacturers other than for its own consumption under the patent license agreements with the Defendants.

58 (iv) Intel: A semiconductor manufacturer based in the United States, Intel acquired the modem business of Germany's Infineon, which had GSM and WCDMA modems and RF technologies and products, around the end of 2010, and currently supplies modem chipsets and AP-modem integrated chip products that support WCDMA and LTE.37) As of 2015, it held a 1.6% share of the global modem chipset market.

59 (v) VIA: VIA is a Taiwan-based company that began its CDMA modem chipset business around 2002 and has been the only company other than the Respondents to supply CDMA modem chipsets since 2009. As of 2015, it had a 0.2% share of the global modem chipset market and a 16.5% share of the CDMA modem chipset market. However, it does not supply multimode modem chipsets capable of supporting WCDMA and LTE standards and mainly supplies CDMA modem chipsets for low-end mobile phones.

60 (vi) Broadcomm: A U.S.-based company that entered the modem chipset market around 2004 and launched modem chipsets supporting

GSM, WCDMA, LTE, etc. and AP-modem integrated modem chipsets, but withdrew from the modem chipset business in June 2014.

36) Spreadtrum was acquired by Tsinghua Uni Group in July 2013.
37) In 2014, Intel acquired a 20% stake in Tsinghua Unigroup, which has Spreadtrum and RDA as subsidiaries, and jointly develops and markets AP-modem integrated chipset products with Spreadtrum and RDA.

61 (vii) Nvidia: Nvidia, an American semiconductor company, announced in May 2011 that it had acquired the rights to the Ise
Ltd. in February 2013, the company launched AP-modem integrated chips supporting GSM, WCDMA, and LTE, but withdrew from the modem chipset business in May 2015.

62 (vii) Ericsson (ST-Ericsson): Ericsson, a Swedish telecommunications equipment company, spun off its wireless business in February 2009 to form a joint venture with Estee, but Estee-Ericsson was dissolved in August 2013. Ericsson then reacquired the modem chipset business of Estee-Ericsson, but withdrew from the modem chipset business in September 2014.

63 (xi) Eonex: A Korean modem chipset developer and designer founded in 2000, Eonex supplied modem chipsets to LG Electronics and others, but was closed around 2009.

c) Mobile phone market

64 The cell phone market, a sub-market of the modem chipset market, has been dominated by Samsung Electronics, Apple, and Nokia, but recently Nokia's share has been declining and Apple's share has been increasing38).

<Table 20 Global mobile phone market units sold and share by vendor

(Unit: million

volume (Share)

	2010	2011	2012	2013	2014	In 2015
Samsung Electronics	280.2 (20.6%)	327.4 (21.2%)	396.5 (25.1%)	451.7 (26.8%)	405.0 (22.1%)	390.0 (20.7%)
Apple	47.5 (3.5%)	93.0 (6.0%)	135.8 (8.6%)	153.4 (9.1%)	192.7 (10.5%)	231.5 (12.3%)
Huawei	30.9 (2.3%)	53.8 (3.5%)	48.3 (3.1%)	57.0 (3.4%)	76.4 (4.2%)	107.6 (5.7%)

Nokia (MS)	453.0 (33.3%)	417.1 (27.0%)	335.6 (21.2%)	252.4 (15.0%)	199.7 (10.9%)	119.8 (6.4%)

38) Apple does not manufacture its cell phones directly, but rather uses third parties, such as Foxconn, to do so. As a result, Apple does not have a direct patent license agreement with the accused, but instead has an indirect licensing structure in which Apple purchases products from a contract manufacturer (such as Foxconn) that has a licensing relationship with the accused, and the contract manufacturer pays the accused patent license fees, which Apple then reimburses to the contract manufacturer.

LG Electronics	116.7 (8.6%)	88.1 (5.7%)	56.6 (3.6%)	71.0 (4.2%)	78.1 (4.3%)	72.1 (3.8%)
TCL - Alcatel	29.7 (2.2%)	39.7 (2.6%)	39.5 (2.5%)	52.0 (3.1%)	70.3 (3.8%)	71.5 (3.8%)
Xiaomi	-.	-.	5.7 (0.4%)	18.7 (1.1%)	61.1 (3.3%)	72.0 (3.8%)
Zitiyi	50.7 (3.7%)	78.1 (5.1%)	71.7 (4.5%)	59.8 (3.5%)	54.7 (3.0%)	62.7 (3.3%)
Lenovo-Motorola	46.0 (3.4%)	55.6 (3.6%)	59.6 (3.8%)	63.6 (3.8%)	93.6 (5.1%)	74.5 (4.0%)
Other	305.3 (22.4%)	393.2 (25.4%)	430.7 (27.3%)	505.4 (30.0%)	600.0 (32.8%)	681.1 (36.2%)
Total	1360.0 (100%)	1546.0 (100%)	1580.0 (100%)	1685.0 (100%)	1831.6 (100%)	1882.8 (100%)

* Source: Strategy Analytics 'Global Handset Vendor Marketshare for 15 Countries'

65 The domestic mobile phone market is dominated by domestic mobile phone manufacturers such as Samsung Electronics, LG Electronics, and Pantech, and Apple is the only foreign mobile phone manufacturer that has consistently maintained a share of more than 1% in the domestic mobile phone market since 2009.

<Table 21 Domestic cell phone market share

	2009	2010	2011	2012	2013	2014	In 2015
Samsung Electronics	51.6	51.8	52.5	58.8	56.3	56.6	54
Elgie	28.7	20.3	16.7	13.5	20.6	21.4	17.2
Apple	-.	6.5	10.8	6.1	5.8	10.3	17.2
Pantech	13.7	14.3	13.8	19.3	16.6	7.3	1.5
Blackberry	-.	-.	-.	0.2	-.	-.	-.
HTC	-.	-.	-.	1.2	-.	-.	-.
Nokia	-.	-.	-.	0.2	-.	-.	-.
Other	6.1	7.2	6.2	0.6	0.8	4.4	10
Total	100	100	100	100	100	100	100

* Source: Strategy Analytics 'Global Handset Vendor Marketshare for 15 Countries'

2. Determining illegality

A. Admitted facts and grounds

1) The status of the subjects in the telecommunications industry

66 Since its establishment in 1985, the Defendants have conducted business in the telecommunications industry, including base stations, mobile phones, and components, but discontinued and sold the base station and mobile phone businesses in the late 1990s, and since 2000 have focused primarily on the patent licensing business and the manufacture and sale of modem chipsets. Since then, the Defendants have been engaged in the patent licensing business and the manufacture and sale of modem chipsets. 3) above, the Defendants have established a dominant position as vertically integrated companies in both the mobile communication patent licensing and modem chipset markets, and have continued to maintain their dominant position in both the patent licensing market and the modem chipset market even as the mobile communication standards have changed to WCDMA and LTE.

2) Declaration of FRAND commitments to standardization bodies and acquisition of standard essential patents by the subjects.

67 When CDMA was selected as the second generation telecommunications standard in 1993, WCDMA, CDMA2000, etc. were selected as the third generation standard in the late 1990s and early 2000s, and LTE was selected as the fourth generation standard in the late 2000s, Defendants made FRAND commitments to standardization organizations such as ETSI, TIA, ITU, and TTA that they would license their patented technologies on fair, reasonable, and non-discriminatory terms.

¶ 68. Defendants also announced in a March 26, 1999 press release that they would "license standard essential patents to all participants in the industry under FRAND commitments."[39)]

39) At the time, the Respondents refused to sign the FRAND agreement in a confrontation with Ericsson, and the International Telecommunication Union (ITU) announced that it would exclude the Respondents' technology from the 3G standard. The Defendants then issued the above press release announcing that they would "offer licenses under FRAND commitments to all participants in the industry". The full text of the Defendants' announcement is as follows "As part of the agreement, the companies will each commit to the ITU and to other standard bodies to license their essential patents for a single CDMA standard or any of its modes to the rest of the industry on a fair and reasonable basis free from unfair discrimination." (https://www.qualcomm.com/news/releases/1999/03/25/ericsson-and-qualcomm) -reach-global-cdma-resolution)

3) Overview of the subjects' business model

69 In licensing their CDMA, WCDMA, and LTE-related mobile communications standards essential patents, Defendants skip the modem chipset stage and license at the handset stage, charging handset manufacturers a royalty based on the total selling price of the handset. This handset-level licensing policy is linked to the decoupling of licensing policies from modem chipset sales.

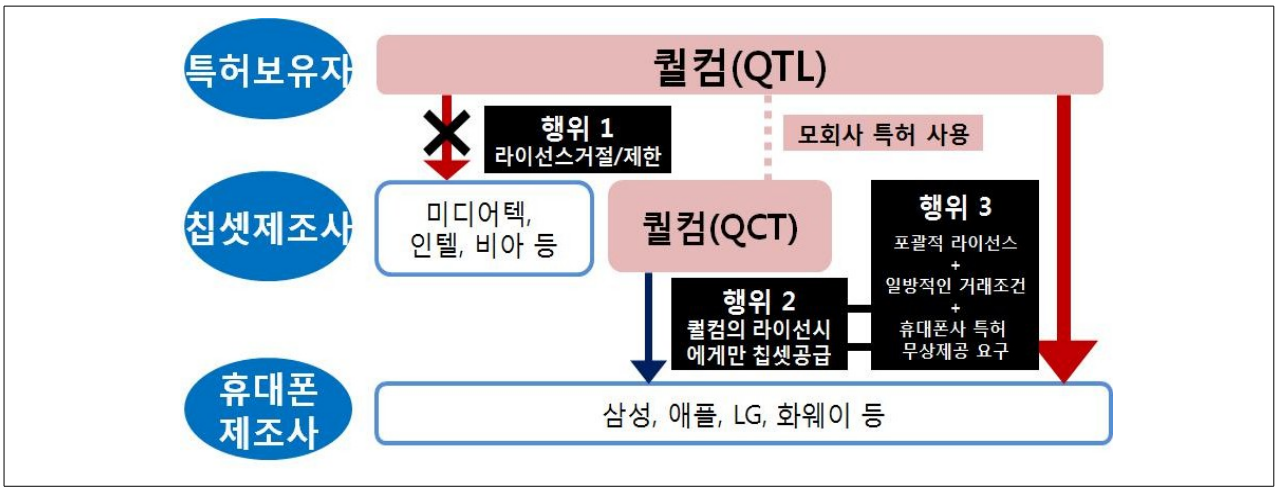
70 Generally, when a patentee legally sells a product, the patentee's patent rights restricting the use of the product are exhausted, and the user or purchaser of the product does not need to enter into a separate license agreement with the patentee.[40] In other words, if a modem chipset manufacturer acquired a license from a standard-essential patent holder for a standard-essential patent embodied in a modem chipset and then manufactured and sold the modem chipset to a mobile phone manufacturer, the mobile phone manufacturer would not have to pay any additional royalties to the standard-essential patent holder because the royalties for the standard-essential patent were already included in the purchase price of the modem chipset.

71 However, the Defendants have a business policy of separating their mobile communication patents from the sale of modem chipsets and entering into separate license agreements at the mobile phone stage, so that mobile phone manufacturers who purchase competing modem chipsets must enter into separate license agreements with the Defendants. After 2000, when the Defendants discontinued and sold their mobile phone business, including base station equipment and mobile phone manufacturing and sales business, and focused on patent licensing business and modem chipset manufacturing and sales business, they entered

into modem chipset supply contracts with mobile phone manufacturers, stating in the modem chipset supply contracts that their patents embodied in the modem chipsets would not be transferred even if they sold the modem chipsets.

40) This is known as the doctrine of patent exhaustion, which states that when a licensee sells a product embodying a patent under the licensee's lawful authority, the rights acquired by the licensee pass through to the licensee's customers, and the rights are exhausted. Thus, the licensee's customers can utilize the product as they wish without the need to enter into a separate patent agreement.

<Table 22 Overview of the subjects' business models



72 The Defendants' business policy is reinforced by the interconnectedness and circularity of each of the acts in the course of its execution, including the refusal to enter into license agreements for its essential telecommunications standard patents with competing modem chipset manufacturers (Act 1), the linking of modem chipset supply contracts to mobile phone manufacturers to patent license agreements (Act 2), and the entering into of patent license agreements with mobile phone manufacturers based on these agreements (Act 3), as follows

73 First, Defendants refuse to fully license their telecommunications standard essential patents to competing modem chipset manufacturers, and refuse or limit the scope of licenses to competing modem chipset manufacturers when they request them. The result is that modem chipsets do not license the Defendants' standard-essential patents, forcing mobile phone manufacturers to enter into separate patent licenses with the Defendants in order to purchase modem chipsets.

74 Second, within this structure, Defendants require cell phone manufacturers that wish to purchase their modem chipsets to first enter into a license agreement with them, and then refuse to supply,

discontinue, or restrict the supply of their modem chipsets to cell phone manufacturers that do not enter into a license agreement with them or fail to properly perform under the license agreement.41 Defendants

41) In addition, even if a mobile phone manufacturer supplies a modem chipset, it cannot use the modem chipset unless it enters into a license agreement under the provisions of the modem chipset supply agreement.

This is stipulated in the modem chipset supply contract, which means that the cell phone manufacturer is required to provide a modem chipset to the

Beyond the situation where a patent license agreement must be entered into separately from the purchase of a demo chipset, a structure has been established that links the supply of a demo chipset to the execution and performance of a license agreement, whereby a patent license agreement must be entered into and performed with the defendants "in advance" in order to receive and use the defendants' demo chipsets.43)

75 Third, in the course of entering into patent licensing agreements with mobile phone manufacturers seeking to supply their modem chipsets, Defendants (i) entered into "blanket license agreements" covering all of their patents, including not only the essential patents of the telecommunications standards implemented in their modem chipsets, but also other patents44), and (ii) charged a royalty of a percentage of the total selling price of the mobile phone45) in return, (iii) cross-licensed from the mobile phone manufacturers the mobile phone manufacturer's patents necessary for the manufacture, use, and sale of their modem chipsets at no cost, and created a "patent umbrella"46) around their modem chipsets by preventing the mobile phone manufacturer from asserting the mobile phone manufacturer's patents against them and against other customers who purchase their modem chipsets (cross-grant). Defendants then used the benefits of the patent umbrella to their advantage in the modem chipset business by passing on the benefits of the patent umbrella to customers who purchased their modem chipsets, thus creating a reciprocal cycle between the patent licensing business and the modem chipset business.47)

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42) Even when Defendants entered into restrictive agreements with modem chipset manufacturers, they limited the distribution of modem chipsets to their licensees, making it difficult for mobile phone manufacturers to purchase not only Defendants' modem chipsets but also those of competing modem chipset manufacturers without entering into a license with Defendants.

43)In their response to the Commission, Respondents stated that "Qualcomm's policy is not to

enter into component supply agreements with unlicensed handset manufacturers." (Exhibit 34, Respondents' Response to Commission's Request for Production).

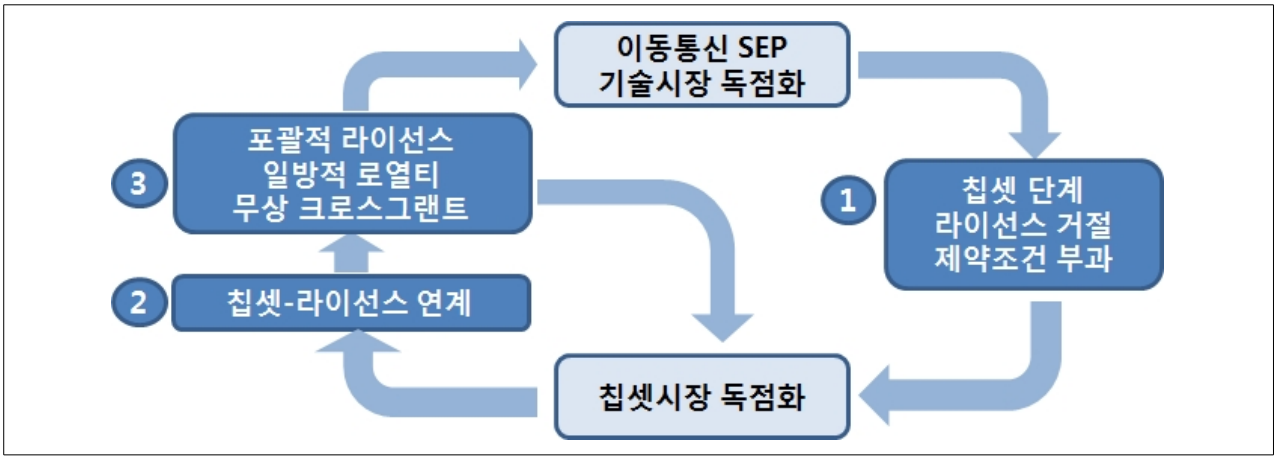
44) During the hearing, Respondents suggested that there are some examples of license agreements (***, ***, ***, ***, ***) that are not comprehensive and are limited to essential patents for mobile communications standards. However, in their rebuttal to the Commission, the Respondents pointed out that "Qualcomm's hundreds of license agreements with mobile phone manufacturers around the world are almost all substantially similar in structure and terms. More specifically, almost all of these license agreements provide rights to Qualcomm's patent portfolio (either by way of license or sublicense) for the manufacture, sale, and distribution of handsets that use 2G and/or 3G and 4G mobile communications technologies." (Exhibit 32 to Respondents' Response to First Request for Information).

45) The fee varies slightly by carrier and phone manufacturer, but is typically around *% of the net selling price of the phone.

46) The huge shield that protects cell phone makers from numerous patent infringements by preventing them from bringing patent infringement claims against the defendants' modem chipset customers.

47) In 2012, Respondent QI separated its patent licensing business (QTL) and modem chipset business (QCT) into separate entities, with the patent licensing business being conducted by itself and the modem chipset business being conducted by its subsidiary QTI.

<Table 23 Organic circle of the subjects' business models



76 The following describes in detail each of the Defendants' conduct 1, 2, and 3 in building, enhancing, and maintaining the above business model.

4) Act 1: Refusing to license or restricting essential telecommunications standard patents to competing modem chipset manufacturers

a) Licenses with limited permissions and other restrictions (before 2008)

77 While Defendants entered into patent licensing agreements with mobile phone manufacturers for their mobile communication standard essential patents, they also entered into limited-scope license agreements with modem chipset manufacturers, including ***, ***, ***, and ***, whereby mobile phone manufacturers received royalties based on the price of the mobile phone and modem chipset manufacturers received royalties based on the price of the modem chipset. In other words, prior to 2008, Defendants licensed at both the modem chipset stage and the mobile phone stage, but the license agreements Defendants entered into with modem chipset manufacturers were subject to the following conditions

78 (a) Limitation of license rights: Defendants excluded the right to

"use" the modem chipsets from the scope of the license.[48]) As a result, competing modem chipset manufacturers have been unable to compete with Defendants.

As mentioned above, even after the separation of the business and the establishment of a separate subsidiary, the modem chipset business and the patent license business are operated in conjunction with one business policy.

48) License rights include rights to make, sell, rent, lease, use, repair, and otherwise dispose of, and are subject to the terms and conditions of the respective

can only manufacture and sell modem chipsets while paying licensing fees, not competing modem chipsets.
The modem chipsets manufactured and sold by the three manufacturers did not include the right to use them. Therefore, in order for a mobile phone manufacturer to purchase and use a modem chipset from a competing modem chipset manufacturer, it had to enter into a separate patent license agreement with the Respondents.

79 Restriction of modem chipset sales outlets: The Respondents restricted the sales outlets of competing modem chipset manufacturers' modem chipsets to mobile phone manufacturers that signed license agreements with them. As a result, competing modem chipset manufacturers could not sell their modem chipsets according to their own business judgment, even if they had entered into a license agreement with the Respondents, but had to check whether the mobile phone manufacturer had entered into a separate license agreement with the Respondents at the mobile phone stage before selling the modem chipset.

80 (3) Obligation to report sales information: Under a contractual structure that provided limited licenses to competing modem chipset manufacturers while simultaneously collecting royalties from mobile phone manufacturers, the defendants required modem chipset manufacturers that entered into license agreements with them to report sales information, including sales volume and purchasers of modem chipsets, sales volume per purchaser, date of purchase, product model, price, etc.

81 Free cross-granting: Defendants requested and received agreements from competing modem chipset manufacturers that limited the rights subject to licensing, but prohibited competing modem chipset manufacturers from offering patent licenses to Defendants and customers purchasing Defendants' modem chipsets or claiming patent infringement.

82 Table 24 below summarizes the terms and conditions of Defendants' agreements with competing modem chipset manufacturers by modem chipset manufacturer and when the agreements were entered into.

The acts constitute separate and distinct patent infringement. However, the license granted by Defendants to the modem chipset manufacturer included the right to manufacture and sell, but excluded the right to use.

49) Although the Respondents did not collect royalties after 2008 because they refused to license to competing modem chipset manufacturers, they continued to be obligated to report such sales information. On the other hand, the Respondents' patent license agreements with mobile phone manufacturers also provided for reporting obligations, but these were limited to information necessary to determine the appropriateness of the royalties (such as the total number and value of sales during the period and the amount of royalty deductions).

<Table 24 Contract terms and time of contract signing by modem chipset manufacturer50)

	Contract terms	Contract name and when it was signed
*** B	- Restrict use permissions in addition to make and sell modem chipset permissions - Restricting Distributors to Licensees - Management information needs such as pricing, sales volume, buyers, etc. - Requiring a free cross-license and sublicense agreement	B ***B ○ Signing date: 2002 *Month*Day
*** B	- Restrict the use permission in addition to the make and sell modem chipset permissions. - Restricting Distributors to Licensees - Management information needs such as pricing, sales volume, buyers, etc. - Requiring a free cross-license and sublicense agreement	B ***B ○ Signing date: 2005 *Month*Day - Amended Agreement: *Month *Date 2010
*** B	- Restrict the use permission in addition to the make and sell modem chipset permissions. - Restricting Distributors to Licensees - Management information needs such as pricing, sales volume, buyers, etc. - Requiring a free cross-license and sublicense agreement	B ***B ○ Signing date: 2004 *Month*Day - Amended Agreement: *Month *Date 2010

*** B	- Restrict the use permission in addition to the make and sell modem chipset permissions. - Restricting Distributors to Licensees - Management information needs such as pricing, sales volume, buyers, etc. - Requiring a free cross-license and sublicense agreement	B ***B ○ Signing date: 1997 *Month*Day - Amended and Restated Agreement: *Month *Date 2006
*** B	- Restrict the use permission in addition to the make and sell modem chipset permissions. - Restricting Distributors to Licensees	B ***B ○ Signing date: 2009 *Month*Day * *March 2013 ***Distributed by contract

50) Defendants are modem chipset manufacturers with written contracts in effect as of August 2014 with ***, ***, ***, ***, ***, ***,
A list of * companies was submitted: ***, ***, ***, ***, ***, ***, ***, ***, etc. Among them, *** and *** are distinguished in that they are *** contracts. In addition to the above companies, the Respondents have also signed contracts with other modem chipset companies such as *** and ***, which are currently invalid.

	– Management information needs such as pricing, sales volume, buyers, etc. – Requiring subtitle covenants for third parties (licensees of subjects)	Delete a restriction
*** B	– Restrict use permissions in addition to make and sell modem chipset permissions – Restricting Distributors to Licensees	B ***B ○ Signing date: 2002 *Month*Day
*** B	- Restrict use permissions in addition to make and sell modem chipset permissions	○ ***51) ○ Signing date: 2009 *Month*Day
***52)	- Restrict use permissions in addition to make and sell modem chipset permissions	B ***B ○ Signing date: 2000 *month*day

83 This fact has been consistently acknowledged by the Respondents throughout the investigation and at the hearing, and is reflected in the *** Patent License Agreement between *** and the Respondents (Exhibit 5 to the Examiner's Report, Exhibit 53),

*** Patent License Agreement (Exhibit 6), dated ***, by and between *** and Defendants; *** Patent License Agreement (Exhibit 6), by and between *** and Defendants.

It is also recognized in the Patent License Agreement (Exhibit 7), dated ***.

b) Refusal to enter into a license agreement or offer terms with non-licensing restrictions (since 2008)

51) The agreement between the Respondents and *** was to permanently settle the patent disputes that the parties have been engaged in worldwide since 2005 for at least three years and to withdraw any reports or complaints filed by *** with competition authorities, courts, etc. worldwide.

of the Settlement Agreement. In consideration for the Settlement, Respondents [redacted] approximately [redacted] U.S. Dollars ($ [redacted]).
However, the situation was the same as with other chipset companies in that the Respondents did not provide *** with a full and unrestricted license, and purchasers of ***'s mobile modem chipsets were still exposed to the risk of patent attacks by the Respondents.

52) *** stated that it requested a license to the CDMA patents from the Respondents in 1997, but that the Respondents refused for more than three years, until December 2000, when they entered into a patent portfolio arrangement instead of a full license agreement, making it impossible for it to compete normally in the modem chipset market.

53) Hereinafter referred to as the "Examination Report" is abbreviated as the "Examination Report".

(1) Licensing at the modem chipset stage due to refusal to sign a license agreement for telecommunications standard essential patents Change a policy

84 Defendants had traditionally entered into patent license agreements with mobile phone manufacturers and infrastructure equipment manufacturers, as well as component manufacturers, subject to limited royalty payments, but between 2006 and 2008, Defendants adopted a business policy that changed their licensing program.

85 The Respondents stated in their business reports until 2007 that they had granted licensees to manufacture and sell modem chipsets through licenses or agreements to modem chipset manufacturers, but in their business reports from 2008 onwards, they deleted all references to licenses or licensees, further clarified that the commitments to modem chipset manufacturers do not grant exhaustive rights to the Respondents' patents, and that even if a mobile phone manufacturer uses a modem chipset from a competing modem chipset manufacturer, it must pay royalties to the Respondents under a separate license agreement for the volume of mobile phones.

86 In their presentation at the Taiwan Conference on February 26, 2008, Respondents stated that they had changed their position to not provide even limited rights to modem chipset manufacturers because they believed that the existing arrangements would exhaust their patents.

87 (3) Defendants publicly disclosed this business policy in their February 20, 2013 "Qualcomm Technology Licensing" document, which classified competing modem chipset manufacturers as a group outside of their standard licensing program.

88 On the other hand, after changing their business policy, unlike before 2008, when they offered licenses, even on a limited basis, Defendants refused to enter into license agreements with competing modem chipset manufacturers, even when they requested to enter into license agreements for essential mobile communication standard patents that are necessary for the manufacture, sale, and use of their modem chipsets, and only offered "limited commitments" rather than license agreements.

54), the same conditions that have traditionally been required in restrictive license agreements.

The defendants were required to include all of the terms of the agreement, including: (1) a requirement to limit sales of the modem chipsets to mobile phone manufacturers with whom they had entered into a license agreement; (2) a requirement to report quarterly to the defendants sales information such as sales volume, purchasers, sales volume by purchaser, and prices; and (3) a requirement to cross-grant patents held by the modem chipset manufacturers at no cost.

89 This fact has been consistently acknowledged by Respondents throughout the investigation and proceedings, and is supported by Respondents' responses to the Tribunal's requests for documentation (Exhibit 32), Respondents' business reports (Exhibits 37 and 38), Respondents' speaker fees for the February 2008 Taiwan Conference (Exhibit 46), and Respondents' presentation of their licensing program (Exhibit 49).

(2) Decline to sign a license agreement and offer a limited commitment with strings attached

: Case A

A first introduced modem chipsets implementing the second generation GSM communication standard in 200* and sold mainly GSM-based modem chipsets until 2008. When the third-generation WCDMA standard spread globally, it requested the Respondents to enter into a patent license agreement in order to expand its business to WCDMA modem chipsets after 2008.

54) Typically, a patentee grants an "exhaustive license" to a licensee, whereby the exhaustive rights acquired by the licensee are passed through to the licensee's customers upon the licensee's lawful sale of the patented product. In contrast, Defendants' proposed restrictive covenant differs from an exhaustive license in the following ways

① Covenant not to sue: A contractual agreement in which the patentee agrees not to assert

patent rights against the licensee, usually adopted as a result of a settlement after a patent dispute. It is unclear whether there is an exhaustive transfer of rights to the licensee's customers.

② covenant to exhaust remedies: This is a contractual agreement that the patentee will not assert patent rights against the licensee first, and the patentee will assert patent infringement against third parties using the licensee's products first, and then assert patent rights against the licensee as a last resort, thus creating a kind of joint and several liability for the licensee. The contract makes it clear that the patentee can assert infringement against the licensee's customers, thus limiting exhaustive assignment.

③ Stand still: This is a contractual agreement that the patentee will not assert its patent rights for a limited period of time (60 or 90 days). Therefore, after a certain period of time, the licensee can file a patent infringement claim at any time. The patentee as well as the licensee's customers are exposed to the patentee's infringement attack after a certain period of time, and the exhaustion right transfer is blocked.

(a) 2009 Subtitle Covenant

90 Since January 2008, Respondents have refused A's requests to enter into a license agreement for the WCDMA mobile telecommunications standard essential patents and have only provided A with a sub-license agreement with restrictions.

91 A requested that Respondents provide the terms of a license agreement for the WCDMA patents via email in January 2008, to which Respondents responded in January 2008.
*On July 1, the Respondents responded that they would provide a draft WCDMA modem chipset license agreement for A to review. However, despite repeated requests by A, the Respondents failed to provide a negotiated license agreement for more than four months without providing any explanation.

<Table 25 Excerpts of emails between A and subjects55)

55) Exhibit 41, Emails Between A and the Subjects

92 However, on *March* 2008, the Respondents proposed to A by e-mail to enter into a side agreement56) with different terms and conditions than the original proposal. Moreover, the proposed sublicense agreement required A to (1) sell the modem chipsets only to licensees designated and notified by the Respondents; (2) report to the Respondents on a quarterly basis the sales volume of A's modem chipsets, as well as sales information related to the sales of the modem chipsets, including the geographic area and purchasers; and (3) provide the Respondents and third parties with sublicenses to the patents held by A at no cost.

93 In response, A arranged a conference call with Respondents in January 2008 and sent an email with approximately 30 questions, including why Respondents were proposing a sublicense agreement instead of a license agreement; why Respondents were conditioning the sale of modem chipsets only to their licensees; whether granting a sublicense agreement without a license to the WCDMA standard-essential patents was consistent with the FRAND commitments, even though A had promised the standardization bodies that it would license on FRAND terms; and whether granting a sublicense agreement without a license to the WCDMA standard-essential patents was consistent with the FRAND commitments, whether A would be at a disadvantage compared to other licensed modem chipset manufacturers if it received only a sub-license, and why A should be required to provide a license to Respondents at no cost to Respondents when Respondents only provide A with a sub-license.

<Table 26 List of questions for the conference call between A and the

subjects57)

56) As discussed earlier, unlike a license agreement, a covenant not to sue is a promise by the defendants not to sue against A, but to sue without restriction against customers who purchase A's modem chipsets.

57) Exhibit 44, a list of questions from the conference call between A and the subjects.

94 However, the Respondents did not respond to A's request and reaffirmed their position that they could not accept A's request. Subsequently, on *March* 2009, A entered into a limited sublicense agreement58) with the Respondents59) that was not a license agreement, but rather a limited sublicense agreement with the terms proposed by the Respondents.

58) Exhibit 2, A's 2009 subtitle agreement with the defendants.

59) In response, A argues that Qualcomm "believed that, based on Qualcomm's past practices and representations, at least certain consumers would believe that it was necessary for A to

enter into a contract with Qualcomm," and therefore rejected Qualcomm's two subheadings.

(b) the 2013[redacted] agreement

95 In the second round of patent license negotiations, which resumed in 2012, Defendants again refused to provide A with a patent license to manufacture and sell modem chipsets.

96 A has been engaged in the WCDMA modem chipset business since the initial agreement in 2009, but recognizes that it is practically impossible to compete on a level playing field with Respondents due to the restrictions imposed by the sub-licensing agreement with Respondents,[60] and in January 2012, less than three years after the 2009 agreement was signed, A again requested Respondents to license its patents on FRAND terms, in compliance with the FRAND commitments declared by Respondents during the standardization process.

97 On *May*, 2012, A noted that the sub-licensing agreement with Respondents contained unfair contractual terms, such as restricting its chipset sales to Respondents' licensees and requiring Respondents to report sensitive business information to A. It questioned why Respondents refused to negotiate royalty rates and licensing terms, saying that if they had made FRAND commitments to standardization organizations, they should be willing to provide licenses under FRAND terms, and suggested that Respondents offer FRAND royalties and enter into a licensing agreement in which A would pay the royalties.

I felt that I had no choice but to make a decision." (A's response to the Commission's request for documents in Exhibit 80).

60) In its submission to the Commission, A stated that "under the First Agreement, A reported sensitive trade information about its modem chipset business to the Respondents, which enabled the Respondents to extensively monitor A's activities, and the Respondents used this information to approach A's customers and interfere with A's sales of modem chipsets. A perceived that it was being reduced to the status of an agent for the Respondents' patents because, as a result of being restricted to selling modem chipsets only to

handset manufacturers that have licenses to the Respondents' patents, A was forced to require handset manufacturers that did not have a license agreement with the Respondents to first enter into a patent license agreement with the Respondents in order to sell modem chipsets. In addition, the mobile phone manufacturers perceived that they could not compete fairly and equitably with the Respondents under the existing subtitle agreement because, if they purchased A's modem chipsets, they would have to enter into separate license agreements with other operators, which would increase their transaction costs and discourage them from purchasing A's modem chipsets, and one customer who used 100% of A's products was threatened by Qualcomm with stricter audits of its mobile phone sales if it did not start using Qualcomm products.(Exhibit A to Commission's Request for Production of Documents, Ex. 80)

<Table 27>. 2012.*.* A's letter to the defendants61)

98 On * * * 2013, approximately five months after the above letter dated * * * 2012,62) Fabian Gonell, Vice President & Legal Counsel of the Respondents, responded to A's letter, stating that the Respondents "never agreed to enter into a license agreement with A and that the Respondents and A have exhausted their rights to the Respondents' patents.

61) Microtuberculosis 43, Letter from A to the Subjects

62) In a letter dated *March *, 2012, A requested a written response from the Respondents to A's proposal within three weeks, but the Respondents did not respond until *March *, 2013,

approximately five months later.

and that this Agreement is not a license agreement."
enemy revealed.

<Table 28> 2013.*.* Respondents' replies to A63)

99 In response, A sent a letter on *March *, 2013, requesting the Respondents to propose the terms of the license agreement and the FRAND royalty rate for A's chipset, noting that the Respondents had not responded.

<Table 29>. 2013.*.* A's second email to the defendants64)

63) Exhibit 43, 2013.*.* Respondents' reply to A.
64) Exhibit 43, A's second email to the subjects.

100 However, in response to A's above request, on *March *, 2013, the Defendants entered into an existing agreement with A whereby the Defendants agreed to pay the

The company said it was open to discussing the possibility of modifying or replacing the agreement, but again declined to provide specific license agreement terms, including the level of fees.

<Table 30 2013.*.* Second letter from the Respondents to A65)

101 In response, by email on *March *, 2013, A reminded the Respondents that it hoped to agree with the basic premise that standard-essential patents should be licensed to parties that agree to FRAND terms at the time of standardization body standard selection, but reiterated its request that the Respondents propose FRAND license terms by *March *, 2013.

<Table 31>. 2013.*.* A's third email to the Respondents66)

65) Microtuberculosis 43, second letter from subjects to A.

66) Exhibit 43, A's Third Email to the Subjects

102 In response, in a letter dated *March* 2013, the Respondents stated that they were not obliged to offer A a license and did not send a draft license agreement with contractual terms and fee levels. The Respondents proceeded to negotiate a different kind of license agreement by, on the one hand, not offering A a FRAND license on FRAND terms, and, on the other hand, stating that they were willing to terminate the existing agreement and replace it with a new one, and asking A to indicate what rights he wanted to obtain in addition to the original agreement.

<Table 32 2013.*.*. Third reply to A from the Respondents67)

67) Soporific 43, the third letter from the Respondents to A.

103 Negotiations to finalize a license agreement between the two parties were subsequently suspended, and instead the

On *March *, 2013, the [redacted] Agreement was executed, replacing the material terms of the prior agreement.[68)]

104 The recitals of the [redacted] agreement clearly set forth the facts that A requested that the Respondents license the Respondents' standard essential patents on FRAND terms, and that the Respondents argued that they were not required to license A. The main [redacted]

<Table 33 2013.*.*. The [redacted] Contract69)

105 This fact has been consistently acknowledged by the Respondents throughout the investigation and proceedings, and is evidenced by the e-mails between A and the Respondents (Exhibits 41 and 43), the draft contract proposed by the Respondents to A (Exhibit 42), the list of questions for the conference call sent by A to the Respondents (Exhibit 44), A's response to the

request for production of documents (Exhibit 80), and the e-mails between the Respondents and A in 2009.

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68) Microtuberculosis,[redacted] contract

69) Subchapter III Exhibit A 2013[redacted] Contract

(Exhibit 2), A's 2013[redacted] agreement with Respondents (Exhibit 2), and Respondents' 2013 agreement with A (Exhibit 3).
No. 3), etc.

(3) Refusal to sign a license agreement and imposition of non-licensing constraints breaks the contract (A) Case B

106 In 1993, B entered into a license agreement with the Respondents to license the patents necessary to use the CDMA standard technology developed by the Respondents. The license agreement granted B the right to manufacture, sell, and use cellular phones (subscriber units) using the Respondents' essential patents for telecommunications standards. However, B was only allowed to manufacture modem chipsets for use in its own mobile phones, and it would be a breach of the agreement if B sold the modem chipsets to other operators.

<Table 34 Modem chipset provisions in the 1993 patent license agreement70)

107 Defendants and B modified the Agreement several times, including in

1997, 2004, 2005, and 2009, but with respect to the sale of modem chipsets to other operators, the Agreement remained the same.

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70) 1993 Patent License Agreement for Subparagraph 11 of Exhibit B

conditions. Therefore, the sale of modem chipsets manufactured by B to other operators is prohibited.
was outside the scope of the license agreement between Respondents and B, so Respondents could have asserted patent infringement against B at any time.

108 In this situation, B has a [redacted] a business division that manufactures APs used in cell phones, etc. Around 2011, B decided that integrated chipsets that combine APs and modem chipsets will dominate the market in the future, and decided to pursue the outsourcing business of modem chipsets, which it had never done before. At that time, B contacted and negotiated with mobile phone manufacturers who could be potential customers for the modem chipset business, but most of the mobile phone manufacturers raised the risk that the Defendants could launch patent attacks if they purchased B's unlicensed modem chipsets. They requested that B provide indemnification agreements to cover damages arising from patent attacks by the accused, or that B provide assurances that they would not be subject to patent attacks by the accused and that B would not be subject to patent attacks by the accused.[71)]

109 In response, B believed that it was essential to secure a license from the Defendants for its external sales business, and requested the Defendants to sign a license agreement allowing the external sale of modem chipsets after *March 2011, but the Defendants refused. The details of this are as follows.

110 First, on June 29, 2011, Cha Young-koo, the then president of Quarkom Korea, gave an interview to an electronic newspaper to the effect that "anyone can sell modem chipsets using the patents of the Respondents." On or about *March* 2011, shortly after the interview was reported, B requested the Respondents to enter into a patent license

agreement to manufacture and sell modem chipsets, but the Respondents refused.72)

<Table 35 Executive confirmation of B's license request73)

71) Exhibit 62, B's Statement of Rejection of the Respondents' Chipset License Deal
72) Exhibit 60, Acknowledgement of License Request for Respondents

111 When the Respondents rejected his request for a license without negotiation, B asked the Respondents for an agreement that was at least as good as that offered to other modem chipset manufacturers.

112 In response, on *March* 2011, the Respondents sent B a draft agreement that, unlike the agreements with other competing modem chipset manufacturers, provided for Exhaust Remedies.[74] The Exhaust Remedies Agreement provided for a form of joint and several liability whereby B, as well as the handset manufacturer, would bear the risk of being sued by the Respondents if the handset manufacturer that purchased B's modem chipset breached the agreement, such as by failing to pay the Respondents.

113 In addition to the conditions previously required of A, including (1) restrictions on where the modem chipsets could be sold, (2) a requirement to report sales information, and (3) a cross-grant of B's patents, the supplemental license agreement also provided that B would be jointly and severally liable with the cell phone manufacturer in the event of a patent dispute between the cell phone manufacturer and the accused because the cell phone manufacturer purchased B's modem chipsets and failed to pay patent fees to the accused.

<Table 36 Status of Qualcomm contract for B modem chip sales75)

74) Exhibit 63, Draft Patent Agreement between Respondents B and C
75) Status of the contract with Q Company for the s a l e of m o d e m chips in connection with the small pox 59

114 Nevertheless, considering that it was essential for B to secure an irrevocable written agreement from the Respondents in order to pursue its modem chipset distributorship business, it continued to negotiate, exchanging several amendments to the agreement over the next year or so to iron out details such as the specific terms and conditions of the supplemental exercise agreement and the scope of the products covered by the patent license.

115 By the end of January 2013, B decided to accept the supplemental exercise terms that the Respondents continued to demand, although they were not sufficient for the modem chipset business, and continued to negotiate with the Respondents on other detailed contract terms.

<Table 37 Report from Qualcomm's IC offsite meeting written by B (1)[76]

76) Microtelangiectasia 64, Qualcomm reports results of IC out-of-circuit meeting 2013. *. *.

116 B then discussed the terms of the agreement with the Respondents until mid-March 201377), but at a meeting with B on *March * of that year, the Respondents negotiated a supplemental enforcement agreeme[redacted] more unfavorable to B than the standstill, [redacted] if the subjects [redacted] are At a meeting on *March* of the same year, the defendants again proposed a change to the terms of the agreement to reserve patent infringement claims against B for only ** days, but to hold B liable at any time after that period.78) At a subsequent meeting on *May* of the same year, the defendants informed B that they would not be able to enter into a written agreement to sell the modem chipsets outside of Korea if B did not accept the terms of the limited reservation of claims.79) Thereafter, discussions between the parties about entering into a written agreement to sell the modem chipsets outside of Korea ceased.80)

<Table 38> Report of Qualcomm's IC external sales meeting prepared by B (2)81)

77) B and the subjects had meetings or emails on or about 2011.*.*., 2012.*.*., 2012.*.*., 2012.*.*., 2013.*.*., 2013.*.*., and 2013.*.*. They had meetings or exchanged e-mails.

78) Defendants subsequently filed a motion to dismiss the 2013. *. *. B a n e-mail with the same content.

79) In response, B replied t h a t i t had been n e g o t i a t i n g i n e v i t a b l y for the external sale of the Modem chipset despite the risk to B of the supplemental right exercise agreement proposed by the defendants, b u t t h a t t h e d e f e n d a n t s' new proposal for a limited time to suspend t h e lawsuit w a s i n f a c t a r e f u s a l to accept any of B's demands, and that it was no longer w i l l i n g t o negotiate with the d e f e n d a n t s. (E x h i b i t 6 2 o f B's r e f u s a l t o license the chipset)

80) Since then, B has not been able to p u r s u e the o u t s o u r c i n g business of s e l l i n g its m o d e m chipsets to other cell phone manufacturers in addition to self-consumption.

81) Report on Qualcomm's IC Outsourcing Meeting by B, aka Sigmoiditis 65 (2)

117 Separately, around *March 2010, B received a request from *** to participate in a joint venture company82) to develop a modem called "*** Project," and around *March 2011, B received a

request from *** to participate in a joint venture company called "*** Project.

82) The joint venture was supposed to involve three companies (***, ***, and ***) in addition to ***.

We have entered into a Joint Venture Agreement. The Joint Venture Agreement contains the

The license agreement between *** and the Respondents stated that the signing of the license agreement was a prerequisite for the establishment of the joint venture. However, *** was ultimately unable to obtain a license for the modem chipset from the Respondents, and the joint venture between B and *** to develop the modem was ultimately abandoned.[83)]

(b) Case C

118 In 2009, C requested a license from the Respondents to manufacture and sell modem chipsets for mobile phones, but the Respondents refused to provide a license by imposing contractual terms that were unacceptable to C. In early January 2009, in response to C's request for a license, the Respondents communicated that they were willing to provide non-exhaustive rights, but not a license, for modem chipsets for mobile phones.[84)]

119 In the second round of negotiations held in January 2009, the Respondents maintained their position that they could not license the modem chipsets for mobile phones to C, and demanded that B accept the additional conditions that the Respondents had requested, namely, (i) limiting the sales outlets of the modem chipsets, (ii) reporting the sales information of the modem chipsets, and (iii) providing exhaustive rights to C's patents. Subsequently, negotiations between the two companies to conclude a license agreement for the manufacture and sale of modem chipsets were no longer conducted, and C was unable to secure any rights to the Respondents' standard essential patents.[85)] As a result, C was unable to start its modem chipset business through direct license negotiations with the Respondents.[86)]

83) Materials related to the *** Project submitted by B to the Committee on October 7, 2016 (Attachment 2)

"Failure to secure a QCOM ASIC license : Securing a QCOM ASIC license was contractually a pre-requisite for First Colsing to happen, but NEWCo & *** failed to achieve that and offered no remedy."

84) Meanwhile, the [redacted] You said you have a doctor.

85) In response, C replied that the main reason for the breakdown of negotiations was that Qualcomm refused to provide C with fair and reasonable licensing terms in relation to the components installed in the mobile phones, and that it was impossible for C to conclude a contract with Qualcomm because of Qualcomm's refusal to grant a full license for the modem chipsets for mobile phones and its insistence on selling the modem chipsets only to customers separately licensed by Qualcomm.(C's Response to Interrogatory No. 79).

86) C subsequently entered the modem chipset business in 2011 by acquiring ***, a modem chipset manufacturer (modem chipset patent portfolio agreement entered into with Respondents on *March *, ****), taking over [redacted] of Respondents' telecommunications standard essential patents held by ***.

(c) Case D

120 D asked the Respondents in early 2012 to either enter into a new or amend the existing CDMA2000 modular chipset license agreement87) to include WCDMA modular chipsets, but the Respondents instead proposed to rescind the license agreement for the CDMA2000 standard and enter into a supplemental rights agreement covering both CDMA2000 and WCDMA.

121 In response, D sent an email to the Respondents on *May*, 2012, demanding that the Respondents' proposed Supplemental Rights Exercise Agreement provide virtually no rights and comply with the FRAND commitments88), but the Respondents refused to enter into an agreement that would provide a license other than the Supplemental Rights Exercise Agreement, and thereafter there were no further discussions between the parties to enter into a license agreement to manufacture and sell WCDMA modem chipsets89).

(d) Other cases

122 In addition, modem chipset manufacturers such as *** (2011) and *** (2010) requested license agreements for modem chipsets for mobile communications from the Respondents, but the Respondents consistently refused to provide licenses to modem chipset manufacturers in accordance with their business policies, and no license agreements were signed between the Respondents and these manufacturers to manufacture and sell modem chipsets.90)

87) D had a license to manufacture, sell, and distribute CDMA2000 chips as a result of its 2003 acquisition of ***, a modem chipset manufacturer (which entered into a patent license agreement with Defendants dated *March *, ****).

88) The original text read [redacted]

Emails sent to subjects)

89) D subsequently failed to e n t e r t h e WCDMA modem c h i p s e t market.

90) Exhibit 33, Defendants' June 26, 2015 Reply.

123 This fact has been consistently acknowledged by the Respondents throughout the investigation and proceedings, and is evidenced by the 1993 Patent License Agreement between B and the Respondents (Exhibit 11), the Respondents' responses to the Commission's requests for documents (Exhibits 32 and 33), B's internal report on the status of the Qualcomm contract for the sale of modem chipsets to B (Exhibit 59), the affidavit of B*** (Exhibit 60), and B's submission of the Respondents' reasons for declining to license the modem chipsets (Exhibit 62), A draft patent agreement sent to B by the Respondents in 2011 (Exhibit 63), a report on the outcome of a meeting with the Respondents in 2013 prepared by B (Exhibits 64 and 65), C's response to the Tribunal's data requests (Exhibit 79), an email sent by D to the Respondents in 2012 (Exhibit 40), and B's submission to the Tribunal on October 7, 2016 (submitting data on the *** Project).

5) Act 2: Demanding mobile phone manufacturers to enter into and fulfill patent license agreements as a condition of supplying modem chipsets

124 As discussed in Section 2. a. 4) above, the Respondents have maintained a business policy of refusing to license their patents, including standard essential patents, to modem chipset manufacturers, or to provide them with a full license, and have required mobile phone manufacturers that wish to purchase their modem chipsets to first enter into a patent license agreement with them as a condition of supplying their modem chipsets.

125 First, Defendants entered into modem chipset supply agreements with mobile phone manufacturers after 1993, independent of entering into patent license agreements with mobile phone manufacturers; all of the modem chipset supply agreements between Defendants and mobile phone manufacturers were entered into after the license agreements were signed.91) The dates on which

Defendants entered into license agreements and modem chipset supply agreements with major mobile phone manufacturers are shown in Table 39 below.

91) In their responses to the Commission, the Respondents clarified that, as a matter of policy, they do not enter into component supply agreements with unlicensed handset manufacturers and do not sell modem chipsets to handset manufacturers that do not have a patent license agreement with them.

<Table 39 Dates of Defendants' agreements with major cell phone manufacturers to license and supply modem chipsets92)

Cell phone carrier	Initial license agreement signing date	Modem chipset supply contract signing date
*** B	1993.8.*.	1994.3.*.
*** B	1997.8.*.	1997.9.*.
*** B	1993.8.*.	1996.2.*.
*** B	2000.12.*.	2001.6.*.
*** B	2001.4.*.	2001.7.*.
*** B	2001.9.*.	2001.11.*.
*** B	2003.8.*.	2003.8.*.

* Source: Exhibit 35 of the Respondents' submissions.

a) Linking modem chipset supply to patent license agreement in modem chipset supply contract

126 In their modem chipset supply contracts with cell phone manufacturers, Defendants set forth the following provisions:93)

127 First, the sale of a modem chipset does not include patent rights. A mobile phone manufacturer cannot use or sell a purchased modem chipset in combination with other components without a patent license. A purchased modem chipset may only be used or sold in accordance with the license agreement, and the incorporation of the modem chipset into a mobile phone does not relieve the licensee of its obligation to pay the license fee.94)

92) The Respondents submitted documents relating to the signing dates of license and modem chipset agreements with a total of *** mobile phone manufacturers, but only the dates of agreements with major mobile phone manufacturers were extracted.

93) In their response to the Commission, the Respondents stated that these provisions appear in virtually all of the Respondents' component supply chain agreements and are generally worded in the same manner (Exhibit 34, June 26, 2015 (Respondents' submission)).

94) [redacted]

128 Second, the purchased modem chipset can only be used to develop and manufacture mobile phones, The sale and use of the phone is subject to the terms of the patent license agreement.[95)]

129 Third, if the buyer breaches the modem chipset supply agreement, breaches the license agreement, or breaches the If not cured within ** days or ** days[96)], Defendants may terminate the Modem Chip Set Supply Agreement or suspend or withhold supply of Modem Chip Sets.[97)]

130 The main examples of such contracts for the supply of modem chipsets by the accused in conjunction with patent license agreements are shown in Table 40 below.

[redacted]

95) [redacted]

96) [redacted]

97) [redacted]

<Table 40 Major modem chipset supply agreements associated with license agreements

Cell phone carrier	Contract name	Execution Date
*** B	Parts Supply Agreement (***)	2004.9.*.
*** B	Custom Integrated Circuit Supply Agreement (***)	2000.9.*.
*** B	Parts Supply Agreement (***)	2001.3.*.
*** B	Parts Supply Agreement (***)	2001.6.*.
*** B	Parts Supply Agreement (***)	2003.8.*.
*** B	Parts Supply Agreement (***)	2001.11.*.

* Source: Respondents' submissions (Exhibit 26 through Exhibit 31)

b) Specific examples

(1) Example E

131 Since Defendants and E entered into a CDMA license agreement in 1993, and since 2003, E has been using the

In the process of concluding an amended license agreement in connection with the conclusion of the WCDMA license agreement around 2004, a dispute arose between the Respondents and E regarding the interpretation of the license agreement, including whether WCDMA was included in the existing CDMA license agreement and whether the license fee terms needed to be adjusted due to the Respondents' decrease in the proportion of WCDMA standard essential patents.

132 ① In the course of the WCDMA patent license agreement between the

Respondents and E from 2003 to 2004, E informed the Respondents in *March 2003 that "the existing CDMA patent license agreement does not apply to the WCDMA patent, so no royalties can be paid for the WCDMA patent".

133 ② In response, the Respondents sent an e-mail on *March *, 2004, notifying E that "since the existing **** CDMA patent license agreement also includes the WCDMA patent, the non-payment of royalties is a breach of the license agreement, and E's breach of the license agreement is also a breach of the modem chipset supply agreement (***), the Memorandum of Understanding (MOU)98) regarding modem chipset rebates will be terminated and all rebates99) will cease to be paid."100)

134 In response, on *March* 2004, E requested that the Respondents "not assert a breach of the Modem Chip Set Supply Agreement based on a breach of another contract, namely the License Agreement, and that any dispute regarding the interpretation of the License Agreement be handled by arbitration as provided for in the Patent License Agreement."101)

135 ④ However, the defendants did not immediately reach an agreement between the parties on *March *, 2004, but instead "terminated the Memorandum of Understanding on *March *, 2004, which merely terminated the modem chipset rebate, but reserved the right for the defendants to terminate the modem chipset supply agreement.

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98) Memorandum of Understanding (MOU) between Respondents and E for the supply [redacted] of [redacted]

99) According to internal re[redacted]

100) The text of the email sent by the defendants to E was [redacted]." (Fingerprinting 70)

101) The text of the email sent by E to the defendants is as [redacted] fol[redacted]"(Microtelangiectasia 71)

Reminded.[102]

136 E became concerned that its mobile phone business would suffer if the current situation worsened[103], and when the Respondents subsequently notified E of specific plans to stop supplying WCDMA standard modem chipsets to E on * * * 2004, including stopping purchase orders for WCDMA standard modem chipsets, stopping transportation, withdrawing related engineering resources, and returning software,[104] E became concerned that its mobile phone business would suffer as a result of the interruption and delay in supplying modem chipsets,[105] and entered into the Amended License Agreement as required by the Respondents on * ** 2004.[106]

(2) Case F

137 In 2009, F started a data modem business for PCs using the Respondents' modem chipsets in connection with data modems for PCs that were not clearly covered by the existing patent license agreement with the Respondents.

138 On *March* 2009, Korea Qualcomm's modem chipset division *** sent an email to F informing F of an imminent request to stop supplying modem chipsets because F had produced unlicensed products using the modem chipsets, and demanded that F sign the data modem license agreement under negotiation to avoid a serious situation.[108]

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102) [redacted]

103) [redacted]

104) [redacted] " (Exhibit 69, E's internal report at ****. *. *.)

105) Ex. 167, wherein the defendants claim that the 2004. *. *. E by fax and e-mail sent via DHL.

106) In this regard, E testified that in his license negotiations [redacted]

" (Exhibit 69, Qualcomm Negotiation Status and Plans at E), and E, who was present at the meeting with the defendants.

*** is 2004. *. *. " [redacted] " (Email from E*** at Exhibit 69, 2004).

107) Exhibit 15, Amended 2004 Patent License Agreement between E and Respondents.

108) The text of the email is as follows " [redacted]

139 On the other hand, the above *** reported the above facts to the licensing division of the Respondents' headquarters on the same day and asked whether to stop supplying all modem chipsets to F or only to the relevant division (computer division).[109]

140 ④ Respondents and F subsequently entered into an amended license agreement covering the Product on *March *, 2009.

141 This fact has been consistently acknowledged by the Respondents throughout the investigation and the proceedings, and is evidenced by the Respondents' contracts for the supply of modem chipsets to ********** (Exhibits 26 through 31), the Respondents' responses to the Tribunal (Exhibits 34 through 36), the Respondents' 2007 business report (Exhibit 38), E's opinion (Exhibit 67), E's description of its negotiations with the Respondents in 2003 and 2004, and my sub-reports (Exhibits 68 and 69), E's opinion letter (Exhibit 67), E's 2003-2004 negotiations with the Respondents and internal reports (Exhibits 68 and 69), e-mails exchanged between the Respondents and E in 2004 (Exhibits 70 through 73), and the 2004 Patent License Amendment Agreement between E and the Respondents (Exhibit 15).

6) Act 3: When concluding a patent license agreement with a mobile phone manufacturer, the act of presenting contract terms including a comprehensive license term, a license fee term set by the company, and a free cross-grant term

142 Under the same contractual basis and structure as in 2. a. 5) above, the Respondents entered into license agreements for mobile phone manufacturing and mobile communication standard essential patents[110], including (i) license

[redacted]"(Email from 2009.*.*. ***)

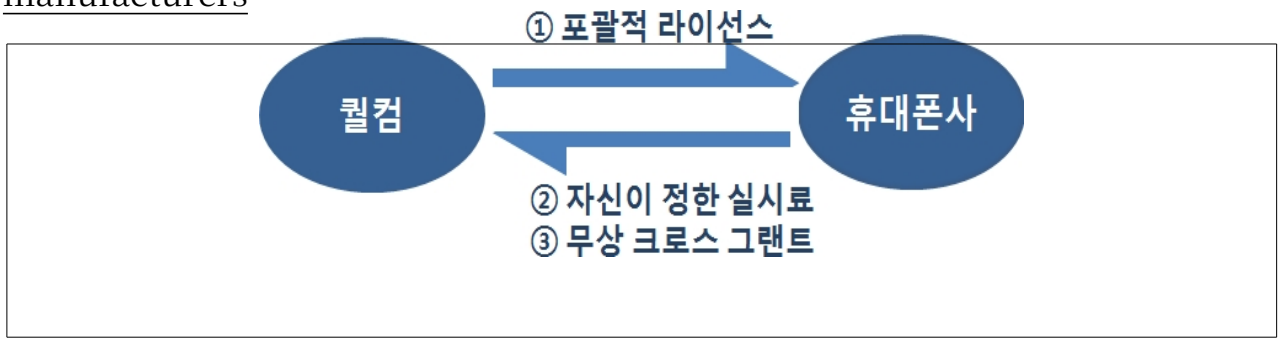
109) The full text of the email looks like [redacted]?"(2009.*.*. ***The this " mail) electronic

110) Around the year 2000, a new third generation standard was adopted by standardization organizations and the WCDMA standard was replaced by the former

As WCDMA began to proliferate globally, negotiations to amend existing patent license agreements in relation to WCDMA took place. Act 3 is understood to have been executed in earnest during this process from about 2000 to 2004, when, according to the individual cell phone manufacturers, the Defendants began to negotiate with each other in 2002. *. *. *** and 2004. *. *. *** and, 2004. *. *. ***, each of which renewed patent license agreements covering their respective WCDMA standards.

You can target all of your patents without specifying or distinguishing the scope of the targeted patents.
(the "Comprehensive License"); and (2) unilaterally setting a royalty fee based on the total price of the mobile phone for the Comprehensive License without going through a good faith negotiation process to evaluate and calculate the value of the patent (the "Self-Determined Fee Condition"), In exchange for licensing his patents, he proposed, and ultimately entered into, a license agreement that included a condition that the other party, a mobile phone manufacturer, would provide patents owned by him and his modem chipsets to customers who purchased his modem chipsets free of charge or would be barred from asserting patent infringement in the future (the "Free Cross-Grant").

<Table 41 License structure between the defendants and phone manufacturers



a) the terms of the license agreements that the defendants entered into with the phone manufacturers

143 As of September 2015, Defendants had entered into and maintained license agreements with 195 mobile phone manufacturers worldwide111), the contractual structure and terms of which were substantially similar in the following respects112)

(1) Universal License Terms

111) By September 18, 2015 (the Reference Date), Defendants had entered into patent license agreements with a total of *** mobile phone manufacturers, of which license agreements with ** operators had terminated as of the Reference Date.

112) Defendants respond that "the hundreds of license agreements that they have entered into with mobile phone manufacturers around the world are substantially similar in structure and terms, and that all of their license agreements provide rights to Qualcomm's patent portfolio to manufacture, sell, and distribute handsets that use 2G, 3G, and 4G mobile communications technologies." (Defendants' Response to Commission's Request for Production, Exhibit 32).

144 Patents may be cellular SEPs, non-cellular SEPs, or other standard-essential patents.

Non-cellular SEPs and non-standard-essential patents (Non-SEPs) that are not directly related to standards. However, in their patent licensing standards agreements and patent license agreements with actual handset manufacturers, Defendants specify that the licensed subject matter is "all of their patents,"[113] which "means both Qualcomm's technically essential intellectual property and Qualcomm's other intellectual property."[114] Accordingly, Defendants proposed, and were granted, blanket license agreements to mobile phone manufacturers covering all of their patents, without distinguishing between patents that are essential to mobile communications standards[115] or patents that are essential to each mobile communications standard, such as CDMA, WCDMA, and LTE.

(2) Set your own fee terms

145 Defendants' patent license standard agreements provide for royalties to be calculated as a percentage of the Net Selling Price[116] of the handset[117] and, in practice, Defendants have entered into license agreements with handset manufacturers with royalty rates of approximately *% for handsets using CDMA and WCDMA standards, *% for handsets using LTE standards only, and *% for handsets using LTE multimode, the same as CDMA and WCDMA.[118] The actual terms of a sample of the license agreements that Defendants entered into with major handset manufacturers are shown in Table 42 below.

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113) Defendants refer to this as "QUALCOMM's Licensed Patent Claims" in the Standard Agreement and "QUALCOMM's Intellectual Property" in individual agreements.

114) The text of the Defendants' standard agreement reads as follows "QUALCOMM's Licensed Patent Claims" means QUALCOMM's Technically Necessary IPR and QUALCOMM's Included Other IPR." (Exhibits 9 and 10)

115) "Other Patents" includes "Non-Cellular SEPs" and "Non-SEPs," which are patents that are not essential to the manufacture and sale of mobile phones that implement mobile communication standards, or that can be used, circumvented, or avoided by licensing alternative technologies from third parties.

116) The wholesale selling price of a mobile phone from the licensee, the mobile phone manufacturer, less certain deductions.

117) Section 5.2 of Defendants' stan[redacted]

(Standardized Contract for Individuals with Microcephaly 9)

118) Generally speaking, even LTE phones are LTE multimode modem chipsets that use not only LTE but also CDMA or WCDMA standards, so they are subject to a *% implementation fee, not *%.

<Table 42 Fee terms of the license agreements the subjects signed with cell phone manufacturers

	Periodicity	Pre-Release	Implementation Fee Rate
*** B	Wholesale cell phone prices119)	*** B	***120)
*** B	Wholesale cell phone prices121)	*** B	***122)123)124)
*** B	Wholesale cell phone prices125)	*** B	*** B
*** B	Wholesale cell phone prices	***126)	***127)
*** B	Transfer value of cell phones supplied to **128)	*** B	***129)

119) For *** phones sold after the effective date of the January 2009 Amendment, the net selling price of the phone for purposes of calculating the Implementation Fee Rate shall be capped at ** dollars (Section 4.1.3. of the January 2009 Amendment, ****).

120) The license fee rate for the *** was [redacted], as amended at **** [redacted] **** contract is a [redacted]

121) [redacted]

122) In a license agreement with **** in the year ***, the [redacted] **** as an amendment to the [redacted]

123) Single-mode LTE is characterized by using only the LTE network for both voice and data communications, unlike current dual-mode LTE, which uses the LTE network for data communications but crosses 3G and LTE networks for voice calls. Therefore, due to the nature of mobile communication technology that requires the use of previous generations of standards for network compatibility, there are virtually no cell phones that use LTE single-mode modem chipsets.

124) Defendants and *** separately entered into an OFDM agreement in **** that applies to cell phones implementing only the LTE standard, which provides for an implementation fee rate of *% for single-mode LTE cell phones, with a maximum implementation fee of * dollars per cell phone (* dollars for LTE cell phones sold on or after *March *, ****).

125) In the contract between the Respondents and ***, the upper limit of the performance fee is

126) The **** CDMA contract between Defendants and *** provides for the sale [redacted] of [redacted]

127) [redacted]

* dollars.

(3) Apply the same CDMA usage fee conditions to WCDMA and LTE

146 Defendants applied the same license fee terms to the CDMA essential patents as they applied to the third and fourth generation mobile communication standards. Specifically, Defendants did not enter into separate license agreements for each generation standard, regardless of the number or proportion of Defendants' patents in the WCDMA or LTE standards compared to CDMA130), or the value or contribution of the patents, but instead included the WCDMA or LTE standards in the scope of agreements based on existing CDMA or WCDMA licenses, so that the royalty payment terms of the existing license agreements would apply to the next generation of mobile communications standards technology131). For example, Defendants argue that

Amendment to Patent License Agreement with ****, dated ***, amending the scope of patents covered by the license.

In addition to the existing CDMA, CDMA2000, and WCDMA standards, the patents for other standards, such as LTE, were included in the patent portfolio, and thus, the same royalty rate was applied to LTE, where the defendants' share of standard-essential patents was 16%, as to CDMA and WCDMA, where the share of patents was 90% and 27%, respectively.

(4) Long-term and perpetual contracts

147 Defendants have license agreements with cell phone manufacturers that are either perpetual or have a term of 15 years or less.

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128) The transfer price of the cell phones supplied to *** by **, a contract manufacturer of ***, includes the cost of the components and implementation fees paid by ** and the fees for the

provision of services by **.

129) The License Agreement with ** above, which contains the fee terms typical of Defendants' standard agreements, was executed by ***,

Other cell phone manufacturers, such as ***, have generally the same terms as **.

130) The share of all standard-essential patents held by the accused in each generation is over 90% for CDMA, 27% for WCDMA, and 16% for LTE.

131) On the other hand, the defendants also entered into separate license agreements for the LTE standard, which reduced the actual sample rate from *% to *%, but only for the LTE single-mode variant. However, due to the nature of mobile communications technology, which requires the use of previous generations of standards for network compatibility, virtually no cell phones use LTE single-mode modem chipsets, and therefore the *% royalty rate of the CDMA or WCDMA license agreements applies to most cell phones with multimode modem chipsets.

of the agreement. Accordingly, during the term of the agreement, the patents of the defendants are invalidated,
The level of royalties remained the same even when the patent expired132) or when the obligation to pay royalties was waived after a certain period of time under the payment completion clause in the individual agreements133). The terms of the license agreements between the Respondents and the major mobile phone manufacturers are shown in Table 43 below.

<Table 43 Length of license agreements between the defendants and phone manufacturers

Business name	Initial contract signing date	Contract end date	Fee rates during the contract period
*** B	1993.*.*.	2023.*.*.134)	Fixed135)
*** B	1993.*.*.	Persistent	Fixed
*** B	1997.*.*.	Persistent	Fixed
*** B	2003.*.*.	Persistent	Fixed
*** B	2005.*.*.	Persistent	Fixed
*** B	2001.*.*.	Persistent	Fixed
*** B	2000.*.*.	Persistent	Fixed
*** B	1992.*.*.	Persistent	Fixed
*** B	1996.*.*.	Persistent	Fixed
*** B	1990.*.*.	2013.*.*.	Fixed

(5) Free cross-grant

148 In exchange for licensing their patents, Defendants received a royalty, and at the same time, they requested that cell phone manufacturers provide them and their customers with standard essential patents and other patents held by them at no cost.136) In other words, Defendants were required to provide cell phone

132) Article 88 of the Patent Act stipulates that the term of invention patent rights is 'from the date of registration of the patent right to the date that is 20 years after the date of filing

the patent application'.

133) For example, in the **** and **** agreements with ***, t[redacted] The defendants, however, gave *** the same

You're paying for the implementation.

134) The License Agreement, dated ****, by a[redacted]

135) In some of their contracts with cell phone manufacturers, the Accused have tiered royalty rates based on cell phone sales. However, these tiered royalty rates are not adjusted based on changes in the value of the patents over the term of the contract, and are therefore denoted by **.

Patents owned by the phone manufacturer that are subject to a patent license agreement with the manufacturer.

The license agreements contain provisions that either require Defendants to exhaustively license patents necessary for the manufacture and sale of modem chipsets from Qualcomm Inc. or provide for two-tiered sublicensing arrangements137) for Defendants and their customers who purchase modem chipsets from Defendants. As of September 18, 2015, Defendants have license agreements with 195 mobile phone manufacturers worldwide, with all but *** containing such provisions, and Table 44 below shows the cross-grant or sublicense agreements that Defendants have with major mobile phone manufacturers.139)

<Table 44 Cross-grant or sub-licensing arrangements that subjects have with cell phone manufacturers

Business name	Cross-grant terms	Contract signing date
*** B	Exhaustive License [**** changed to Sub-Title Covenant to Qualcomm (including its Affiliates) and Qualcomm's Customers as Amended].	1993 *Month
*** B	Exhaustive License [**** changed to Sub-Title Covenant to Qualcomm (including its Affiliates) and Qualcomm's Customers as Amended].	1993 *Month
*** B	License to Qualcomm (including its affiliates) and sublicensee covenants to Qualcomm when licensed by Qualcomm	2002 *Month
*** B	License to Qualcomm (including its Affiliates) and Sub-Title Covenants to Qualcomm's Licensees [**** changed to Sub-Title Covenants to Qualcomm and Qualcomm's Customers in the Amended Agreement].	The year 2000 *Month

136) In their comments to the Commission, Respondents stated that "Qualcomm seeks a cross-grant from all Licensees to protect its components business and component customers from Licensees' patent assertions. (emphasis added) Specifically, Qualcomm typically initially seeks cross-licenses for components, which typically involve (i) cross-licensing current and future patents for Licensees' telecommunications standard-essential patents, and (ii) cross-licensing other patents of Licensees, such as non-standard-essential patents and non-telecommunications standard-essential patents (e.g., WiFi and video encoding standards), filed within a limited capture period." (Docket No. 32, Defendants' Response to Request for Production).

137) In the case of an exhaustive license, the exhaustive rights acquired by the licensee are transferred to the licensee's customers when the licensee sells the product embodied in the patent under its lawful authority. The licensee's customers can then use the product as they wish without paying any patent license fees.

138) A covenant not to assert is a contractual agreement by the patent holder not to assert the patent against the licensee. In addition, if a patent holder is offered a "two-tier covenant not to sue" that also promises to sue itself and its customers, such as the accused, the patent holder will not be able to assert infringement claims against both the accused and the customers who purchase the accused's products, so a "two-tier covenant not to sue" can have the same effect as an exhaustive license in terms of the patent holder's risk of being attacked by infringement claims.

139) The subjects, on the other hand, were told that [redacted] (Microtelangiectasia 25)

*** B	Licenses to Qualcomm (including its affiliates) and Licenses from Qualcomm Subtitle pledge to the city	Year 2000 *Month
*** B	License to Qualcomm (including its Affiliates) and sublicense to Qualcomm's licensees [**** states that the license provided to Qualcomm in the Amended Agreement is an exhaustive right provided to Qualcomm's customers].	2005 *Month
*** B	CDMA License Agreement: Exhaustive License WCDMA Memorandum of Understanding: Subtitle Commitments to Qualcomm (including its Affiliates) and its Customers	CDMA contract : WCDMA MOU in *March 2003 : 2004 *Month

149 Furthermore, Respondents received cross-grants from cell phone manufacturers without any consideration being calculated. In this regard, Respondents' standard license agreements provide that the cross-grants they receive are "royalty-free,"140) and, in fact, Respondents' 2004 agreement with ***141) , 2003
The CDMA License Agreement with ***142) and the WCDMA Memorandum of Understanding with *** in 2004 expressly provide that the cross-grant to handset manufacturers is "royalty-free. In addition, the Respondents themselves explain why they receive royalty-free cross-granting, stating in their response to the Commission that "many of Qualcomm's licensees either do not hold any such patents at all, or if they do, they hold patents that have no significant value. In general, these licensees do not request that Qualcomm adjust the terms of their licenses, nor are they required to do so, given the unnecessary controversy that would arise over the relative value of patents that are of minimal value to Qualcomm at best."

140) Section 6.1 of Defendants' standard contract provides that [redacted] (Microtelangiectasia 9 and 10)

141) During negotiations with the Respondents to renew the license agreement on *March *, 2003, *** pointed out that the Respondents were using ***'s patents without payment and requested compensation for the value of the patents. However, the Respondents refused to pay

compensation to ***, stating that ***'s patents were licensed to them free of charge under the existing agreement, and that they had never paid any compensation for cross-granting, even to operators such as *** and ***. (Exhibit 69, Internal Report of *** on the Status of Negotiations in 2003-2004)

142) The CDMA license agreements entered into by Defendants and *** were amended in 2004, 2008, and 2013, and the no-cost cross-grant remained unchanged in the amendments.

b) The process of entering into licenses between the defendants and the phone manufacturers.

150 In the process of entering into patent license agreements, Respondents did not specify which of their patents were infringed by mobile phone manufacturers, nor did they provide the mobile phone manufacturers with the basis for license negotiations, such as a list of the patents to be licensed, their key claims, an analysis of the key claims, and their relevance to the standards. In this regard, (i) *** entered into license agreements with Respondents for CDMA and WCDMA standard-essential patents in 2003-2004, stating that "Respondents never provided a list of patents during the license negotiations. Accordingly, we have never had any specific discussions with the Respondents regarding infringement of any patents."143); (ii) *** responded that in the course of the license agreement with the Respondents for the WCDMA standard-essential patents in 2004, "the Respondents never provided a list of claims from which the nature of the patents could be determined, nor did they provide a clear basis for calculating the royalties, despite repeated requests by ***, the licensee."144). (iii) Furthermore, *** has been negotiating with the Respondents since 2005 to enter into a license agreement,145 and responded that "*** would like to be provided with a list of patents by the Respondents to assess the true value of each of the Respondents' patents and not to license patents that it does not need, but the Respondents have not provided a list of patents and have not allowed *** to do so." 146)147).

151 In response to the Commission's question as to how the royalty rate was determined in the license negotiations with the Respondents, *** stated that (i) "the royalty rate is the standard and non-negotiable terms of the Respondents.

143) Reply of *** to the submissions of the Commission in Case No. 77
144) First Amicus Brief of *** in Response to the Subpoena Duces Tecum 67 Committee's Request for Production
145) *** did not have a direct license agreement with the Respondents, but rather relied on the

license agreements between the CMs and the Respondents by sourcing its mobile phones through contract manufacturers (CMs) such as ***, ***, ***, and *** who had license agreements with the Respondents. *** had entered into license agreements with a number of other standard essential patent holders in the mobile telecommunications industry and was seeking to enter into license agreements directly with the Respondents.(Exhibit 75, ***'s Response to the Commission's Request for Production)

146) Amicus Curiae ***'s Response to the Subpoena duces Tecum 75 Committee's Request for Production

147) The actual subjects were 2006. *. *. In response to ***'s request to provide patent inforr[redacted]for the purpose of entering into a license agreement [redacted](Sjogren's Syndrome 101, emails between *** and the subjects)

We have no choice but to accept it and cannot negotiate on it."148); (ii) the Respondent

During the course of the 2004 license amendment negotiations between the Inmates and ***, *** represented to the Defendants that ***.

*** requested that component prices unrelated to telecommunications standard technology be excluded from the calculation of the implementation fee, but the Respondents stopped disc[redacted] stated that "the Respondents have never provided an estimate of the implementation fee rate,

[redacted]"150).

152 In this regard, (i) Defendants' representative Steve Altman stated in an interview with Deutsche Bank in 2005 that "as long as we have one patent, one claim that applies to their product, they (the handset manufacturers) will have to pay us the same royalties" 151); and (ii) Defendants' representative Paul Jacobs stated at a June 25, 2005 press conference that, when asked how WCDMA royalties are calculated, "WCDMA is CDMA after all, so the royalties are the same for any manufacturer" 152).

153 이와 같은 사실은 피심인들이 조사단계부터 심의절차에 이르기까지 일관되게 인정하였으며, 피심인들의 표준계약서(소갑 제9호、10호증), 피심인들이 ***、***、***Patent License Agreements with *** (Exhibits 11 through 24), Respondents' responses to the Tribunal (Exhibit 32), and documents relating to ***'s negotiations with Respondents in 2003 and 2004 and internal reports (Exhibits 68 and 69), ***'s responses to the Tribunal's requests for production (Exhibits 67, 75, and 77); Steve Altman's conversation with Deutsche Bank's Brian Modoff (Exhibit 91); and Paul Jacobs's June 25, 2005 press conference responses. June 25, 2005 press conference answers by Paul Jacobs (Exhibit 92).

B. Define the relevant market

148) Second Response to Exhibit 77*** of the Complaint
149) Exhibit 67 *** Opinion Letter, Exhibit 68 *** Explanatory Memorandum on 2003-2004 Negotiations, Exhibit 69
*** Internal report on the status of negotiations from 2003 to 2004
150) Exhibit 75, ***'s first opinion letter, Exhibit 76, the draft contract that the defendants presented to *** in 2006
151) A Conversation with Steve Altman, Tiger 91, and Brian Modoff, Deutsche Bank
152) Paul Jacobs' response to the June 25, 2005, press conference on smallpox 92

154 Taken as a whole, Defendants' conduct demonstrates that Defendants refused to enter into license agreements with competing modem chipset manufacturers, or limited the scope of license agreements with competing modem chipset manufacturers, for their essential patents in CDMA, WCDMA, and LTE, which are essential to the manufacture and sale of modem chipsets (Conduct 1), and required mobile phone manufacturers who wished to purchase their modem chipsets to enter into license agreements in order to purchase and use their modem chipsets, tying the supply of modem chipsets to the execution of license agreements (Conduct 2), Based on these contractual practices and business structures, the Defendants caused mobile phone manufacturers to enter into license agreements with the Defendants without engaging in the arm's length negotiations that are reasonably required when entering into license agreements, providing only blanket license terms and demanding and accepting royalty terms based on criteria set by the Defendants without engaging in the arm's length consideration process that is generally required, and cross-granting mobile phone manufacturers' patents to themselves and their customers at no cost (Act 3), each of these acts constituting a pattern of conduct that, in the course of its execution, interconnectedly linked and circulated and affected each other.

155 Conduct 1, 2, and 3 therefore all have in common that the relevant markets are the telecommunications standard-essential patent license market and the modem chipset market. Specifically, in conduct 1, the market in which the dominant position is established is the market for licensing telecommunications standard-essential patents and the market in which the dominant position is exercised is the market for modem chipsets because the accused refuse or restrict licenses to competing modem chipset manufacturers. With respect to Acts 2 and 3, the market in which a dominant position is formed is the modem chipset market and

the market in which the dominant position is exercised is the mobile communication standard essential patent licensing market, as the defendants link their modem chipset supply contracts with mobile phone manufacturers to license contracts with mobile phone manufacturers. Therefore, the relevant markets for both Acts 1, 2, and 3 are the mobile communication standard essential patent licensing market and the modem chipset market.

1) Telecommunications Standards Essential Patent Licenses Related Markets

a) Product market for telecommunications standard essential patent licenses

156 Considering the following, the relevant product market for licensing is the market for licenses of all standard-essential patents held by the Accused that are included in each of the telecommunications standards CDMA, WCDMA, and LTE.

157 Each of the telecommunications standards, including CDMA, WCDMA, and LTE, cannot be fully implemented without the use of the Accused's standard-essential patented technologies contained in each standard, and there is no compatibility between the standards' technologies. In particular, because each standard differs in the configuration, functionality, and communication methods of the technology required to implement it, patented technology that implements one standard cannot implement another standard.

158 On the other hand, the accused's individual patented technologies that comprise each of the cellular standards are not substitutes for each other because they perform independent functions. Thus, while it is possible that each of Defendants' individual standard-essential patents in each standard, such as CDMA, WCDMA, and LTE, could be defined as a separate market in which Defendants have a dominant position, the implementation of a particular cellular standard requires the complementary use of all of Defendants' relevant technologies for that cellular standard, Given that in practice, the Respondents do not license each patented technology separately, but rather license the entirety of their patented standards, the relevant product market is the market for licensing the entirety of the Respondents' standard-essential patents for each standard, including CDMA, WCDMA, and LTE.

b) Geographic markets for telecommunications standard-essential patent

licenses

159 The geographic market for a license is the global market, g i v e n t h a t

160 The CDMA, WCDMA, and LTE standards have been adopted by standardization organizations such as ETSI and ITU as standards for the mobile communications industry, and members with or intending to use mobile communications technologies throughout Asia, Europe, and the Americas participate in the standards-setting process.

161 Users of standard-essential patents are licensed to practice without consideration of the geographical location of the patented technology or the technology holder, and the standard-essential patent holder enters into license agreements without consideration of the geographical location of the licensee.

162 After each telecommunications standard is selected as a standard by international standardization organizations, it spreads to individual countries or regions as each country adopts a particular standard as its industry standard. For example, third-generation standards such as TD-SCDMA or CDMA2000 are recognized as standards in a relatively narrow geographic area, whereas third-generation WCDMA or fourth-generation LTE are recognized as standards in a much broader geographic area worldwide. However, in this case, the defendants' narrower geographical market scope does not make a difference to their market power, given that the entirety of the patents that are essential to implement each of the mobile communication standards is a product market, and the concept of standard-essential patents precludes the existence of alternative technologies or competitors in the product market.

2) Modem Chipsets Related Markets

a) Modem chipset related product market

163 Considering the following, the commodity market for modem chipsets is the market for modem chipsets for each standard: CDMA, WCDMA, and LTE.

164 As mentioned earlier, each mobile communication standard is not

technically interchangeable with other standards due to the different technologies applied, such as the frequency bandwidth used, signal transmission and modulation methods, and the number of antennas. Since it is not possible to send or receive information between base stations and mobile phones that support different communication standards, it is not possible to use modem chipsets that implement different standards without changing the base station network.

Id. The subjects themselves, as suppliers of modem chipsets, reflected this, and called CDMA,

Recognize modem chipsets for each communication standard such as WCDMA and LTE as separate products and employ different competitive strategies for each modem chipset market. In addition, mobile phone manufacturers such as ***, ***, ***, and ***, which are consumers of modem chipsets, recognize modem chipsets implementing each communication standard as different, non-substitutable products.

166 As such, modem chipsets for each standard - CDMA, WCDMA, and LTE - are not substitutable for each other in terms of supply and demand, and thus constitute separate and distinct commodity markets.[153)]

b) Geographic markets for modem chipsets

167 Given the physical characteristics, trade structure, supply and demand substitutability, perceptions and behaviors of market participants, and progress in the adoption of telecommunications standards in each region, the geographic market for modem chipsets for each telecommunications standard is the global market in which the standard is adopted and operated.

168 First, modem chipsets are small semiconductor products that are not prone to spoilage, deterioration, or damage due to cross-border transportation and long-term storage, and therefore the cost of long-distance transportation is low. Therefore, geographic differences do not have a significant impact on market planning because transportation costs are low compared to the price of modem chipsets.

169 In addition, modem chipset manufacturers have production facilities around the world and sales activities that extend beyond a specific

geographic region. In particular, in recent years, competition among mobile phone manufacturers has intensified globally, and consumers of modem chipsets are choosing modem chipsets based on whether the modem chipset reliably implements communications standards, the risk of patent disputes, and the price and service of the modem chipset, rather than the geographic location of the modem chipset manufacturer.

153) However, some LTE modem chipsets are compatible with previous generations of standards. There is demand substitutability because buyers can use a single modem chipset that is compatible with multiple standards or use a chipset that implements only the LTE standard in combination with chipsets that implement other standards, and there is supply substitutability because suppliers can produce and sell any number of modem chipsets that implement only the LTE standard. Therefore, LTE modem chipsets that are compatible with 2G and 3G standards and LTE modem chipsets that only implement the LTE standard can be defined as a single product market.

170 Indeed, the Defendants have recognized competitive pressure from competitors in other parts of the world in the modem chipset market due to the high prices of the Defendants' products relative to their competitors, and have responded by implementing strategies to reduce product prices. This perception and management decision-making behavior is similar for other modem chipset manufacturers.

171 Furthermore, with the evolution of telecommunications technology, telecommunications standards are becoming more globally harmonized, especially for third- and fourth-generation standards, where a single standard has been widely adopted virtually worldwide. This gives geographic markets more scope to expand globally than ever before.

172 Basically, each country adopts and operates the telecommunication standards selected by standardization organizations to ensure compatibility of telecommunications and efficient utilization of limited frequency resources.

173 In Korea, only CDMA and modem chipsets capable of implementing it were used in the past for the second generation of standards. For the third generation, WCDMA and CDMA2000 were used together by carriers, and since the late 2000s, they have been integrated into the fourth generation LTE standard.

174 Looking at the global adoption of mobile communication standards, in the past two generations, the United States and North America adopted the CDMA standard, while Europe and Japan selected GSM and PDC based on TDMA, respectively, and operated different standards by region.

175 However, in the case of the third generation standards since 2000,

regional standards have been consolidated, and the third generation WCDMA, which integrates GSM and CDMA, has been widely adopted worldwide, including in Korea, and the fourth generation in the late 2000s, LTE has been widely deployed as the de facto single communication standard.

<Table 45 Share of the global market for each communication standard

Separation	Communication Standards	Key features	Region	Market Share	Who the subjects are licensed to
2nd Generation	GSM	Telephony over wireless networks (voice/text)	Europe, etc.	81	X
	CDMA		South Korea, United States	19	O
3rd Generation	WCDMA	Voice/Video, Data Communications	Most of the world	85	O
	CDMA2000		South Korea, United States, Japan	13	O
	TD-SCDMA		China	2	X
4th Generation	LTE	Superfast data communications	全球	100	O

* Source: Strategic Analysis (based on market share: 2nd generation 2002, 3rd generation 2011, 4th generation 2013)

176 Given the physical characteristics of telecommunications modem chipsets, the ease of transportation, the proliferation of single standards around the world, and the global scope of the trading relationships of modem chipset demanders and suppliers, the geographic market for modem chipsets can be considered a global market.

C. Illegality

1) Act 1: Refusing to license or restricting essential telecommunications standard patents to competing modem chipset manufacturers

a) Applicable laws and regulations

Antitrust and Fair Trade Act154)

Article 3.2 (Prohibition of Abuse of Market Dominant Position) ① A market dominant business shall not engage in any act falling under any of the following subparagraphs (hereinafter referred to as "abuse").

3. Unreasonably interfering with the business activities of other operators

154) as amended by Act No. 14137 of March 29, 2016, hereinafter referred to as the " Act".

Enforcement Decree of the Act on Monopoly Control and Fair Trade155)

Article 5 (Types or Criteria of Abusive Behavior) ③ Unreasonable interference with the business activities of another business pursuant to Article 3 (2) (Prohibition of Abuse of Market Dominant Position), Paragraph 1, Item 3 of the Act shall be defined as directly or indirectly making it difficult for another business to conduct its business activities by engaging in acts falling under any of the following subparagraphs.

1.~2. (omitted)

3. refusing, interrupting, or restricting the use of or access to elements essential to the production, supply, or sale of goods or services of another business without justifiable cause
4. Acts notified by the Fair Trade Commission as acts that make it difficult for other businesses to conduct business activities in an unfair manner other than items 1 through 3.

Abuse of dominant position screening criteria156)

IV. Specific Types and Criteria for Abuse of Dominance

3. Unreasonable interference with the business activities of another business (Article 3(2)(1) of the Act)

Directly or indirectly making it difficult for another business to conduct its business activities by engaging in acts falling under any of the following subparagraphs (Article 5 (3))

a. through b. (omitted)

c. Refusing, interrupting, or restricting the use of or access to elements essential to the production, supply, or sale of goods or services of another business without justifiable reasons (Article 5 (3) (3))

(1) The term "essential elements" (hereinafter referred to as "essential elements") shall include tangible and intangible elements such as networks and infrastructure, and shall meet the requirements of the following subparagraphs.

(a) The production, supply or sale of goods or services cannot be carried out without the use of the element, so that it is impossible to participate in a certain field of trade, or a significant competitive situation persists in the field of trade.

(b) the element is exclusively owned or controlled by a particular entity; and

(c) the person using or attempting to access the element reproduces the element, or

155) As amended by Presidential Decree No. 27534 of September 29, 2016, hereinafter referred to as the "Enforcement Decree".

156) As amended by KFTC Notice No. 2015-15, dated October 23, 2015, hereinafter referred to as the "Examination Guidelines".

It is not practically, legally, or economically feasible to substitute a different element.

(2) For the purposes of this section, "other business entity" means a business entity that is engaged in a trade in which the Essential Element Holder or its affiliates are engaged or are expected to be engaged in the foreseeable future.

(3) "Refusal, interruption, or restriction" means an act that substantially refuses, interrupts, or restricts, or causes the same effect, including in the following cases

(a) offers an unreasonable price or terms that make access to the essential element practically or economically impossible; or

(b) offers unfair terms, such as significantly discriminatory pricing, exclusivity, or tiering, compared to existing users of the Essentials.

(4) In determining whether there is "good cause," we will consider whether any of the following applies

(a) the just reward for the investment of the provider of the essential element is substantially impaired. However, a reduction in profits due to increased competition shall not be deemed to be an impairment of just compensation.

(b) it is not feasible to provide the Essential Element without materially reducing the amount of provision to existing users; or

(c) the provision of the essential element would significantly degrade the quality of the service already provided.

(d) it is technically impossible to provide the required elements due to non-compliance with technical standards, etc.

(e) If there is a risk to the life or physical safety of customers using the service. In addition to this, the following acts that make it difficult for other business operators to conduct their business activities (Young

Article 5, paragraph 3, subparagraph 4)

(1) Unreasonably refusing to do business with a specific business entity or significantly restricting the quantity or content of goods or services to be transacted

(2) Offer terms to a counterparty that are not reasonable in light of normal trade practices or unfairly discriminate in price or terms of trade.

(3) ~ (6) (omitted)

b) impose terms on a counterparty that are unreasonable in light of normal trading practices, or

Whether it unfairly discriminated in price or terms of trade157)

(1) Related jurisprudence

177 Article 3 (2) (1) (3) of the Act prohibits a dominant undertaking from unduly interfering with the business activities of other undertakings, and Article 5 (3) (4) of the Enforcement Decree of the same Act and Criteria for Abuse of Dominant Position IV. 3. of the Enforcement Decree of the same Act prohibits a market-dominant business entity from offering terms and conditions to a counterparty that are not reasonable in light of normal trade practices or unfairly discriminating in price or terms of trade.

178 Therefore, in order for the defendants' conduct to fall under this provision, (1) the defendants must be in a dominant market position; (2) the conduct must be such that the defendants impose conditions on their counterparties that are unreasonable in light of normal trade practices; and (3) the defendants' conduct must impede the business activities of other businesses. In determining the case of difficulty in the business activities of another business, the production, financial, and sales activities of the other business shall be considered comprehensively, but shall include cases where there is a risk of difficulty in business activities.

179 In this context, normal trade practices are judged in principle by reference to the prevailing trade practices in the industry, but depending on the specific case, they may also mean practices that are consistent with the desirable competitive order, which may not always correspond to actual trade practices.158)

180 In addition, the meaning of "unreasonable" or "unfair" in the

context of making it difficult for other undertakings to conduct business by imposing conditions that are not reasonable in comparison to normal trade practices is not limited to the fact that a particular undertaking is individually disadvantaged by the imposition of unreasonable conditions, but also to the intention or purpose of maintaining or strengthening a monopoly in the market, i.e., the intention or purpose of artificially affecting market order by restricting free competition in the market.

157) Article 3 (2) (1) (3) of the Act, Article 5 (3) (4) of the Enforcement Decree, and Criterion Ⅳ. 3. Presenting terms and conditions to a counterparty under (2) that are unreasonable in light of normal trade practices or unfairly discriminating in price or terms of trade

158) Seoul High Court 2009 No. 33777, Seoul High Court 2008 No. 2868, Seoul High Court 2008 No. 2462, etc.

Conduct that is recognized and objectively likely to have anticompetitive effects.

It should be able to.[159)]

181 Therefore, if it is proved that the imposition of conditions that are unreasonable in light of normal trade practices has in fact had the effect of restricting competition, such as increasing the price of goods, decreasing output, stifling innovation, reducing the number of viable competitors, or reducing diversity, it may be presumed that the conduct was likely to result in a restriction of competition at the time of the conduct and that the conduct was intended or aimed at doing so. Otherwise, it shall be determined whether there was an intention or purpose to restrict competition by comprehensively considering the circumstances and motives of the act, the sun, the characteristics of the relevant market, the degree of disadvantage suffered by the counterparty, changes in prices and output in the relevant market, and whether innovation is stifled and diversity is reduced.

182 On the other hand, a standard-essential patent holder's violation of FRAND commitments can be presumed to have a subjective anticompetitive intent or purpose in light of the above. When a technology is selected as a standard through the standardization process within a standardization organization, the patentee is in a position to easily acquire monopoly power ex post as a result of the collective decision of the standardization organization rather than through market competition. In such a situation, if the patent holder of a standard-essential patent demands unreasonable fees or contract conditions and discriminates against licensees, the efficiency-enhancing effects of standardization, such as increasing consumer benefits and promoting technological innovation, will be suppressed, and the harms of restricting competition, such as increasing fees due to the selection of a monopoly technology, excluding competitors in the relevant market, and hindering future technological

innovation, will remain.

159) Regarding the refusal to deal among the abuses of market dominance, the Supreme Court stated, "The act of refusal to deal, which is one of the abuses of market dominance prohibited by Article 3(2)(1)(3) of the Fair Trade Act, is 'an act in which a market dominant business undertaking unfairly refuses to deal with a specific business undertaking, thereby making it difficult for the business undertaking to conduct its business activities. (However, in all cases where a market-dominant undertaking refuses to deal with a specific undertaking that is the counterparty to an individual transaction with an unfair intention or purpose, or where the refusal has caused a specific undertaking to suffer a disadvantage, such as that it has suffered or is likely to suffer a disadvantage in its business activities, the mere fact that the specific undertaking has suffered a disadvantage is insufficient to recognize the unfairness of the refusal, especially if the intention or purpose is to maintain or strengthen its monopoly in the market, The unfairness may be recognized when the refusal to deal is of such a nature that it has the intention or purpose of artificially affecting the market order by restricting free competition in the market, and can be objectively assessed as an act that may have the effect of restricting competition." (Supreme Court, Nov. 22, 2007, Decision 2002D8626, en banc, POSCO Abuse of Dominant Market Position)

Therefore, to avoid this risk after standardization, standardization bodies use the
Given that FRAND covenants are required of patent holders, and given the purpose of FRAND covenants, violation of FRAND covenants can be viewed as conduct that is likely to have anticompetitive effects per se, and such conduct may be presumed to have a subjective anticompetitive intent or purpose.

(2) Whether you are a dominant player

(a) the total market for licenses of patented technology held by the Accused for each of the telecommunications standards, including CDMA, WCDMA and LTE

183 Defendants are dominant in the overall market for licensing their patented technology for each of the telecommunications standards CDMA, WCDMA, and LTE, given the following.[160)]

184 First, due to the nature of standard-essential patents, each of the Respondents' patents constitutes a separate relevant market because they are essential to the implementation of a technical standard and cannot be substituted by competitors' technologies. Therefore, the Respondents' market share in the licensing market for standard-essential patents for each of CDMA, WCDMA, and LTE is 100% and they are presumed to be in a dominant position under the law.

185 Second, moreover, Respondents do not offer sublicensing rights to third parties for their telecommunications standard essential patents, and there are no competing operators other than Respondents.

<Judgment on the claims of the relevant defendants ①>

160) The Seoul High Court ruled that the KFTC's decision in the case entitled "Case on Abuse of Dominant Market Position of Qualcomm Incorporated, Qualcomm Korea Inc. and Qualcomm CDMA Technology Korea" (Dec. 30, 2009) (KFTC Decision No. 2009-281) was upheld. Korea Fair Trade Commission Decision No. 2009-281)" (2010 No. 3932), the court considered the relevant product market to be "the market for patented technologies held by Qualcomm among the patented technologies included in the CDMA standard" and presumed that Qualcomm was a dominant player in this market because of its 100% market share.

186 In this regard, Respondents argue that the mere fact of holding a standard-essential patent does not establish a dominant position.

It argues that the Respondents are not in a dominant position because the ability of the Respondents to maintain, determine or change terms and conditions of trade and to exercise exclusionary rights is constrained by the FRAND commitments and by non-Respondents' holders of mobile telecommunications standard features patents.

187 In light of the fact that (i) the accused have a 100% market share of the market for the accused's patented technology in each of the telecommunications standards, namely CDMA, WCDMA and LTE, and (ii) the downstream market for each telecommunications standard is locked in and no alternative standard exists,

The Respondents' argument is without merit, given that standardization organizations require standard-essential patent holders such as the Respondents to commit to FRAND as a means of preventing abuse of the dominant market position created by standard-setting, but not as a basis for not recognizing a dominant market position; and 3) the patented technologies included in each telecommunications standard perform different functions and are not substitutable for other patented technologies.

(b) the modem chipset market for each standard, including CDMA, WCDMA, and LTE.

188 Given the following, Respondents are market dominant operators in the modem chipset market for each of the standards CDMA, WCDMA, and LTE.161)

189 First, the modem chipset market requires significant initial investment and certain production conditions such as networks and manpower162) to enter the market; second, the mobile phone market, which is a

downstream market for modem chipsets, has a short product replacement cycle, so mobile phone manufacturers develop products using modem chipsets from existing operators that have proven their performance and reliability in the market rather than modem chipsets from new operators.

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161) The Seoul High Court ruled in a lawsuit (2010 No. 3932) against Qualcomm Incorporated, Qualcomm Korea Inc. and Qualcomm CDMA Technology Korea for abuse of dominant market position (KFTC Decision No. 2009-281, Dec. 30, 2009) that Qualcomm is presumed to be a dominant player in the relevant product market as the market for CDMA2000 modem chips, and that Qualcomm's market share is much higher than 50%.

162) Modem chipsets are semiconductor products that require state-of-the-art technology and require a highly qualified workforce and a huge amount of investment.
It needs to be funded and requires continuous investment to keep up with technological advances. It is said that it takes about 15 billion to 20 billion won to design an actual modem chip.

Given the general preference for minimizing turnaround time, and the fact that cell phone carriers are not required to use the

In order to supply mobile phones to a carrier, there are barriers to entry for new modem chipset operators to enter the modem chipset market, including the need to pass multiple levels of compatibility testing and telecommunications network certification tests.

190 Second, in these markets, where there are barriers to entry due to the nature of the products and businesses, the Respondents hold the presumptively dominant market shares in the modem chipset market for each of the following cellular standards: CDMA, WCDMA, and LTE, as shown in Table 46 below.

<Table 46 World market share of modem chipsets by standard among respondents

Standard	2008	2009	2010	2011	2012	2013	2014	In 2015
CDMA	98.4	97.6	96.4	94.3	92.4	93.1	91.6	83.1
WCDMA	38.8	47.4	45.7	55.0	50.4	53.9	48.8	32.3
LTE	-.	-.	34.2	58.8	94.5	96.0	84.8	69.4

* Source: Strategy Analytics 'Baseband Market Share Tracker'

191 Third, the market for CDMA modem chipsets shows that the accused infringe the CDMA standard-essential patents.

We have been in the CDMA modem chipset market since the beginning of our business.

The market share of *** has consistently been over 90% until 2014, and *** is the only competitor of the Respondents. However, ***'s CDMA modem chipsets are mainly installed in low-end mobile phones, and *** requested a license agreement for WCDMA modem chipsets from the Respondents in 2012 to manufacture and sell WCDMA modem chipsets, but the Respondents refused and failed to enter the market.

192 Fourth, Defendants' dominance of the CDMA modem chipset market has had an impact on the current deployment of fourth-generation LTE through the backward compatibility of modem chipsets, solidifying their dominance of the LTE modem chipset market. In order for handset manufacturers to supply handsets to carriers163) that had adopted the second generation CDMA standard, a "CDMA-LTE multimode" modem chipset was required.

163) Examples include Verizon and Sprint in the United States, China Telecom in China, and KDDI in Japan.

Johande, modem chipset manufacturers that supply 'CDMA-LTE multimode' modem chipsets are effectively avoided.
Sims are the only ones.

193 Fifth, in the LTE modem chipset market, the Respondents have been the dominant player in the LTE modem chipset market since the beginning of the LTE modem chipset business, with a 96% share in 2013 and a 70% share in 2015. In particular, as LTE continues to evolve to enable faster processing of larger amounts of data, premium LTE modem chipsets are becoming increasingly important and dominant in the market, and the Respondents have virtually no competitors in this premium LTE modem chipset market.164)

194 Sixth, in addition to being the number one player in the modem chipset market, Defendants are also vertically integrated companies that, unlike competing modem chipset manufacturers, also license modem chipsets, and the organic and reciprocal effects of linking the supply of modem chipsets to licensing agreements reinforce their dominance in both markets.

<Judgment on the claims of the relevant defendants ②>

195 In this regard, the Respondents argue that they are not in a dominant position in the modem chipset market because there are alternative suppliers of modem chipsets to the Respondents' modem chipsets, the major cell phone manufacturers have significant purchasing power in the modem chipset market, and the Respondents' share of the WCDMA and LTE modem chipset markets has been declining in recent years.165)

196 In sum, even if Respondents' share of the WCDMA and LTE modem chipset markets has declined somewhat in recent years,

Respondents are still dominant operators as a matter of law.

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164) In the internal data prepared by the Respondents in September 2014, the Respondents classified ***, ***, ***, ***, etc. as their competitors in the mid- and low-end product categories for LTE modem chipsets, and only *** in the high-end premium product category (Exhibit 56, September 2014 Respondents' internal data). However, *** is not allowed to sell modem chipsets externally due to its patent license agreement with the Respondents, so it can be said that there are virtually no competitors of the Respondents in the premium modem chipset market.

165) Respondents also argue that their conduct does not have anticompetitive effects for the same reasons, but we reject this argument for the same reasons below.

166), and (b) the defendants' share of the WCDMA modem chipset market by 2005.

The fact that the market share of the WCDMA modem chipset market was less than 20% in 2008 and 2014, but has steadily increased since then, and that the Respondents have maintained a dominant position in the market, with a market share of more than 50% from 2011 to 2013; ③ The fact that the Respondents' share of the WCDMA modem chipset market has declined since 2015, partly due to the market shift towards the fourth generation LTE standard167 and partly due to the fact that the Respondents have not introduced any new model modem chipsets for the WCDMA standard since 2013, ④ Given the backward compatibility between generational mobile communications standards, the fact that Respondents' share of the WCDMA modem chipset market has declined in some cases does not mean that Respondents lack dominance in the WCDMA modem chipset market;168)

The accused's market share in the LTE modem chipset market is due to the fact that cell phone manufacturers do not sell their own modem chipsets.

The Respondents' arguments are without merit in light of the fact that, excluding self-consumption by those who produce and procure,169 the Respondents' share of modem chipsets, which are considered to be in a substantially competitive situation, was still 78.5% in 2015; and ⑥ the Respondents' arguments are unlikely to overcome the statutory presumption.

(c) Sintering

197 Taken together, Respondents are dominant players in both the telecommunications standard essential patent license market and the modem chipset market.

166) Article 4(2) of the Act stipulates that a market dominant operator is presumed if the combined market share of three or fewer operators is 75% or more (but excluding those with a

market share of less than 10%). In 2015 (according to Strategy Analytics), the Respondents were (i) the second largest operator in the CDMA (83.1%) and LTE (69.4%) modem chipset market share exceeds 50%, and (2) in WCDMA modem chipset, the defendants are presumed to be a dominant player because they are the second largest player (32.3%) after MediaTek (35.9%) and the third largest player (spread rum 23.9%) combined is more than 75%. The same is true as of Q3 2016, where (1) CDMA (71%) and LTE (54%) modem chipset market share is over 50%, and (2) WCDMA modem chipset market share is 10% after the top two vendors (MediaTek 44% and Spreadtrum 40%), and the combined market share of the top three vendors is over 75%, so it is estimated to be the dominant vendor.

167) In 2015, global WCDMA modem chipset revenues were one-quarter to one-third of LTE, and by the third quarter of 2016, they were down to one-tenth.

168) In this regard, *** (Professor, Faculty of Economics, *** University), who participated as an expert for the interested parties in the Commission's deliberations, stated that (1) WCDMA modem chipsets have effectively lost their share in the market, and therefore the 'WCDMA only chip' share is meaningless, as claimed by the Respondents; instead, if the 'WCDMA+LTE modem chipset' share is measured by sales, the Respondents' share has increased from 38.8% in 2008 to 43.2% in 2016, and they are the number one operator, much higher than MediaTek (25.4%) and Spreadtrum (10.5%); ② LTE modem chipset market share is still above 50% (53-55%) when measured by revenue rather than sales volume; and ③ Although the Respondents do not mention it, their CDMA modem chipset market share has been consistently above 90%.(August 17, 2016, 2nd deliberation of the KFTC)

169) Samsung Electronics and Hysilicon are two of the leading operators that produce virtually exclusively modem chipsets for internal procurement, with a combined global LTE modem chipset market share of 7.9% for Samsung Electronics and 3.7% for Hysilicon in 2015 (Table 17).

198 In this regard, we note that Defendants have not shown that licensing the Telecommunications Standard Essentials Patent

In achieving their dominance in the market, they artificially created the position of standard-essential patent holder by agreement among carriers to standardize rather than compete, while at the same time declaring FRAND commitments to license on a fair, reasonable, and nondiscriminatory basis. In addition, the Respondents are vertically integrated companies that refuse to license their mobile telecommunications standard-essential patents to modem chipset manufacturers, while simultaneously licensing them to handset manufacturers and participating directly in the modem chipset market. Moreover, Defendants are currently the only dominant vertically integrated entities with market power in both the mobile communications standard-essential patent license market and the modem chipset market. Furthermore, a dominant vertically integrated operator such as the Respondents would be subject to the conditions set forth in 1. c) above. 2) b)

(4) Restricting competition in both markets in violation of FRAND commitments, as discussed in (b)

While the incentive to do so increases, the anticompetitive effects are significant.

199 Accordingly, in determining whether Respondents' conduct 1, 2, and 3 in this case constitutes an abuse of a dominant position under Section 3.2 of the Act, the test is whether the conduct is that of (i) a dominant firm that has achieved its dominant position through competition rather than through standardization and FRAND commitments; (ii) a typical dominant firm that is not vertically integrated; or

(iii) The standard for determining whether the conduct of a typical vertically integrated undertaking that does not have dominance in both markets constitutes an abuse of dominance will not be the same as the standard for determining whether the conduct of a typical vertically integrated undertaking that does not have dominance in both markets constitutes an abuse of dominance, and appropriate standards will need to

be established to determine whether the conduct is appropriate to curb the anticompetitive effect and prevent its harmful effects. This will be discussed in more detail below.

(3) Whether you offered terms that were not reasonable in light of normal trading practices.

200 The following points from the admitted facts and evidence show that, in light of the applicable law, it is reasonable in the light of normal trade practices for Respondents to have refused to enter into license agreements with modem chipset manufacturers for the standard essential patents they hold, or to have limited the scope of their license rights by failing to provide full license rights to manufacture, sell, rent, use, repair, and otherwise dispose of modem chipsets, and by limiting the scope of their license rights.

This is equivalent to presenting a condition that does not exist.

(a) what are normal trade practices with respect to modem chipset tier licenses

201 Normal trade practices are those that are consistent with good competitive order, and the determination of whether a practice constitutes a normal trade practice is based on the prevailing trade practices in the relevant market, but does not necessarily correspond to actual trade practices, and even if there are actual trade practices, it is also necessary to consider whether the practice is a good trade practice consistent with good competitive order.

202 Therefore, in order to determine what is normal trade practice with respect to licensing at the modem chipset stage, it is necessary to consider what practices currently exist in the industry with respect to standard-essential patents, and whether licensing of standard-essential patents at the modem chipset stage is necessary in light of the concept and nature of standard-essential patents, Whether there is a reasonable expectation that the license of the standard-essential patent will be provided at the modem chipset stage in light of the basic purpose of the FRAND commitments, and whether it is not practically possible to provide the license of the standard-essential patent at the modem chipset stage.

203 We will discuss this in more detail below, but first, we summarize what we believe to be normal trade practices for modem chipset level licenses.

204 First, the Respondents themselves have obtained the essential patents needed to manufacture and sell their modem chipsets from other

standard-essential patent holders, and there is a current industry practice of licensing standard-essential patents at the modem chipset stage or component stage, such as modem chipsets, with respect to standard-essential patents.

205 Second, in light of the concept and nature of standard-essential patents and the Respondents' own recognition of the fact that they need to obtain licenses from the Accused for their modem chipset business

At the modem chipset stage, licenses for standard-essential patents need to be provided.

206 Third, given the process and purpose of declaring a FRAND commitment to a standardization body and being recognized as a standard-essential patent, the positions of standardization bodies and competition authorities on FRAND commitments, and the positions that Respondents have expressed on FRAND, there is a reasonable expectation that a modem chipset manufacturer that is an active enabler would be offered a license to the standard-essential patent on FRAND terms.

207 Fourth, the existence of the Accused's standard essential patents that are implemented at the modem chipset stage is acknowledged by the Accused, and licensing of components such as modem chipsets, including modem chipsets, is in practice, so that licensing the modem chipset stage is not per se difficult or impossible, even if it entails some cost to the Accused.

208 Taken together, therefore, it is consistent with normal trade practice for a FRAND-committed standard-essential patent holder to grant a license to a modem chipset manufacturer that is willing to pay a fair price for its standard-essential patents to enter into a license agreement with the modem chipset manufacturer to provide unrestricted access to and use of its standard-essential patents on FRAND terms.

209 On the other hand, as discussed above, a vertically integrated player in both the telecommunications standard-essential patent licensing and modem chipset markets, such as the Respondents, and a player with a dominant market position in both markets, would be required to observe and enforce the FRAND commitments more strictly and in good faith because the harm to competition would be significant if it were to refuse to license standard-essential patents to competing modem chipset

manufacturers in violation of the FRAND commitments, contrary to its normal trade practices.

210 In the following, we detail what we consider to be normal trade practices for modem chipset tier licenses.

Existence of modem chipset tier licensing practices

211 Is there a current practice in the telecommunications industry to license essential patents at the modem chipset stage?

212 First, Defendants themselves have licensed 297 standard-essential patent holders around the world, including mobile phone manufacturers, modem chipset manufacturers, base station equipment manufacturers, and patent management companies, as of May 2015, for standard-essential patents necessary for the manufacture and sale of their modem chipsets.

213 Second, 35 operators, including ***, ***, ***, ***, ***, ***, ***, ***, ***, ***, and ***, participate in a patent pool that licenses their patents, including essential patents for telecommunications standards, for all stages of the modem-chipset stage.[170)]

214 Third, modem chipset manufacturers such as ***, ***, ***, ***, and *** have cross-licensed all of their standard-essential patents, including telecommunications standard-essential patents, to competing modem chipset manufacturers.[171)]

215 Fourth, it is known that standard essential patents are licensed at the component level for the manufacture and sale of other components installed in mobile phones, such as wi-fi, application processors (APs), memory, and cameras.[172)]

216 Thus, we would say that there is a trade practice of licensing standard-essential patents to component manufacturers, such as modem chipsets, rather than to modem chipset manufacturers or end devices.

170) SOP 87 Introduces ViaLicensing's LTE SEP Licensing Program

171) The details of such agreements shall not be disclosed to protect the trade secrets of such operators.

172) ***, who participated as an interested party in the Panel's deliberations, stated that "chipset-level licensing is common in the semiconductor industry, including baseband chipsets" (***'s presentation at the Panel's third deliberations, September 5, 2016), and ***, who worked as a licensing executive at ***, submitted an opinion that "licenses for modem technology that provides Internet connectivity to computers ... are licensed at the chip level" (***'s expert opinion submission, September 9, 2016).

<Judgment on the claims of the relevant defendants ③>

217 In this regard, the Respondents argue that licensing at the handset level is a practice that has existed in the market since the Respondents entered the market, and that it is normal trade practice to license at the handset level, as *** and **** also license to handset manufacturers, and that the Respondents' conduct in following this practice is not at issue.

218 In sum, Respondents' arguments are without merit given that

219 First, the individual and specific conduct of the Respondents must be viewed in the context of the individual and specific characteristics of the conduct in order to determine whether it is within the scope of normal trade practices, and the existence or existence of licensing practices at the mobile phone stage does not make the individual and specific conduct of the Respondents normal trade practices.

220 (i) The Respondents' conduct in this case concerns "standard-essential patents," not ordinary patents. The Respondents have become "standard-essential patent holders" who have gained dominance over their patented technology through an artificial process of standardization rather than competition, and in return have entered into FRAND commitments to waive the rights of "exclusion" and "discrimination" that a typical patent holder would have173). Therefore, it is difficult to view the Respondents' conduct in refusing to license standard-essential patents to competing modem chipset manufacturers, i.e., "excluding" and "discriminating" against them, as a normal trade practice, given the way other businesses do business.

221 (ii) Respondents are dominant vertical integrators in both the telecommunications standard-essential patent and modem chipset

markets. The conduct of a dominant vertically integrated undertaking may be considered to be harmful if it has an adverse effect on the competitive situation in the relevant market as compared to conduct that would normally result from the position of a standard-essential patent holder or conduct that would normally result from the position of a non-dominant vertically integrated undertaking.

173) In their 2011 FRAND comments to the U.S. Federal Trade Commission (FTC), the Respondents stated that "a patentee who makes a RAND commitment has given up the right to refuse to license." [Qualcomm.Inc., Comment to U.S. Federal Trade Commission, Project No. P11-1204, (June 13, 2011)].

The degree will be significant, so a more stringent standard is required. Therefore, some other general business
The mere fact that a person has engaged in the same or similar conduct as the Respondents does not mean that the Respondents' conduct is consistent with normal trade practices.174)

222 (iii) The standard-essential patents for which the Respondents refuse to license at the modem chipset stage, despite requests for licenses from competing modem chipset manufacturers, but license at the mobile phone stage and collect royalties for doing so, are "standard-essential patents implemented in modem chipsets". Therefore, unless there is a special reason, such as that the modem chipset manufacturer does not want to license it or the mobile phone manufacturer wants to license it freely, it is not a normal trade practice to refuse or restrict the license of a standard-essential patent at the modem chipset stage or to force the mobile phone manufacturer to license it at the mobile phone stage without reasonable and justifiable reasons against the will of the patent user, as it may lead to the result that the standard-essential patentee has no choice but to pay an unreasonable implementation fee.

¶ 223. Second, contrary to Respondents' assertions, licensing of standard-essential patents is widespread, even at the component level, such as modem chipsets and modem chipsets. The first example is the Respondents themselves. Currently, Defendants have licensed the Standard Essential Patents from 297 different operators around the world to manufacture and sell their modem chipsets. 또 한 피심인들은 과거 *** 、*** 、*** 、*** 등과 사용권한이 제한된 형태의 모뎀칩셋 특허 라이선스 계약을 체결하여 실시료를 수취해 온 사실도 있다. In addition, the Respondents are not the only ones who license standard-essential patents at the modem chipset stage through participation in patent pools or cross-licensing arrangements. In addition to the modem chipset, other components used in mobile phones, such as Wi-Fi, application processors (APs), memory, and cameras,

are also licensed at the component stage. Therefore, licensing at the mobile phone stage can be used to license mobile communication standard essential patents.

174) In this regard, the EU Commission has decided that the fact that a behavior that constitutes an abuse of a dominant position is an industry practice that happens to be practiced by other undertakings does not change the finding of illegality. The text of the decision is as follows "In any case, there are no grounds under Article 82 of the Treaty according to which instances of behaviour by a dominant company which accord with industry practice should be screened out. Even if it were therefore the case that other vendors did not disclose similar interface information, this would not mechanically exculpate Microsoft. (Commission decision of 24.03.2004, Case COMP/C-3/37.792, footnote 877)

This is not the sole or usual course of dealing with licenses from the Company.

224 Third, early cell phones at the time of the Defendants' entry into the market were essentially cellular telephones for voice communication, and the core functions of mobile communications were concentrated in the modem chipset. In contrast, modern mobile phones serve as comprehensive IT devices, and the price of a modem chipset accounts for only about 4% of the price of a mobile phone, and given the changes in the modem chipset market and mobile phone market conditions, it is not possible to conclude that the Respondents' conduct in this case was in accordance with normal trade practices simply because of past licensing practices at the mobile phone level or similar licensing practices of some other operators.

Modem chipset tier license required

225 Whether a modem chipset manufacturer is required to obtain a license from Respondents for the CDMA, WCDMA, and LTE standard essential patents held by Respondents in order to manufacture and sell modem chipsets,

226 Unlike non-standard-essential patents, standard-essential patents, by their very concept and nature, must be infringed by modem chipset manufacturers in order to manufacture and sell modem chipsets,175 and there is no dispute between Respondents and interested parties that some of Respondents' CDMA, WCDMA, and LTE standard-essential patents are necessary for the manufacture and sale of modem chipsets, and that such standard-essential patents exist.176)

227 Therefore, if a modem chipset manufacturer manufactures and sells a modem chipset without a license to the standard-essential patent, the modem chipset manufacturer may be subject to a claim for infringement of the

175) In this regard, the accused entered into license agreements at the modem chipset stage until 2008, and in a 2006 lawsuit against Broadcom, the accused alleged that Broadcom's modem chipsets infringe the accused's standard essential patents. (Broadcom v. Qualcomm, Case NO. 08cv1607, Southern District of California)

176) However, there is disagreement among the Respondents and other interested parties as to the scope and percentage of the Respondents' standard-essential patents that are implemented in modem chipsets. However, at least the Respondents themselves admit that 4.23% of their standard-essential patents have all claims directed to modem chipsets.

Prohibited from suing for damages for patent infringement at any time177) or selling or using such products

Provisionally exposed to the risk of claims.

228 Moreover, Defendants themselves have taken the position that the modem chipset business requires a license from Defendants, stating in their business report (K-10) that they "recognize that companies that manufacture and sell products that use CDMA technology must license our patents."178) The Respondents have also stated that they have a "forbearance policy" toward competing modem chipset manufacturers that requires them to license their patents, which implies that if a modem chipset manufacturer uses a standard-essential patent without a license, it infringes the patent, but that they are looking the other way, which in itself presupposes that the modem chipset infringes their standard-essential patent.

229 Therefore, in light of the above, it is imperative for modem chipset manufacturers to obtain a license to the essential patents of CDMA, WCDMA, and LTE standards held by the defendants in order to engage in the business of selling modem chipsets, and in order to start a modem chipset manufacturing and sales business with high barriers to entry, it is necessary to have strong financial capabilities, technical capabilities, and business plans, as well as thorough preparations to avoid being involved in patent disputes.

<Judgment on the claims of the relevant defendants ④>

230 In this regard, the Respondents argue that because modem chipset manufacturers have allowed the Respondents to practice the standard-essential patents held by the Respondents, the modem chipset manufacturers can engage in the business of manufacturing and selling modem

chipsets without a license agreement with the Respondents, and there is no particular uncertainty about patent disputes, including patent infringement litigation.

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177) Especially in the U.S., punitive damages can be applied in damages judgments for patent infringement, so the burden on the licensee (in this case, the modem chipset manufacturer) is significantly higher if infringement is found.

178) In its 2014 business report, Defendants stated that "The mobile communications industry generally recognizes that a company seeking to develop, manufacture and/or sell products that use CDMA technology will require a patent license from us."

231 However, Respondents' argument is without merit given that

232 First, in light of the purpose and terms of the FRAND commitments, the expected normal course of dealing with a request for a license to practice a standard-essential patent is to grant an irrevocable license under the terms of FRAND, not mere inaction, as Respondents contend. Furthermore, the inaction alleged by Respondents is merely a de facto discretionary withholding of rights by the patentee, not an unwillingness of the patentee to exercise any rights. There is a fundamental difference in the risk of patent infringement for a party requesting a license compared to a party entering into a license agreement.[179] Therefore, it is difficult to conclude that it is within the scope of normal business practices in the relationship with a modem chipset manufacturer to refuse a request for a license to practice a semi-essential patent and merely state to the other party that it is neglecting to practice.

233 Second, the Respondents have not publicly disclosed their alleged laissez faire policy to the level of an irrevocable commitment, and (i) although the Respondents' General Manager of Licensing *** initially accepted B's request to "document in a specific contract that the Respondents will not prevent B from selling the modem chipsets externally" in a phone call with B's Executive Vice President *** on *March* 2011, he later reversed his position and claimed in an email on *March* 2011 that "B will not need a separate contract to sell the modem chipsets externally. but later changed its position and asserted in an email on *March* 2011 that 'B would not need a separate agreement to sell the modem chipsets externally';[180] (ii) Furthermore, Respondents' General Manager of Licensing *** stated in an email to his employees on *March* 2011 that "We have not done anything to allow E to develop and sell the modem chipsets to third parties. If E sells the modem chipset to a third party, we can assert our patents against E or E's customers."[181] In light of the fact that the Defendants have not

done anything to authorize E to develop and sell the modem chipset to a third party, it is clear that the Defendants have the right to terminate the license at any time due to a change in circumstances or at their sole discretion.

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179) In this regard, Mr. *** (Professor, *** University School of Law, *** University), who participated in the deliberations of the Committee as an expert, stated that "a contract is an agreement between two parties to exchange consideration, and an act that does not fulfill the conditions for the formation of such an agreement cannot be recognized as a contract. If a party makes a unilateral promise not to sue another party for patent infringement, the promise does not constitute a contract unless there is an exchange of consideration and a meeting of minds. A promise that does not reach the stage of a contract is not an enforceable covenant, and thus the promise alone does not comply with the FRAND covenant." (*** Opinion submitted to the KFTC on September 5, 2016)

180) Microtelangiectasia 60

181) Microtelangiectasia 47

to reverse its leniency policy and assert its standard-essential patent infringement claims against modem chipset manufacturers.
Since it cannot be said that there is no possibility at all that this could be the case, it cannot be said that Defendants' alleged declarations of leniency have eliminated the uncertainty of patent disputes in the modem chipset business for modem chipset manufacturers.

234 Third, Defendants (i) entered into royalty-bearing license agreements with five competing modem chipset manufacturers through 2008, and (ii) have alleged in litigation with Broadcom since 2005 that Broadcom infringed Defendants' patents in the manufacture of modem chipsets and that licenses were necessary, In 2003, Defendants filed a patent termination claim against Texas Instruments (TI) for breach of contractual confidentiality provisions183), and Defendants have asserted patent infringement claims against competing modem chipset manufacturers. In light of this, it is difficult to conclude that Respondents have been consistently negligent in preventing modem chipset manufacturers from practicing their patents, and therefore that there is no particular uncertainty regarding patent infringement litigation.

③ Reasonable expectation of licensing telecommunications standard essential patents at the modem chipset stage

235 The Respondents have attained the status of standard-essential patent holders by virtue of the commitments made by various standardization organizations to grant FRAND terms to those who wish to practice their standard-essential patents. Therefore, the defendants are obligated to negotiate in good faith to provide modem chipset

183) [redacted] ." (Sigmoiditis 90, *** [illegible])

spirit of standardization and FRAND commitments unless there are special

circumstances. Therefore, a modem chipset manufacturer that actively seeks to implement the standard essential patents necessary for the manufacture and sale of modem chipsets can obtain a license to implement the standard essential patents from the defendants on FRAND terms.

182) Broadcom v. Qualcomm, Case NO. 08cv1607, Southern District of California

It would be reasonable to expect that there might be.

<Judgment of the claims of the relevant defendants ⑤>

236 In this regard, the Respondents argue that the refusal to license standard-essential patents to modem chipset manufacturers cannot be said to violate the FRAND commitments because the IPR rules of ETSI, one of the standardization organizations, do not clearly define the subject matter of licensing as the component modem chipsets, but rather can be interpreted as defining the end product.

237 However, leaving aside the issue of ETSI's individual rules, in light of the concept and recognition process of standard essential patents, the purpose of the FRAND commitments, the positions of other standardization organizations and competition authorities, and the Respondents' own past positions on the FRAND commitments, it is reasonable to conclude that it is within the meaning of normal trade practice to license standard essential patents essential to the manufacture and sale of modem chipsets to a willing modem chipset manufacturer for a fair consideration, and therefore the Respondents' argument is without merit.

238 First, a standard-essential patent is conceptually and inherently required to be licensed for the manufacture and sale of products that embody the standard-essential patent, and those who wish to manufacture and sell such products are exposed to the risk of patent infringement claims from the standard-essential patent holder if they do not enter into a license agreement, which is no different in the case of a modem chipset manufacturer who wishes to manufacture and sell modem chipsets. To prevent such harm, standardization organizations require patent holders to declare a FRAND commitment before adopting a technology as a

standard, and the Respondents have already made written commitments to standardization organizations that they will license their essential patents on FRAND terms.

239 Second, FRAND commitments are not limited in scope to component manufacturers, such as modem chipsets, or end device manufacturers, such as cellular phones, and have been recognized by leading competition authorities, standardization organizations, and courts as

The position.[184][185]

240 (i) The European Commission explains in its Horizontal Agreement Guidelines that standard-essential patent holders must declare in writing an irrevocable FRAND commitment to license on FRAND terms to "all third parties."[186]

241 (ii) The U.S. Department of Justice (DOJ) has explained that FRAND commitments should apply to "all implementers," whether or not they are members of a standardization body.[187]

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184) "The need to provide anyone licenses stems from the nature of standardization," said Professor *** University School of Law ***, who participated in the deliberations as an expert, "standardization has the effect of removing from the market technology that is not selected as a standard by the concerted action of multiple participants, and allowing only the patentee of the standardized patent to practice the technology exclusively. To prevent such artificial monopolization, the FRAND covenant contains a whoever requirement. If the patentee of a standard-essential patent is able to select the licensee, the subject invention becomes the exclusive property of a few or the patentee alone, and competition is harmed. The anyone requirement is necessary to prevent this from happening. In that sense, the anyone requirement is the most important of the FRAND commitment's many requirements. If the anyone requirement is not complied with, there would be no implementation agreement, and therefore no opportunity to discuss the other requirements of the FRAND commitments. Therefore, the violation of the anyone requirement should be treated more severely than the violation of other requirements." (Expert *** submission to the third session of the KFTC plenary meeting, September 5, 2016)

185) Other scholars have stated that there are no restrictions on who can be licensed under FRAND commitments.
is called.

Yoo, "Examining the Possibility of Unreasonable Restrictions on the Exercise of Standardized Patent Rights," at 57 ("One device for restricting the exclusive exercise of patent rights by a standardized patent holder so that anyone can use them is a 'FRAND' declaration.").

Ho-Young Lee, "A Study on the Enforcement of Antitrust Laws Against FRAND Covenant Violations by Standard Essential Patent Holders",
"Commercial Law Review, Vol. 31, No. 4, 2013, p. 256 ("The anyone requirement does not prevent a standard patentee from refusing to enter into an implementation agreement, and there is no question about that.")

③Jay P. Kesan and Carol M. Hayes, *Patent Transfers in the Information Age: FRAND Commitments and Transparency*, p. 5 ("During the standard setting process, a board or committee may request that members disclose relevant patents, and may also seek agreements from patent holders to either license these patents on royalty-free (FRAND-RF) or FRAND terms to anyone who requests a license.")

④Brian T. Yeh, *Availability of Injunctive Relief for Standard-Essential Patent Holders*, 2012, in summary ("when those companies have previously committed themselves to license their patented technology to anyone (corporate partners or competitors) on FRAND terms.")

⑤Stanley M. Besen, *Why Royalties for Standard Essential Patents Should Not Be Set by the Courts*, 15 Chi.-Kent J. Intell. Prop. 19, 20 (2015) ("The policies often require or encourage members of the standards setting organizations to identify patents that are essential to a proposed standard

and to agree to license their essential patents on reasonable and nondiscriminatory ('RAND') terms to anyone who requests a license.")

186) "In order to ensure effective access to the standard, the IPR policy would need to require participants wishing to have their IPR included in the standard to provide an irrevocable commitment in writing to offer to license their essential IPR to all third parties on fair, reasonable and non-discriminatory terms(FRAND commitment). That commitment should be given prior to the adoption of the standard." (EU Commission, Horizontal Cooperation Agreement Guidelines 285)

187) "we have encouraged SSO to : Make it clear that licensing commitments made to the standards body are intended to bind both the current patent holder and subsequent purchasers of

242 (iii) Major standardization organizations such as ITU, ISO, IEC, TIA, and ARIB have signed the "FRAND Commitment.

."[188] The IEEE has stated that once a FRAND commitment has been made, it is not possible to exclude a particular stage or a particular operator from licensing.[189]

243 (iv) The U.S. Court of Appeals for the Ninth Circuit has held that "there is no limit on the qualifications or number of applicants that can be licensed under a FRAND commitment,"190) and the U.S. District Court for the Northern District of Illinois190) and the U.S. Court of Appeals for the Federal Circuit191) have ordered that standard-essential patents be licensed on FRAND terms to anyone191) without tiered discrimination192).

244 Third, the Respondents themselves have acknowledged that they are obligated to license on FRAND terms at the modem chipset level since they made FRAND commitments in their 1998 litigation with Ericsson, in their 2005 investor meeting materials, in their 2008 court filings in the Elsevier vs. Qantas litigation, and in their 2011 filings with the U.S. FTC, as shown below.

245 (i) Defendants acknowledge that in their litigation with Ericsson on October 6, 1998, Ericsson asserted that the

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the patents and that these commitments extend to all implementers of the standard whether or not they are a member of the standards body." (Department of Justice, "The Role of Standards in the Current Patent Wars")

188) The Licensing declaration of the ITU, ISO, and IEC's FRAND commitment states that "Patent holders is prepared to grant a license to a restricted number of applicants on a worldwide non-discriminatory basis and on reasonable terms and conditions to make, use and sell implementations of the above document."

189) "The Update obligates patent holders band by the IEEE RAND commitment to license their patents for 'any compliant Implementation' meaning that a patent holder making an IEEE RAND Commitment cannot refuse to license its patents for use in IEEE SA standards at certain levels of production" (U.S. Department of Justice Position on the IEEE Intellectual Property Changes in Subchapter 86)

190) Microsoft Corp. v. Motorola, Inc. 696 F.3d 872, 884 (9th Cir.2012)

191) Apple, Inc. v. Motorola, Inc., 869 F. Supp. 2nd, 901, 914 (N.D. Ill 2012) "By committing to license its patents on FRAND terms, Motorola committed to license the '898 to anyone willing to pay a FRAND royalty and thus implicitly acknowledged that a royalty is adequate compensation

for a license to use that patent."
In re Innovatio IP Ventures, LLC Patent Litig., No. 11 C 9308, 2013 WL 5593609 (N.D. I ll. Oct. 3, 2013) "considering the profit of the chip manufacture on the chip, rather than the profit margins of the Manufactures on the accused products, is appropriate because a RAND licensor such as Innovatio cannot discriminate between licensees on the basis of their position in the market."

192) Ericsson, Inc. v. D-Link Sys. Inc., 773 F.3d 1201, 1231 (Fed. Cir. 2014) "Ericsson promised that it would 'grant a license under reasonable rates to an unrestricted number of applicants on a worldwide basis with reasonable terms and conditions that are demonstrably free from unfair discrimination.'"

193) The original text is "to anyone" or "to an unrestricted number of applicants on a worldwide basis".

obligated to license standard-essential patents to licensees on FRAND terms. On March 26, 1999, the defendants issued a press release194) ending the legal dispute with Ericsson, stating that they would license their standard essential patents on FRAND terms "to the rest of the industry".

246 (ii) At Respondents' 2005 investor conference, Respondents' Chief Executive Officer, Steve Altman, stated that "Qualcomm has never refused to license the WCDMA standard-essential patents to anyone," in response to allegations that Respondents were excluding competing modem chipset manufacturers from the market.196)

247 (iii) Respondents stated in a legal brief filed with the U.S. Supreme Court in the 2008 case between LG Electronics and Qantas197) that modem chipsets are products that implement standards, and that they have been licensing their standards-essential patents under FRAND commitments to licensees seeking to produce products that implement standards, and that such licenses include companies that produce various components such as modem chipsets.198)

194) Ericsson Inc. et al vs. Qualcomm Inc. Civil Action No. 2-96-CY183 (E.D.Tex)

195) "As part of the settlement, the companies will declare their commitment to the ITU and other standardization bodies to license to all other operators in the industry, on a fair, reasonable and non-discriminatory basis, the defendants' essential patents for a single CDMA standard and all modes of the standard." (https://www.qualcomm. com/news/releases/1999/03/25/ericsson-and-qualcomm-reach-global-cdma-resolution)

196) "They claim that we're trying to exclude competing manufacturers of chipsets from the market, or preventing them from entering. (Emphasis added.) We've never refused to license our WCDMA essential patents to any company." (Exhibit 82, Defendants' 2005 Investor Meeting Materials)

197) Quanta Computer, Inc. v. LG Electronics, Inc. 553 U.S. 617, 635 (2008); LG Electronics is the patentee of a method patent for efficiently managing various memories and buses through microprocessors and chipsets, and LG Electronics licensed the patent to Intel under the condition that Intel must obtain permission from LG Electronics to manufacture computer systems in combination with memories or buses manufactured by LG Electronics or other parties other than Intel, and Intel's customer, Quanta, claimed patent infringement against Quanta for violating the terms of the license. In response, the court ruled that (1) the doctrine of patent exhaustion applies to method patents as long as the method patent is embodied in a product, and (2) if the product essentially embodies the method patent, the sale of the product exhausts the method patent. In other words, if the product embodies the core features of the patent under the rightful authority, the patent right is exhausted by the sale, even if the license agreement sets the authority narrowly.

198) "Qualcomm is a member of many standards setting organizations, which typically request that

a member commit to license its patents that are actually essential to a standard on terms that are Fair, Reasonable and Non-Discriminatory ("FRAND"). (middle) Qualcomm has offered licenses to its portfolio of technically necessary patents on FRAND terms to entities desiring such a license to produce products that implement a given standard. Qualcomm has entered into over 200 patent license agreements covering all or substantial portions of its patent portfolio. (emphasis added) These portfolio licenses have enabled numerous entities to manufacture and sell a wide variety of CDMA components, products and combinations thereof, including wireless chips and handsets." (Amicus Curiae Brief for Respondents in LG Electronics-Quanta case)

248 (iv) In their 2011 comments on FRAND filed with the U.S. FTC, Defendants refer to the "RAN

D199) that a patentee who makes a commitment has waived its right to refuse a license."200).

249 On the other hand, the Respondents, relying solely on the regulations of ETSI among the major standardization organizations, argue that there is no obligation to license the manufacturers of modem chipsets because, according to Article 15 of the ETSI Regulations, "manufacture" means the production of "equipment" and "equipment" means "any device fully conforming to a standard" and modem chipsets are not final devices and therefore are not devices that fully implement a standard. However, (i) Article 6.1 of the ETSI Regulation expressly provides for the licensing of 'customized components';201; (ii) the phrase 'any device fully conforming to a standard' in the Regulation is understood to mean 'a device that fully complies with a standard', and given that modem chipsets comply with telecommunications standards, it would be difficult to argue that the concept of 'equipment' does not include modem chipsets. (iii) Moreover, the Respondents' contracts with competing modem chipset manufacturers and mobile phone manufacturers, which themselves describe modem chipsets as "electronic devices implementing the standards",202 suggest that the Respondents also understood equipment to include components, and therefore the Respondents' arguments regarding the interpretation of the ETSI Regulations are untenable.

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199) Some use the term " RAND" instead of "FRAND" because they believe that the concept of fair can be subsumed by the concepts of resonable and non-discriminatory.

200) "a patent-holder who gives a RAND commitment gives up the right to refuse to license" [Qualcomm.Inc., Comment to U.S. Federal Trade Commission, Project No. P11-1204, (2011.6.13.)]

201) ETSI's Intellectual Property Right Policy 6.1 is as follows (ETSI Intellectual Property Right Policy 6.1)

6. Availability of Licenses

6.1 When an ESSENTIAL IPR relating to a particular STANDARD or TECHNICAL SPECIFICATION is brought to the attention of ETSI, the Director-General of ETSI shall immediately request the owner to give within three months an irrevocable undertaking in

writing that it is prepared to grant irrevocable licenses on fair, reasonable and non-discriminatory ("FRAND") terms and conditions under such IPR to at least the following extent:

- MANUFACTURE, including the right to make or have made customized components and sub-systems to the licensee's own design for use in MANUFACTURE;
- sell, lease, or otherwise dispose of EQUIPMENT so MANUFACTURED;
- repair, use, or operate EQUIPMENT; and
- use METHODS.

202) Pursuant to Article ** of the Patent Portfolio Agreement between Defendants and **, and the definitions in Appendix 1 of the Patent Agreement between Defendants and ***, Defendants define "integrated circuit" to include a modem chipset as a "device," and given that ASIC generally refers to an application specific integrated circuit, a modem chipset would be a device.

250 Therefore, based on the above considerations, the modem chipset manufacturers are listed in Table
Respondents' argument that refusing to license quasi-essential patents cannot be said to violate the FRAND covenant is without merit.

④ Feasibility of licensing the modem chipset stage

251 As to whether it is feasible for the Respondents to license their essential patents to modem chipset manufacturers, (i) the Respondents themselves have entered into license agreements with ** modem chipset manufacturers through 2008 and received royalties, (ii) as of May 2015, the Respondents have been licensing the essential patents necessary for the manufacture and sale of their modem chipsets from 297 standard-essential patent holders worldwide, including mobile phone manufacturers; (iii) under normal patent licensing practice, it is possible to license standard-essential patents at the component stage, such as modem chipsets203). (iv) On the other hand, even if licensing at the modem chipset stage would result in a multi-stage license and incur some costs to the Respondents, as the Household Respondents argue,204) it is more reasonable to conclude that these costs are within the range of costs that a patentee must bear in order to collect royalties.205) Therefore, based on the above, it cannot be said that it is impractical or impossible for the Respondents to license their standard-essential patents to modem chipset manufacturers.

<Judgment on the claims of the relevant defendants ⑥>

252 In this regard, Defendants argue that because their patents are not fully implemented at the component level, such as in a modem chipset, but only at the mobile phone level, they are not entitled to a license at the mobile phone level.

203) In this regard, *** (Vice President of ***), *** (Professor of Electronics and Telecommunications Engineering at *** University), and *** (Patent Attorney), who participated as experts in the committee's deliberations, stated that "the modem chipset license is in line with the patent practice agreement, and it is possible to divide the patent into stages and enter into a license agreement at the modem chip and mobile phone stages respectively, and if the claims cover multiple products, the product scope can be determined by each claim."

204) Defendants argue that even if they were to license at the modem chipset stage, some of their patents would still be implemented at the handset stage, requiring them to enter into additional licensing agreements with handset manufacturers, making a multi-stage (modem chipset stage + handset stage) license unavoidable.

205) Moreover, as discussed above, Defendants themselves had already engaged in multi-level licensing of modem chipsets and mobile phones prior to 2008.

is the way to ensure that Defendants are fairly compensated for their patents.

253 But let's look at the defendants' arguments,

254 First, it is possible to distinguish the essential patents held by the Accused into those that are implemented in the modem chipset stage and those that are implemented in the mobile phone stage, as the Accused themselves admit that there are essential patents that are implemented in the modem chipset stage, and the white papers sent by the Accused to mobile phone manufacturers describe the modem chipsets as implementing their telecommunications essential patents206), it is reasonable to license the essential patents implemented in the modem chipset to the modem chipset manufacturer, and it is practically possible to determine whether the essential patents are implemented in the modem chipset.[207)]

255 Second, the Defendants themselves submitted an amicus curiae brief in the 2008 litigation between LG Electronics and Qantas stating that "modem chip set stage licensing is a way to more fully compensate for the value of the patent rights," and in fact, prior to 2008, the Defendants had been entering into dual multi-stage license agreements and collecting royalties for both the modem chip set and mobile phone stages, If it is possible to conclude a license agreement and collect royalties at each stage, such as the modem chipset stage and the mobile phone stage, even if the license is provided at the modem chipset stage, if there is a patent practiced in a part other than the modem chipset, the defendants themselves can prove it and collect royalties at the stage other than the modem chipset, such as the mobile phone, so it cannot be said that it is impossible to justly compensate the defendants for the standard essential patent.[208)]

206) The original claim reads "Qualcomm's chipsets including components implementing the CDMA2000 family of standards, the WCDMA family of standards and/or the OFDM/OFDMA

family of standards." (Exhibit 39, Defendants' promotional materials for the benefits of using third party patents)

207) In this regard, ***, an expert for *** who participated as an interested party in the Tribunal's deliberations, submitted his experience when he worked for A. He stated that "A organized its licensing program in such a way that it was able to derive value from A's patent portfolio. It divided its patent portfolio into two groups, (i) system patents and (ii) chip patents, and introduced different licensing programs for each group of patents due to the significant difference in the way the royalties for those patents were calculated, i.e., the difference between (i) system-based royalties and (ii) chip-based royalties. In order for A to have separate licensing programs for system and chip patents, it would naturally need to know which patents were substantially implemented in chips and which were system-wide. Thus, A has adopted the very classification process that Qualcomm argues is impossible." (Expert Opinion filed Sept. 9, 2016).

208) In this regard, *** (Vice President of ***), *** (Professor of Electronics and Telecommunications Engineering at *** University), and *** (Patent Attorney), who participated as experts in the Commission's deliberations, submitted the opinion that "licensing the patents to a modem chipset company does not cause Qualcomm to lose the value of its patents because not all patent rights are exhausted at the modem chipset stage, but only those claims that are implemented in the modem chipset are exhausted and those that are not are not." (Aug. 17, 2016.

256 Third, the stakeholder209) and expert210) comments submitted during the deliberation phase and the responses to those comments.

In light of the debates between one Respondent and stakeholders and experts211), it is difficult to accept Respondents' claim that only 4.23% of the semi-essential patents are implemented in modem chipsets.

257 Thus, in light of the above, it is difficult to conclude that Respondents' mobile communications standards patents are implemented only in the mobile phone stage, and it is difficult to conclude that licensing only the mobile phone stage is the only way for Respondents to receive fair compensation for their patents.

(b) proposed terms that are not reasonable in light of normal trading practices.

258 As discussed in (a) above, it is normal practice for a modem chipset manufacturer that wishes to enter into a license agreement for a telecommunications standard-essential patent to license the standard-essential patent to the standard-essential patentee on FRAND terms, and to provide full, unrestricted license rights as a condition of the license.

259 However, in light of the admitted facts and the following, Respondents' refusal to enter into a license agreement for the Standard Essentials patents covering the modem chipset invention was not in the ordinary course of business.

Second review by all members of the FTC)

209) Mr. ***, an expert participating as an interested party in the Panel's deliberations, stated that 56.5% of the Respondents' standard-essential patents are implemented in the motherboard chipset and only 1% of the standard-essential patents are implemented in the mobile phone.

210) In this regard, *** (Vice President of ***), *** (Professor of Electronics and Telecommunications Engineering at *** University), and *** (Patent Attorney), who participated as experts in the Commission's deliberations, argued that the Respondents intentionally reduced the percentage of standard-essential patents implemented in modem chipsets, on the grounds that (1) a patent consists of multiple claims, and the Respondents' 4.The Respondents' classification is incorrect as infringement of only one claim is sufficient to constitute patent infringement; (2) the Respondents arbitrarily classify

non-standard-essential patents as standard-essential patents and claim them as patents implemented in mobile phones; and (3) the four patents that the Respondents representatively presented as mobile phone stage patents in the first trial are actually mobile chipset stage patents.(Second deliberation of the plenary session of the KFTC on August 17, 2016)

211) At the first hearing before the full Commission, Respondents argued that none of Respondents' patents are practiced exclusively on modem chips.
However, after the expert participant at the second hearing rebutted the above footnote 209, Samsung Electronics Co. withdrew its arguments at the first hearing with respect to three of the four patents it had originally presented before the third hearing. (Sejong No. 16-5397, Sept. 2, 2016, Correction of the Respondents' First Presentation).

line, and that the Defendants, instead of providing licenses to competing modem chipset manufacturers, have instead provided licenses to
Because many of the proposed restrictions are not terms that would be imposed under normal trade practices in response to a request to enter into a license agreement for a standard-essential patent, Respondents' conduct amounts to imposing terms that are unreasonable in light of normal trade practices.

260 First, in light of the standardization organization's intellectual property rules on the scope of the license212) and the patent law rules on practice213), the meaning of "providing a license" for a standard-essential patent " includes, at a minimum, the right to manufacture and the right to sell, rent, dispose of, repair, use, operate, and use the manufactured apparatus, and the right to use the method", WHEREAS, each of the acts of manufacturing, selling, renting, disposing of, repairing, using, operating, etc. constitutes patent infringement, so a complete license agreement must be one that freely grants the right to do all of the acts listed above, and not one that limits some of them to provide only a limited scope of rights.

261 Second, however, prior to 2008, Respondents licensed their telecommunications standards patents to competing modem chipset manufacturers, including ***, ***, ***, **, and ****, under terms that (i) excluded competing modem chipset manufacturers from using their modem chipsets (ii) limiting the distribution of the competing modem chipset manufacturer's modem chipsets to handset manufacturers that

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have entered into license agreements with the Respondents214), (iii) requiring the competing modem chipset manufacturer to report to the Respondents sales information, including the price of the competing modem chipset, the purchaser, the volume of sales per purchaser, the date of purchase, and the product model215), (iv) requiring the competing modem

212) Article 6.1 of ETSI's IPR Policy states that the scope of the license "includes, at a minimum, the right to manufacture and the right to sell, rent, dispose of, repair, use, operate and use the manufactured devices and methods."

213) Article 2 of the Patent Act (Definitions) 3. "Practice" means an act under each of the following headings.

a. In the case of the invention of an object: produce, use, transfer, rent, or import the object, or offer to transfer or rent the object (including display for transfer or rental, as follows)

B. If you invented the method: The act of using the method

C. In the case of the invention of a method of producing goods: In addition to the acts listed in the preceding paragraphs, the act of using, transferring, renting, or importing goods produced by the method or making a contract for the transfer or rental of the goods

214) As a result of these conditions (i) and (ii), (1) competing modem chipset manufacturers will be forced to sell modem chipsets only to mobile phone manufacturers that have entered into licenses with the Respondents, thereby restricting their business activities; (2) mobile phone manufacturers will be reluctant to purchase modem chipsets from competing modem chipset manufacturers that have not been provided with full license rights (in this regard, LG Electronics considers the presence or absence of licenses and patent-related guarantees as the main factors in selecting modem chipsets. Internal Document on Considerations from Intellectual Property Rights Perspective in Selecting Chipsets in Exhibit 66 ***), and ③ Even if a mobile phone manufacturer purchases a modem chipset from a competing modem chipset manufacturer other than the Defendants, it will be forced to enter into a license with the Defendants.

215) Through this condition (iii), the defendants have access to sensitive business information, including sales information and market strategies of competing modem chipset manufacturers.

The patents held by the chipset manufacturers have been infringed by Defendants and their modem chipsets without fair consideration.
It has only entered into restrictive license agreements, such as those that require it to provide to its purchasing customers or not to assert patent infringement claims,[216)]

262 Third, after 2008, the company (i) made it clear in its business reports, external presentations, and otherwise, that it would not license its patents, including standard-essential patents, to competing modem chipset manufacturers; and (ii) in accordance with this business policy, ****, ****, ***,
*** consistently refused to enter into license agreements with competing modem chipset manufacturers when asked to do so for standard-essential patents, (iii) offered to enter into subheadings rather than license agreements or to enter into supplemental exercise or limited covenants not to sue, and (iv) reported sales information about modem chipset sales217), ⓐ provided subheadings to the Respondents and to customers purchasing modem chipsets from the Respondents for patents held by the modem chipset manufacturers, to sell modem chipsets only to mobile phone manufacturers that have entered into license agreements with the Respondents; and ⓓ to hold competing modem chipset manufacturers jointly and severally liable with mobile phone manufacturers in the event of a patent dispute between a mobile phone manufacturer that purchased a modem chipset from a competing modem chipset manufacturer and the Respondents.

(4) (a) Intent and purpose of restraint of

competition (b) Unreasonably

interfering with the activities of other

undertakings

263 Respondents' position as vertical integrators in the modem chipset market and the telecommunications standards-essential patent licensing market, and Respondents' awareness of the need to comply with FRAND commitments,

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216) This condition (iv) allows Defendants to create a so-called "patent umbrella" that provides purchasers of their modem chipsets with licenses or sublicenses to patents held by competing modem chipset manufacturers.

217) In this regard, B states that "even after the reporting obligations were relaxed, B was still obligated to report (to Qualcomm) the total number of sales in each category. In particular, B was required to identify the number of B's chipset customers who were not licensed by Qualcomm to implement B's chipsets and the total number of chipsets sold to these non-licensed customers. By providing such information, B would be compelled to disclose trade secrets and information about B's customer base that its competitor, Qualcomm, should not and could not otherwise have access to." (Exhibit 80, B's Response to the Commission's Request for Production).

The modem chipset business, as evidenced by the Defendants' business reports and public presentations, is a major part of the

Based on the Defendants' perception of the importance of securing essential telecommunications patents and the manner in which the Defendants' patent licensing arrangements are structured, it appears that the Defendants have refused or restricted the ability of competing modem chipset manufacturers to license essential telecommunications patents with the intent to limit competition in the modem chipset market and to secure, maintain, and enhance their modem chipset dominance and to implement and establish a licensing policy model at the mobile phone level.

(1) Incentives to restrict competition as a dominant vertically integrated business

264 As discussed in paragraphs 2. c. above. As discussed in paragraphs 1) and (2), as the holders of the essential patents for each of the telecommunications standards, namely CDMA, WCDMA, and LTE, the Respondents are 100% market dominant in the relevant licensed markets and are also the dominant players in the relevant modem chipset markets as modem chipset operators that manufacture and sell modem chipsets using the essential patents. As such, Defendants are a dominant vertically integrated business and therefore have an incentive or motivation to exclude competitors from the modem chipset market and further consolidate their dominance in the modem chipset market by refusing to license their standard-essential patents to competing modem chipset manufacturers and to manufacture and sell their own modem chipsets using the unrestricted license, and to use their dominance in the modem chipset market to maximize profits by entering into license agreements with mobile phone manufacturers and collecting royalties.218)

265 Indeed, a 2013 email219) between ***, the general manager of Respondents' modem chipset sales in China, and ***, the vice chairman of Qualcomm Asia, states that "maintaining the competitiveness of Respondents' chipset segment (QCT) is important for maintaining a strong revenue stream for the licensing segment (QTL)."220

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218) As discussed above, the Respondents' refusal to license their patents to competing modem chipset manufacturers does not reduce the Respondents' royalty income because the patent law's "exhaustion doctrine" does not allow the Respondents to collect royalties for the same patent at both the modem chipset and mobile phone stages, and the Respondents have entered into licenses with mobile phone manufacturers and collect royalties from mobile phone manufacturers based on the total price of the mobile phone. Considering that the price of the modem chipset accounts for only 4% of the total price of a cell phone, the inducement or motivation of the accused in the above aspect is even greater, as recent cell phones are not just for voice calls but also serve as comprehensive IT devices.

219) The emails were also shared with key executives of the Respondents, including *** and ***, the Senior Vice Presidents.

220) "[redacted]" (Exhibit 51, 2013. *. *. Emails between employees of the Respondents)

This is because the defendants themselves were not aware of the competitiveness of their modem chipsets to cell phones.
The evidence shows that the defendants had an incentive to restrain competition and that the defendants themselves were aware of it, as they perceived patent license agreements as a lever to enforce against manufacturers221).

Declaring a FRAND commitment but violating it

266 The Respondents declared a FRAND commitment to license their patented technology to the standardizing body on fair, reasonable and non-discriminatory terms to all participants in the industry as a condition of having their patented technology selected as a standard. In addition, in past litigation with Ericsson, the defendants have alleged that Ericsson has obligated them to license Ericsson's standard-essential patents, and they themselves have stated that they will license their standard-essential patents to "all participants in the industry" on FRAND terms. Thus, it is arguable that the defendants were fully aware that rejecting and restricting requests from competing modem chipset manufacturers to enter into patent license agreements would be impermissible in violation of their FRAND commitments. Nevertheless, the Respondents' refusal to grant a limited scope license to competing modem chipset manufacturers' requests to enter into patent licensing agreements before 2008, and their refusal to grant licenses after 2008, can only be viewed as an attempt to limit competition in the modem chipset market and to secure their dominance in the modem chipset market and to secure favorable licensing terms at the mobile phone stage.

③ Recognizing the importance of having a complete license in the modem chipset market, we refuse to license or restrict standard essential patents to competitors.

267 In light of Defendants' business reports, public presentations by Defendants' representatives or officers, and other public statements, Defendants have taken steps to protect their products from patent attacks by patent holders, including

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221) [redacted]

"(Polydactyly 50, 2013. *. *. Emails between employees of the defendants)

and mobile communication standards are required to secure design freedom, which is a key element of the modem chipset business.
It is clear that Defendants were fully aware of the importance of securing licenses for their patents. Nevertheless, Respondents' refusal and restriction to license their essential telecommunications patents to competing modem chip manufacturers, while securing licenses from other patent holders and offering them to customers purchasing their modem chipsets, is consistent with an intent to restrict competition in the modem chip market.

268 (i) Eric Reifschneider, Respondents' patent licensing executive, stated in a February 2013 presentation that Respondents' use of cross-granting to secure licenses from other patent holders protects them from patent attacks and provides long-term design freedom, a key competitive factor in the modem chipset business.[222)]

269 (ii) While Respondents stated in their business reports from 2002 to 2007 that they license to competing modem chipset manufacturers because they "believe that this will promote the global diffusion of CDMA-based systems and technologies and will result in business growth, even if Qualcomm's market share declines."[223)] Since 2008, Respondents have removed this statement from their business reports and have rejected requests from competing modem chipset manufacturers to enter into patent license agreements.

④ Ambivalence about modem chipset tier licenses

270 Defendants have refused to enter into license agreements for essential telecommunications patents with competing modem chipset manufacturers, yet as of August 2015, 195 mobile phone manufacturers worldwide have licensed telecommunications patents necessary to manufacture and sell their

m o d e m chipsets.

222) S.B. 49, Qualcomm's Technology Licensing Program PT Materials

223) The full text of this material is as follows "We license our CDMA intellectual property to the competitors of our QCT segment to support the deployment of CDMA-based systems and technologies worldwide in order to grow our royalty revenues from customers licensed to sell CDMA phones and equipment. We believe that, if the use of CDMA expands sufficiently, QCT's business will also grow, even if we lose market share."

which is being cross-granted by the government. It also cross-granted In the course of negotiations with ***224) and ***225) to conclude patent license agreements, the Respondents offered full licenses for non-mobile communication terminals but refused to provide licenses for mobile phone modem chipsets. In addition, the Respondents refused licenses to competing modem chipset manufacturers, such as B, which required a license to sell modem chipsets externally, and C, which required a license to start a new modem chipset business, but offered restrictive terms, such as requiring them to cross-grant the licenses held by their competitors at no cost, which prevented the conclusion of the license agreements. It is difficult to find any rational explanation for the Respondents' duplicitous attitude other than that the Respondents intended to use their essential patents for telecommunications standards to create a structure that favors them and makes it difficult for competing modem chip set manufacturers to compete fairly and equally with them.

⑤ The stronger the competitor, the more unfavorable the terms of trade.

271 The Respondents proposed a form of contract with more restrictive terms for a potential competing modem chipset manufacturer. (i) Initially, Defendants recognized B as a strong competitor based on internal data226), and when B requested to enter into a license agreement in 2011, Defendants initially offered a "supplemental enforcement agreement" as a negotiation proposal, which was later changed to a "limited reservation of claims". The original supplemental enforcement agreement provided for a patent infringement claim against the cell phone manufacturer first, and then against B as a last resort. [redacted]

224) During patent license negotiations with *** in 2009, Respondents refused to grant *** an exhaustive license for modem chipsets for mobile phones and modem chipsets for computers equipped with communication functions. Thereafter, Respondents revised their

negotiations to include a

It has maintained its position that it does not intend to obtain exhaustion rights in any case for modem chipsets for cell phones (Exhibit 79, ***'s Response to the Commission's Request for Production).

225) In 2009, Respondents settled a patent dispute with *** and entered into a patent license and sublicense agreement (the "Settlement, License and Sublicense Agreement"). In this agreement, in exchange for [redacted] (Exhibit 81, Commission's Exhibits ***'s response to your request)

226) The internal documents created [redacted] and more. by the subjects included

(Sogap No. 56, internal data of Korea

Qualcomm)

In contrast, a limited stay of proceedings only lasts for a certain period of time, from ** days to ** days.

(ii) In addition, in 2009, the Respondents provided A with a non-suit covenant instead of a license related to the WCDMA standard, which was tied to a condition restricting the sales outlets to the Respondents' licensees, and imposed a penalty for violation of the sales outlets, which was not the case for other modem chipset manufacturers. In view of the fact that A was emerging as a top player in the GSM standard-related modem chipset market, surpassing ***, ***, ***, etc. at the time, the Respondents' imposition of stronger restrictions on A was intended to limit A's business in the WCDMA modem chipset market. (iii) On the other hand, when C, its only competitor in the CDMA modem chipset market, requested to enter into a patent license agreement related to the WCDMA standard in 2012, the Respondents required C to change its existing CDMA license agreement to a supplementary right exercise agreement, even though ***, ***, ***, and ***, which had a weak position in the modem chipset market, had already entered into agreements related to the WCDMA standard, which also demonstrates that the Respondents imposed more unfavorable contractual terms on their strong competitors.

(b) Anti-competitive effects

272 Based on the above findings of fact and the following, the Respondents' refusal and restriction to license essential telecommunications patents to modem chip set manufacturers (i) directly creates a business interruption risk for competing modem chip set manufacturers due to the cessation of modem chip set sales; (ii) raises the costs of competing modem chip set manufacturers; (iii) creates a barrier to entry for potential competitors; (iv) interferes with the business activities of competitors through refusal to license and unfair contractual terms and conditions; and (v) creates an advantageous position for competitors, (iv) interferes with competitors' business activities and gains

an advantage over competitors through refusal to license and unfair contractual terms; (v) as a result, many competitors are driven out of the market or cease to operate, while Defendants' overall modem chipset market share increases and their position in the market consolidates; and (vi) as a dominant player and vertically integrated business, Defendants abuse their dominance by violating FRAND commitments.

227) For the record, neither the supplemental exercise covenant nor the limited right of first refusal offered by Defendants are less favorable terms than a typical exhaustive license.

This makes it even more difficult to control the market for modem chipsets by carrier standard, and thus eliminates competition in the modem chipset market.
or conduct that is likely to have an anticompetitive effect.

Risk of competing modem chipset manufacturers discontinuing sales of modem chipsets

273 The most immediate problem that arises for competing modem chipset manufacturers as a result of not being able to license the Standard Essential Patents from the Accused is the risk that at any time, the Accused may attack them for infringement of their patents, which could result in the cessation of modem chipset sales. Moreover, the risk of such patent infringement is substantial because (i) the standard essential patents held by the Respondents must be infringed in order to manufacture and sell modem chipsets; (ii) the business risk of patent infringement is that the entire modem chipset business could be halted at once, and the risk is substantially greater than the risk of ordinary business uncertainty, (iii) competing modem chipset manufacturers cannot even predict whether they will be subjected to patent attacks because they have not been provided with written agreements from the Respondents to license the patents; and (iv) on the other hand, mobile phone manufacturers are reluctant to purchase modem chipsets that do not resolve the licensing issue228), which is more serious and unreasonably hinders the business activities of competing modem chipset manufacturers.

Rising costs of competing modem chipset manufacturers

Increased costs due to the risk of patent attacks by defendants

ⓐ Mechanism of cost escalation effect due to risk of patent attack by defendants

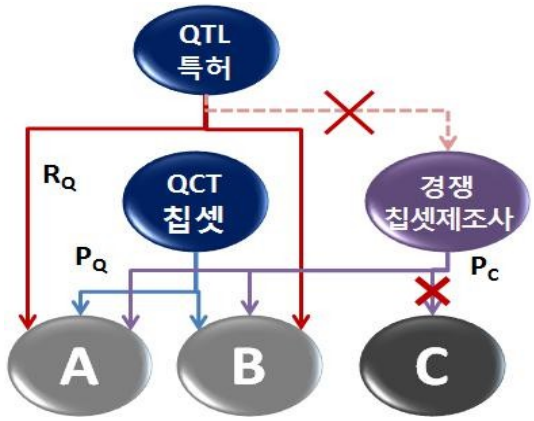
274 Defendants License Standard Essential Patents to Competing Modem Chipset Manufacturers

228) Although B's modem chipset division has attempted to contact several mobile phone manufacturers to pursue a modem chipset outsourcing business, the mobile phone manufacturers have been unwilling to purchase B's modem chipsets because they believe that B cannot be free from the Respondents' patent attacks without securing a license from the Respondents, and in the case of ***, the Respondents filed a lawsuit seeking a declaratory judgment to stop ***'s modem chipset sales, and the mere fact of the filing of the lawsuit caused his customers to abandon the business (Exhibit 90 at ***).

rejected, competing modem chipset manufacturers would face increased costs due to the risk of patent attack.
is raised.

275 As shown in Table 47 below, if the mobile phone manufacturer C does not license the standard essential patents embodied in the modem chipset from the accused, the competing modem chipset manufacturer will immediately be at risk of patent infringement suits from the accused if it sells its modem chipset to C (since neither the competing modem chipset manufacturer nor the mobile phone manufacturer C has a license to the accused's patents), and the competing modem chipset manufacturer will therefore abandon the sale of the modem chipset to C. If the competing modem chipset manufacturer wishes to sell the modem chipset to C, a non-licensee of Defendants' patents, C will require the competing modem chipset manufacturer to pay the costs incurred by the competing modem chipset manufacturer in connection with any patent infringement lawsuit brought by Defendants against C for purchasing a modem chipset that is not licensed from Defendants (the "Indemnification Costs"), The indemnification fee will be based on the customer's damages for patent infringement, and given that the Defendants are charging a fee based on the total price of the cell phone, the indemnification fee that competing modem chipset manufacturers will have to bear is likely to be more than their modem chipset sales margins can cover.

<Table 47 Increased costs for competing modem chipset manufacturers due to the risk of patent infringement by the accused



* A and B: cell phone manufacturers that are licensees of the Respondents
* C: The subject's non-licensor cell phone manufacturer
* R_Q: License agreements with deponents
* P_Q: Purchase of modem chipsets from subjects
* P_C: Purchase a modem chipset from a competing modem chipset manufacturer

276 In addition, a competing modem chipset manufacturer entered into a patent license agreement with the accused.

Even if you sell modem chipsets to phone manufacturers (A and B), you must pay the cost of competing modem chipset manufacturers.
The synergistic effect is not completely eliminated. First of all, from the perspective of a competing modem chipset manufacturer, if a patent dispute arises between A and B and the Respondents regarding the payment of royalties in the future, there is a risk that a competing modem chipset manufacturer that sold modem chipsets without securing a license from the Respondents may also be involved in the patent dispute between them.

277 On the other hand, from the point of view of mobile phone manufacturers A and B, if a competing modem chipset manufacturer that has not received a patent license from the defendants is subjected to patent attacks by the defendants, the mobile phone business of A and B that uses the modem chipset of the competing modem chipset manufacturer may be disrupted, so A and B may request the competing modem chipset manufacturer to share the costs or reduce the price of the modem chipset, which will increase the cost of the competing modem chipset manufacturer. In other words, if the Defendants provided a full and exhaustive license to the competing modem chipset manufacturers without refusing or limiting the license as it is now, not only mobile phone manufacturers A and B (who are licensors of the Defendants) but also C (who is not a licensee of the Defendants) who purchase modem chipsets from the competing modem chipset manufacturers would not be at risk of patent disputes with the Defendants and would not have to pay additional indemnification fees or cost sharing fees to the competing modem chipset manufacturers in addition to the modem chipset price.

ⓑ Examples of rising costs due to the risk of patent attacks by defendants

278 The following are examples of the importance of securing

licenses for actual modem chipset manufacturers, and the cost of not securing licenses increases due to patent disputes and risks.

279 (i) G: ****, written by G. *. *. The document 'Considerations and Results Sharing from IP Perspective in Baseband Chipset Selection' confirms that mobile phone manufacturers are considering whether the modem chipset is licensed and whether patent-related guarantees are available as important factors when selecting the actual modem chipset from the perspective of intellectual property rights.

<Table 48> Considerations from an IPR perspective when selecting a chipset229)

280 (ii) H: H, a competing modem chipset manufacturer, stated in its submissions to the Commission[230]) that after the Respondents filed a lawsuit on **** seeking a declaratory judgment that would have halted the sale of H's modem chipsets[231]), H's major customers became concerned about whether H could legally continue to supply them with modem chipsets, and consequently stopped doing business with H.

281 (iii) I: I, a competing modem chipset manufacturer, negotiated with a cell phone manufacturer to indemnify Respondents and other third parties for patent infringement in the course of supplying modem chipsets to the cell phone manufacturer in ****.[232])

229) IPR Considerations in Selecting a Chipset for the G Internal Documentation
230) Form 90, H Submission
231) H states that Respondents filed an action against H in the Court of Chancery of the State of Delaware on *** 2003 seeking a declaratory judgment that Respondents could terminate H's rights under the Patent Portfolio Agreement while maintaining their own

r i g h t s under the Patent Portfolio Agreement. On *March *, 2005, the Delaware Supreme Court unanimously affirmed the Court of Chancery's d e c i s i o n that Respondents had no grounds to terminate the Patent Portfolio Agreement with H. (Exhibit 9 0 , H's brief).

282 (iv) J: J, a competing modem chipset manufacturer, has entered into a modem chipset supply contract with a cell phone manufacturer and has agreed to pay a substantial portion of the patent dispute costs incurred by the cell phone manufacturer based on the purchase of its modem chipsets (capped at *% of the modem chipset purchase price).233)

283 (v) K: K, a competing modem chipset manufacturer, sought to sell its modem chipsets to a mobile phone manufacturer without obtaining a license to the essential mobile communication standard patents from the Respondents. However, the cell phone manufacturer suggested that K could only purchase K's modem chipsets if K paid for the costs of the patent dispute, and in 2009. *. *. K entered into a patent dispute indemnification agreement with the cell phone manufacturer that exempted K from all costs incurred due to patent infringement.234)

Raising competitors' costs due to the "patent umbrella" constructed by the accused

ⓐ Mechanisms for raising competitors' costs under the "patent umbrella" constructed by defendants

284 As discussed in Section 2. a. 3) above, the Defendants did not only refuse to license patents to competing modem chipset manufacturers, but also linked the supply of modem chipsets to mobile phone manufacturers with the entering into of patent license agreements and the enforcement of unfair patent license agreement terms. In doing so, the Defendants secured cross-grant agreements from mobile phone manufacturers for free and without consideration,235 and created a "patent umbrella" that shielded them from patent infringement claims by mobile phone manufacturers, as well as their customers who

purchased their modem chipsets.

232) [redacted]
[Submitted by I to the Committee on September 26, 2016].

233) Exhibit 58 provides J's immunity from patent disputes

234) Exhibit 57 provides K's immunity from patent disputes

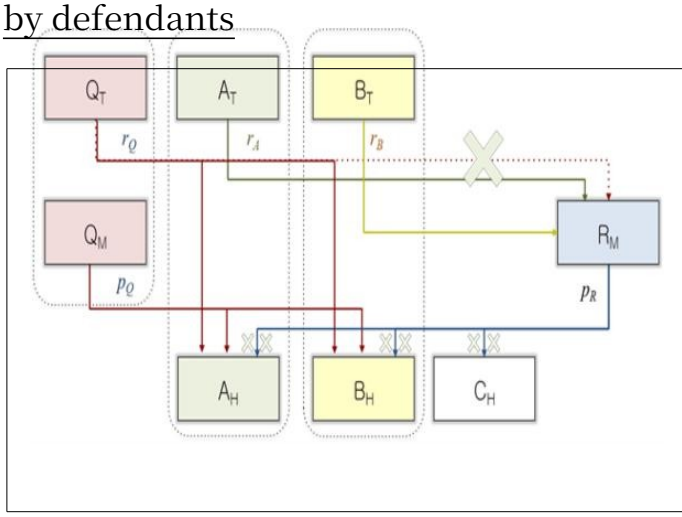
235) Here, the cross-grant secured by the Defendants was obtained against the will of the mobile phone manufacturers by creating a structure in which they refused to license to the modem chipset manufacturers at the modem chipset stage and collected royalties from the mobile phone manufacturers at the mobile phone stage, and required mobile phone manufacturers who wished to purchase their modem chipsets under this structure to first enter into a license agreement. This is discussed in more detail in Conduct 3.

285 The effect of these patent umbrellas is illustrated in more detail below. The table below

As shown in Ex. 49, Defendants refuse or limit licenses to modem chipset manufacturers while simultaneously requiring and obtaining sublicensing commitments from mobile phone manufacturers A_H and B_H that have licenses with Defendants to other mobile phone manufacturers that purchase Defendants' modem chipsets (A_H to B_H and B_H to A_H , respectively), effectively preventing mobile phone manufacturers from exercising their rights to the standard essential patents held by Defendants.236)

286 As Defendants thus secure sub-licensing agreements from handset manufacturers for their own modem chipset customers, they create a "patent umbrella"237) consisting of patents from Defendants and handset manufacturers that have license agreements with Defendants. Defendants use this to further increase the costs of competing modem chipset manufacturers (R_M), amplifying the exclusionary effect.

<Table 49 Competitors' cost increases due to patent umbrellas constructed by defendants



* $_{QT}$: Subjects (license business)
* Q_M : Examiners (modem chipset business)
* $_{RM}$: Competing modem chipset manufacturer
* $_{AT}$, $_{BT}$: Patentee
* A_H , B_H : Hewlett-Packard, the phone manufacturer that licenses the Defendants.
* $_{CH}$: Mobile phone manufacturers that do not have licenses with the defendants

287 In Table 49 above, if A_H and B_H purchase a modem chipset from the defendants (Q_M), they only need to pay P_Q +r_Q for the modem chipset and

the implementation fee. However, if the cell phone manufacturer

236) The Respondents argue that the mobile phone companies' inability to receive royalties for sublicensing to the Respondents' customers is reflected in the cross-license and royalty terms of the cross-licenses, which compare the technology provided by the Respondents to the technology provided by the mobile phone manufacturers. However, the Defendants do not make any assessment of the total amount or value of patents held by individual mobile phone companies, and in particular, they do not properly pay counter-payments for providing sub-title commitments to their customers even if a particular mobile phone company holds a large number of standard essential patents. Both Japan's Fair Trial and Appeal Commission in 2009 and China's National Development and Reform Commission in February 2015 found these practices to be illegal and issued injunctions against them.

237) The "patent umbrella" established by the defendants not only strengthens the effect of excluding competitors from the modem chipset market, but also reduces the incentive for technological innovation by preventing mobile phone companies that possess the technology from receiving fair compensation. However, the effect of restricting the exercise of standard-essential patent rights to mobile phone companies by restricting competition in the technology license market and reducing the incentive for technological innovation will be discussed later.

If A_H and B_H purchase a chipset from a competing modem chipset manufacturer R_M , then A_H will use '$P_R + r_Q + r_B$ '
$_{HRQA}$Thus, if A_H and B_H purchase chipsets from R_M , they will have to pay an additional r_B and r_A , respectively, compared to if they purchase from the accused (Q_M). Thus, even if R_M , a competing modem chipset manufacturer, can supply modem chipsets at the same price as the Accused ($P_R = P_Q$), it will incur additional costs to R_M equal to r_B and r_A .

288 It can also be seen as having a price discriminatory effect between the Accused (Q_M) and competing modem chipset manufacturers. As a result of Q_T securing cross-licenses or sub-licensing agreements (which extend to their customers) from the other patent holders (A_T , B_T), Q_M does not pay any implementation fees to the other patent holders (A_T , B_T). On the other hand, R_M , a competing modem chipset manufacturer, will pay a total of $r_Q + r_A + r_B$ in royalties not only to Q_T but also to A_T and B_T .238) Thus, R_M will have to pay an additional $r_A + r_B$ in royalties compared to the Accused, excluding the royalties attributable to the difference in technology between the two companies (r_Q).

289 This can be interpreted to mean that the defendants forced the handset manufacturers to discriminate between themselves (or their affiliates) and their competitors through the "sub-agreement pool," i.e., the handset manufacturers did not charge the defendants a sub-agreement fee while charging normal sub-agreement fees to their competitors, thereby discriminating against sub-agreement fees and raising the costs of their competitors.

ⓑ Examples of competitor cost increases due to "patent umbrellas" constructed by accused companies

290 The following examples demonstrate that the Respondents' "patent

umbrella" gives them an advantage over competitors in the modem chipset market, and that this "patent umbrella" in turn increases costs for competitors.

291 (i) Defendants have told cell phone manufacturers on more than 240 occasions since 2004, and as recently as the present, that the modem

238) Of course, the Accused are no longer collecting implementation fees from competing chipset companies for their patents after 2009. Therefore, after 2009, competitors do not pay the implementation fees (rQ) owed to the Respondents.

The chipset promotional materials were published and distributed in the form of a white paper,239) in which the defendants claimed that the

After categorizing the modem chipset procurement paths available to mobile phone manufacturers as Qualcomm Chipset Customer, competitor (Thirty Party Chipset Customer), and self-procurement (Vertically integrated Licensee), the complaint emphasizes that only by purchasing Qualcomm's modem chipsets are customers protected from both Qualcomm's patents and third party patents. 240 This shows that customers who purchase competing modem chipsets are not protected from third-party patents and must negotiate and pay patent fees directly to third parties, which is relatively more expensive than using the Accused's modem chipsets.

<Table 50 Business model descriptions written by subjects

Options for QUALCOMM Licensees	QUALCOMM Patents	3rd Party Patents (no additional royalty or fee to licensee)
QUALCOMM Chipset Customer	✓	✓
Third Party Chipset Customer	✓	May have to be negotiated directly with 3rd party
Vertically Integrated Licensee	✓	

292 (ii) ***, a competing modem chipset manufacturer, stated in its answer to the Commission that "***'s principal original equipment manufacturer (ODM)241) told *** that if it used ***'s chipsets, it would not be able to compete with ***,

He said that only when you buy Qualcomm's chipsets do you lose the intellectual property protection that you get from Qualcomm, including the intellectual property of ***, ***, and other companies, and even if you buy the modem chipsets cheaply, the value of these intellectual

property rights (reportedly *$100 per phone) is large enough to make a difference in your decision to buy a chipset,

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239) Exhibit 54, a business model description written by the subjects.

240) After 2009, when Defendants ceased to offer even limited sublicensing arrangements to modem chipset manufacturers, cell phone manufacturers would not be protected from Defendants' patents if they purchased competing modem chipsets. As early as 2006, when the above material was written, Defendants were entering into limited license agreements with modem chipset manufacturers and collecting royalties.

241) Original Development Manufacturing.

Obviously, this means that if an ODM decides they don't want to buy the *** chipset
. was a factor."[242] This clearly demonstrates that the patent umbrella constructed by Defendants is an important factor in determining the parent chipset.

293 (iii) Furthermore, the patent license agreements entered into by the Respondents with the handset manufacturers have a term of more than ** years or ***, and furthermore, the agreements with ***, ***, ***, ***, ***, etc. stipulate that the cross-grant provided by the Respondents will be retained even if the patent license agreement is terminated[243]). Thus, the cross-grant secured by the Respondents from the handset manufacturers has the effect of excluding competitors by raising their costs through the patent umbrella effect in the long term and in perpetuity.

294 In addition, as the following examples illustrate, Defendants recognized and actively promoted the patent umbrella they had established as a key factor in obtaining, maintaining, and enhancing the competitiveness of their modem chipsets by protecting their modem chipset customers from attacks by other patent holders, reducing the cost to their customers of negotiating with other patent holders, and saving their customers the implementation fees they would otherwise have to pay if they purchased competing modem chipsets.

295 (i) When Defendants sought to spin off the modem chipset business in 2000, they cited in their 10-K the business risks that would result from the separation of QCT, stating that a complete separation of QCT from Defendants would result in the inability to pass-through patent rights held by Defendants' licensees to customers purchasing Defendants' modem chipsets, and that Defendants' modem chipsets would no longer be safe from patent attacks by other patent holders, resulting in a loss of patent protection.

242) Exhibit 80, ***'s Response to the Commission's Request for Production of Documents

243) The Patent License Agreement between Defendants and *** is set forth below. [redacted]

Defendants were concerned that the weakening would limit their ability to sell modem chipsets.[244)]

296 (ii) Steve Altman, a representative of Respondents, emphasized the importance of the patent umbrella in the modem chipset business at an investor meeting in the United Kingdom in November 2005, stating that "when you buy a modem chipset from Respondents, you have access not only to Respondents' IP, but to the IP of over 100 companies, which prevents stacking of royalties. However, if one were to purchase a competitor's modem chipset, one would have to negotiate individually with each of the other companies and potentially risk paying separate royalties."245 This emphasizes the superior patent protection afforded to the accused modem chipsets by the patent umbrella.)

297 (iii) In a white paper titled "Third Party IP Rights Benefiting" sent to mobile phone manufacturers, the Respondents explained that as a result of the Respondents' patent licenses from a total of *** companies246) (as of November 2008), purchasers of the Respondents' modem chipsets could save substantial royalty savings over the royalties payable for purchasing modem chipsets from competing modem chipset manufacturers,[247)] This represented that the patent umbrella gave the Accused's modem chipsets a comparative advantage over competitors in terms of actual price.

Limit the entry of potential competitors

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244) The text of the business report reads as follows: (Exhibit 83, Respondents' 2000 Business Report) "EXHIBIT 99.1

set forth below are certain risk factors included in the registration statement on form s-1 (no. 333- 42138) filed by qualcom spinco, inc. with the securities and exchange commission on july 25, 2000, and prepared as of such date. risks related to our separation from qualcom. (middle) AFTER OUR SEPARATION FROM QUALCOMM, WE WILL NO LONGER BE ABLE TO PASS THROUGH OR SUBLICENSE SELECTED INTELLECTUAL PROPERTY RIGHTS OF QUALCOMM'S LICENSES, WHICH MAY HARM OUR ABILITY TO MARKET OUR PRODUCTS AND SUBJECT US TO

LIABILITY. (middle) Following our separation from QUALCOMM, we will no longer be able to pass on those benefits to our customers, which may harm our ability to market our products and may subject us to claims for indemnification by our customers if they are sued by the holders of the intellectual property licensed to QUALCOMM."

245) Exhibit 82, Exhibitors' 2005 Investor Meeting Materials

246) A similar communication sent by Defendants to cell phone companies as of April 2013 shows

a significant increase in the number of companies eligible for the third-party patent use benefit to a total of *** (Exhibit 39, communication promoting the third-party patent use benefit).

247) Exhibit 39, Promoting the Benefits of Using Third-Party Patent Rights

Mechanisms for building barriers to entry

298 As Defendants publicly announce, and in fact practice, that they will not license essential telecommunications patents to competing modem chipset manufacturers, potential competing modem chipset manufacturers will consider the additional cost burden of not being able to enter into patent license agreements with Defendants when deciding whether to enter the modem chipset market.

299 However, the increase in cost due to ② above lowers the expected profitability of modem chipset manufacturers, which acts as an entry barrier for potentially competing modem chipset manufacturers to enter the modem chipset market. In addition, in order to enter the modem chipset market, large-scale investment is required for product development and commercialization, and if a modem chipset is sold without securing a license to the essential patents of the mobile communication standard from the accused, there is business uncertainty due to the risk of patent attack by the accused, which also acts as an entry barrier.

Real-world examples of barriers to entry

300 (i) B has requested the Respondents to enter into a patent license agreement for the external sale of modem chipsets since January 2011,[248] but the Respondents have refused to enter into a patent license agreement, as shown in the admitted facts in paragraph 2. a. 4) b) (3) (a) above, and B is still only manufacturing modem chipsets for its own mobile phones and has not entered the market for the sale of modem chipsets for external sale.

301 (ii) In addition, B is also a member of the group, as recognized by the facts in 2. a. 4) and (3) above,

In or about *March 2011, *** attempted to form a joint venture with *** to develop modems (the *** Project), but *** was unable to obtain a license for modem chipsets from the Respondents, which ultimately prevented *** from entering the modem chipset market through the joint venture.249)

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248) Defendants acknowledge that in their licen[redacted]" Provide a license to manufacture a modem chipset

Therefore, the contract alone does not allow B to sell the modem chipset to a third party even if it manufactures it.

was.

302 (iii) C requested the Respondents to enter into a patent license agreement in ****, but, as noted in the findings of fact in paragraphs 2. a. 4) and (3) (b) above, the Respondents refused, maintaining that they could not license modem chipsets for mobile phones, but instead proposed terms that were unacceptable to C, such as requiring an exhaustive license to C's patents250) and C did not enter the modem chipset market until 2011.251)

303 (iv) D had a significant share of the CDMA standard modem chipset market, but in 2012, it requested the Respondents to enter into a license agreement for the WCDMA standard essential patents, but as discussed in the admissions in paragraphs 2. a. 4) and (3) (c) above, the Respondents refused, and it has not entered the WCDMA modem chipset market since then.

Interfering with competitors' business activities and gaining advantageous positions through license refusal and unfair contract terms

304 Respondents refused to license to competing modem chipset manufacturers and entered into or offered restrictive agreements. Specifically, Defendants restricted the competing modem chipset manufacturers' modem chipset counterparties (distributors) to cell phone manufacturers with whom Defendants had patent license agreements, cross-granted the competing modem chipset manufacturers' patents, and required the competing modem chipset manufacturers to report to Defendants on a quarterly basis business information, including new business trends, modem chipset distributors (customers), and sales volumes per customer.252)

249) At that time, B believed that *** would not be able to effectively sell its modem chipsets in the market due to the threats made by the Respondents as a result of ***'s failure to secure a license from the Respondents. The text is as follows

"[redacted]

". [B's Oct. 7, 2016, submission to the Commission] *** .

Project related materials (Attachment 2)]

250) In addition, Defendants have provided C with a sales outlet restriction of sales to Defendants' licensees only, quarterly sales information contractual terms, such as reporting.

251) C subsequently abandoned seeking licenses directly from Defendants and instead acquired ***, which had previously held licenses from Defendants, only in 2011 (*** entered into a patent license agreement with Defendants at the modem chipset stage in ****).

252) Defendants continued this reporting obligation even after 2008, when they waived the implementation fee for competing modem chipset manufacturers.

305 Among other things, the Respondents' refusal to license to competing modem chipset manufacturers and the restrictions on distributors prevent competing modem chipset manufacturers from selling modem chipsets to mobile phone manufacturers that do not have a patent license agreement with the Respondents, or require mobile phone manufacturers to enter into a patent license agreement with the Respondents before selling modem chipsets. This gives the Defendants the opportunity to gain an advantage over their competitors in the modem chipset business by having first contact with cell phone manufacturers who are potential customers of competing modem chipset manufacturers, which they can then leverage for their own modem chipset business.

306 In the case of mobile phone manufacturers that have already entered into patent license agreements with Defendants, Defendants may also use their patent license agreements with mobile phone manufacturers to interfere with the business activities of competing m[redacted]t manufacturers by intervening in transactions between mobile phone manufacturers and competing modem chipset manufacturers, as evidenced by Defendants' *March *, 2012 letter dated [redacted]'. The report states that, after conducting a fee audit of ***, the Respondents alleged that *** owed the Respondents a fee of *** and demanded that *** not design any smartphones using D's modem chipsets and that ***'s "***+** Project" must be terminated in order to maintain the Respondents' market share of WCDMA smartphones in China and the rest of the world.253)

307 This anticompetitive effect may be equally manifested in the way that Defendants' use of sales information about competing modem chipset manufacturers gives them an unfair competitive advantage. Defendants provided restrictive licenses or agreements to competing

modem chip set manufacturers that required them to periodically report sales information, such as the selling prices, distributors, sales volumes, and product models of competing modem chip set manufacturers' modem chip sets, which enabled Defendants to become aware of the key sales strategies of competing modem chip set manufacturers and to gain an advantageous position in the marketplace.

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for the entire period. On the other hand, the defendants' patent license agreements with mobile phone manufacturers also provided for reporting obligations, but these were limited to information necessary to determine the appropriateness of the royalties (such as the total number and value of sales during the period and the amount of the royalty deduction).

253) Microtelangiec[redacted]

and will be utilized to occupy it. In this regard, we are pleased to announce that the Defendants have filed a motion to dismiss the

Doc. [redacted][254], the defendants were told by E that the

Through the publicly available sales information, it is confirmed that E and F knew in detail the model, price, and quantity of modem chipsets that they intended to supply to G, discussed countermeasures, and used this information to negotiate with G to reduce the impact of competing modem chipset manufacturers even before G's products were launched.[255]

308 On the other hand, when the defendants relaxed the distributor restrictions and sales information reporting obligations for **** in 2015, **'s market share increased.[256]

The exit of a major competing modem chipset manufacturer and the increase in market share of the accused

309 Prior to 2008, ***, ***, who entered into restrictive covenants with the Respondents,

Competing modem chipset manufacturers, such as ***, have been driven out of the modem chipset market,257 and *** has stated that "with large entry and operating costs, Defendants' business model raises competitors' costs, making it difficult for them to achieve profitability, and Defendants' interference with competitors delays or reduces competitors' customer acquisition, squeezing competitors' margins."[258]259) These major competing modem chipset manufacturers have been driven out of the market or have experienced business difficulties.

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254) Microtelangiec[redacted]

255) In its brief to the Commission, E stated that "one of E's customers, which used to

use 100% of E's products, was threatened by Qualcomm with stricter audits of its cell phone sales if it did not begin using Qualcomm products. (emphasis added) E is being forced to disclose trade secrets and customer base that its competitor, Qualcomm, should not have access to and would not otherwise have access to." (Ex. 80, E's Response to the Commission's Request for Production).

256) [redacted]

257) [redacted] in the modem chipset market, respectively Released.

258) Statement of *** at the second hearing of the KFTC's plenary session on Aug. 17, 2016

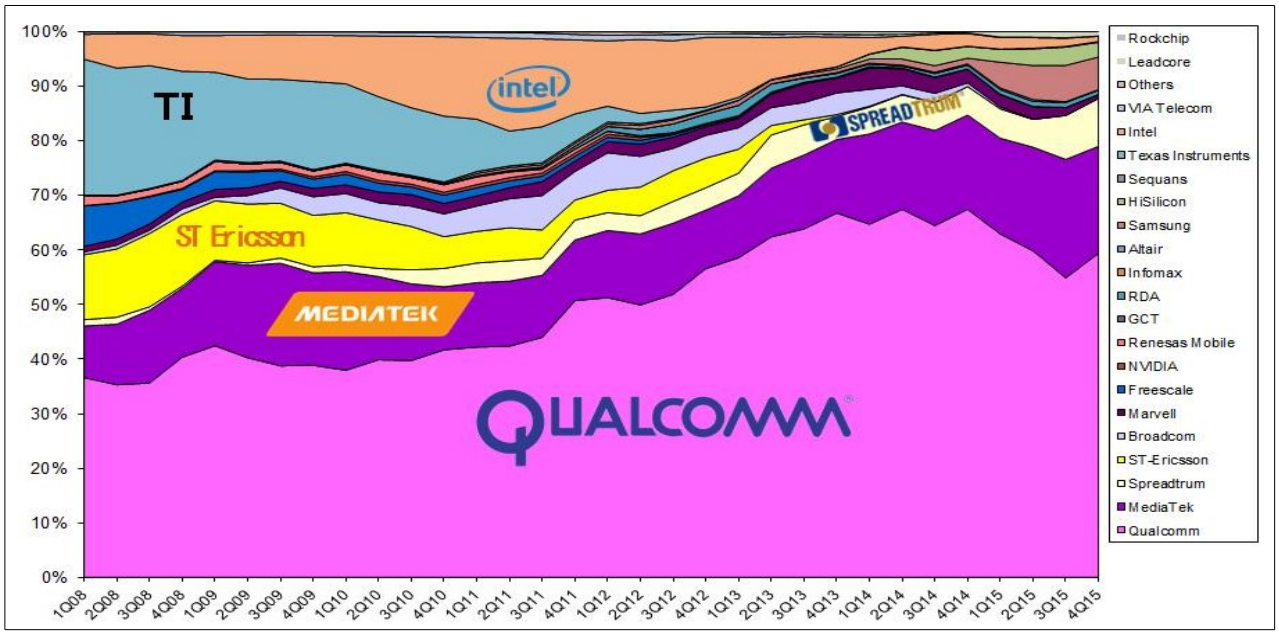
259) In this regard, *** (Professor, Faculty of Economics, *** University), who participated in the deliberations as an expert for the stakeholder side, expressed the opinion that "given the characteristics of the modem chipset business, where the pace of technological innovation is very fast and the competition for R&D investment to launch new products is fierce, there is a causal relationship between Qualcomm's competition-restrictive standard-essential patent license business model and the exit of major modem chipset companies, such that Qualcomm's investment returns are unreasonably high and its competitors' investment returns are unreasonably low."(August 17, 2016, 2nd deliberation of the KFTC plenary session)

During this time, the subject's share of the total global modem chipset market grew from 36.8% in 2008 to

The Herfindahl-Hirschman Index (HHI), a measure of market concentration, rose to 59.4%, while the

From 2,224 in 2008 to 4,670 in 2014.[260)]

<Table 51 Change in worldwide modem chipset market share since 2008 by modem chipset manufacturer



* Source: Strategy Analytics 'Baseband Market Share Tracker'

⑥ Spillover of restrictive effects to neighboring markets

310 Given the recent proliferation of System on Chip (SoCs) in the mobile phone component market, in which the modem chipset and application processor are integrated and implemented on a single chipset, it is likely that the anticompetitive effects of Defendants' conduct in the modem chipset market will spill over into the adjacent application processor market. Indeed, in the global application processor market, the Defendants

260) In this regard, *** (Professor of Economics, *** University), who participated in the committee deliberations as an expert for the stakeholder side, analyzed the long-term

trend of the defendants' modem chipset market share and stated that (i) the overall modem chipset market share of the defendants increased by 2.6 percentage points per year, and (ii) the HHI index increased by 164 per year (2nd deliberation of the plenary session of the Fair Trade Commission on August 17, 206). On the other hand, the Merger Review Criteria stipulates that a merger is likely to substantially restrict competition if the HHI index is greater than 2,500 and the increase in the HHI index is less than 150.

261) While Wi-Fi, Bluetooth (BT), and even GPS functions are sometimes referred to as one-chip, a chip with integrated AP and CP functions is generally referred to as a one-chip, and the concept of one-chip as used in this report refers to a chipset that integrates an AP and modem chipset into one.

's share has grown from 24% in 2008 to over 50% since 2013.

⑦ Competition-restricting effects of breach of FRAND commitments

311 As discussed above, the Respondents' refusal to enter into license agreements with competing modem chipset manufacturers for essential telecommunications patents violates the FRAND commitments. Moreover, as dominant vertically integrated players in both the telecommunications standard-essential patent licensing market and the modem chipset market, Respondents' violation of the FRAND covenant will result in the loss of the benefits of standard-setting and the harms of monopolization because there will be no substitute for a particular standard and no way to control abuses of monopoly power through the competitive structure of the downstream product markets in which the standard is applied.

Stifling competition for innovation in the modem chipset market

312 As Defendants acknowledge262), securing patent licenses is a critical component of providing long-term design freedom in the modem chipset business. Nevertheless, by refusing to enter into license agreements for essential patents for telecommunications standards, or by restricting the ability of competing modem chip set manufacturers to innovate, Defendants have impaired the incentive to do so. The specific anticompetitive effects on innovation are described in the anticompetitive effects of conduct 3.

Reduced consumer choice and diversity

313 As described above, the virtual blockade of competing modem chipset manufacturers from entering the market has resulted in

significantly fewer choices for handset manufacturers and thus fewer choices for handset consumers, and a reduced variety of modem chipset products.

⑩ Acting as a means to force mobile phone manufacturers to enter into unfair license agreements

262) S.B. 49, Qualcomm's Technology Licensing Program PT Materials

314 The Defendants' refusal to license and restrictions on licenses to competing modem chipset manufacturers were used to exclude competitors from the modem chipset market and also as a means for the Defendants to improperly force mobile phone manufacturers to enter into patent license agreements, as described in Acts 2 and 3 below. The detailed facts and anticompetitive effects of these acts are discussed in the following sections on Acts 2 and 3.

<Judgment on the claims of the relevant defendants ⑦>

315 On the other hand, Respondents argue that requiring licensing at the modem chipset stage would force Respondents to charge implementation fees at two levels: the modem chipset manufacturer and the handset manufacturer, resulting in patent disputes and inefficiencies, such as who bears the implementation fees between the modem chipset manufacturer and the handset manufacturer, and between Respondents and the modem chipset manufacturer or the handset manufacturer.

316 However, Respondents' arguments are unavailing for the following reasons.

317 Even if licensed at both the modem chipset and mobile phone levels, the Respondents are only required to license the patents that are implemented in the modem chipset to the modem chipset manufacturer and collect an arm's-length royalty from the modem chipset manufacturer on FRAND terms. Furthermore, for the standard-essential patents held by the defendants that are not implemented in the modem chipset or are implemented beyond the modem chipset, the defendants may enter into separate patent license agreements with

the mobile phone manufacturers through arm's length negotiations on FRAND terms. As a result, the mobile phone manufacturer can claim patent exhaustion for the accused patents already implemented in the modem chipset and object to the payment of a separate license fee, which can be contested in the patent court if necessary.

318 Thus, even if licensing at both the modem chipset and handset levels, Defendants would only have to enter into patent license agreements with the modem chipset manufacturer and the handset manufacturer, and the

Negotiations between parties regarding the calculation of fees that may arise during the process can be avoided by using the
Rather than inefficiency or ineffectiveness, the pro-competitive effect of a standard-essential patent is the result of a legitimate evaluation of its validity or value.[263)]

(5) Sintering

319 Therefore, the Respondents' refusal to license or restriction of essential mobile communication standard patents to competing modem chipset manufacturers in 2. a. 4) above is illegal because it constitutes an act of impairing the business activities of other business operators by other unfair methods in violation of Article 3 (2) (1) (3) of the Act and Article 5 (3) (4) of the Enforcement Decree by imposing conditions on the counterparty that are not reasonable in light of normal trade practices.

c) Whether it has refused or restricted, without justification, the use of or access to elements essential to the production or sale of goods by other operators[264)]

(1) Related jurisprudence

320 Article 3 (2) (Prohibition of Abuse of Dominant Position), Paragraph 1, Item 3, "Unjustified Interference with the Business Activities of Other Businesses" of the Act, Article 5 (Types or Criteria of Abusive Acts), Paragraph 3, Item 3 of the Enforcement Decree, requires the act of "refusing or restricting the use of, or access to, elements essential to the production, sale, or supply of goods or services of another business without justifiable reason" to be established.
The market-dominant entity must have made it difficult for another entity to conduct its business by refusing, suspending, or restricting the use of, or

access to, elements essential to the production, supply, or sale of the other entity's goods or services (essential elements).

321 In determining the case of difficulty in the business activities of other companies, the production, financial, and sales activities of other companies shall be considered comprehensively, but the possibility of difficulty in business activities shall not be considered.

263) In addition, the Actual Defendants had, prior to 2008, been collecting royalties from competing modem chipset manufacturers on a limited basis through licensing agreements, and in some cases at the cell phone level.

264) Article 3 (2) (1) (3) of the Act, Article 5 (3) (3) of the Enforcement Decree

if any.

322 Essential elements include tangible and intangible elements, such as networks and infrastructure, that (1) the production, supply, or sale of goods or services is virtually impossible without the use of the element, resulting in the impossibility of participating in a specified field of trade or the persistence of a significant competitive disadvantage in the field of trade that cannot be avoided (essentiality); (2) the element must be exclusively owned or controlled by a particular undertaking (control); and (3) it must be practically, legally, or economically impossible for those seeking to use or access the element to reproduce it or substitute it with another element (non-substitutability).[265]

323 In addition, refusing, discontinuing, or limiting the use of or access to an essential element means: (1) charging an unreasonable price or terms that make access to the essential element practically or economically impossible; or (2) charging a price or terms that are significantly discriminatory against existing users of the essential element, resulting in substantially the same effect as refusing to use the essential element.

(2) Whether you are a dominant player

324 As discussed above in Section 2. c. 2) As discussed in b. (2), the Defendants are found to be dominant in the overall market for licenses of their patented technology for each of the telecommunications standards, CDMA, WCDMA, and LTE, and in the market for modem chipsets for each of the standards, CDMA, WCDMA, and LTE.

(3) Whether standard essential patents are mandatory for modem chipset manufacturers

265) Regarding the conceptual requirements of essential elements, the Seoul High Court has held that essential elements generally refer to facilities (elements) without which a competitor cannot provide services to its customers, and that if a market-dominant company monopolizes facilities (elements) that are indispensable to a competitor's activities, and it is virtually impossible or economically unfeasible to establish equivalent facilities (elements), denying access to such facilities (elements) will make it virtually impossible or significantly impair the competitor's business. (Seoul High Court, April 17, 2003, Decision 2001 No. 5851, Korea Women's Professional Finance Association and Seven Credit Card Companies for Conspiracy to Interfere with Business Activities)

325 As follows, the standard-essential patents held by Defendants cover the manufacture, sale, and use of modem chipsets.

The accused's standard-essential patents are essential to modem chipset manufacturers because they are essential to the use of the modem chipsets, because the accused have exclusive control over them, and because they cannot be substituted by others.

(a) Necessity

326 A standard-essential patent is a patent that must be used by a business operator in the process of technically implementing a product in accordance with the standard specifications set by a standardization organization, and the implementation of the technical content in the standard document of the standardization organization will inevitably infringe the patent. Therefore, in order for a modem chipset manufacturer to enter the modem chipset market that implements mobile communication functions, it must ensure access to and use of the standard-essential patents for each communication standard such as CDMA, WCDMA, and LTE held by the defendants. If a modem chipset manufacturer manufactures and sells modem chipsets without a license to the essential patents, the modem chipset manufacturer will be exposed to the risk of patent infringement lawsuits or claims of non-infringement from the Defendants. Therefore, the essential patents of the Defendants for each communication standard, including CDMA, WCDMA, and LTE, are essential to the manufacture and sale of modem chipsets.

(b) Control

327 The standard-essential patents for each of the CDMA, WCDMA, and LTE telecommunications standards held by Defendants are official patents filed or registered by Defendants with the patent authorities in each

country, and Defendants are entitled to exclusive possession or control of such standard-essential patents as is customary for a patentee.[266)]

266) With respect to exclusive control, it may be argued that a standard-essential patent is a patent for which a standardization organization has declared a FRAND commitment and therefore has an obligation to license to willing implementers, and therefore lacks the control requirement. However, even a standard-essential patentee that has declared a FRAND commitment can still collect royalties on reasonable and non-discriminatory terms, and can negotiate reasonable royalty levels with willing implementers and receive fair consideration. Therefore, it is reasonable to assume that a standard-essential patent, like any other patent, is exclusively owned and controlled by the patent holder.

(c) Nonfungibility

328 A standard-essential patent is a patent that, in concept, cannot be circumvented or evaded technically or economically in order to implement a specific technical standard, so that infringement of the standard-essential patent is inevitable. Thus, if the defendants refuse to grant a license to practice the essential patents for implementing each of the CDMA, WCDMA, and LTE standards selected by the standardization bodies, it is impossible for a carrier seeking to use the quasi-essential patents to find a substitute technology or, even if a substitute technology exists, to incorporate it into its products and compete effectively in the marketplace.[267)]

(4) Whether you have denied or restricted the use of a required element.

329 First, as shown in the Admitted Facts, prior to 2008, the Accused licensed the Accused's standard essential patents, which are essential to the manufacture, sale, and use of modem chip sets.
***, ***, ***, ***, ***, ***, ***, and other competing modem chipset manufacturers that were willing to actively practice the patent, while granting only licenses to manufacture and sell the modem chipsets and limited rights to use the modem chipsets, imposed conditions that unreasonably restricted the scope of the patent practice, such as limiting sales outlets, requiring free cross-granting, and requiring the provision of sales information.

330 Second, since 2008, when Respondents changed their policy to not license modem chipset manufacturers, Respondents have refused to license the Standard Essential Patents to willing and able licensees, including ***, ***, and ***, who were willing and able to pay the license fee.

331 These Defendants' conduct directly denies or restricts the use of essential elements to competing modem chipset manufacturers and is substantially discriminatory or unfair to other users of essential elements, including Defendants.

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267) A standard-essential patent cannot be deemed essential if there is a competing standard that can replace a particular standard or if the sub-market implementing the standard technology is not locked-in to the standard. However, in the case of standard-essential patents such as WCDMA and LTE held by the defendants, there is no alternative standard, and the downstream modem chipset market, mobile phone market, base station equipment market, etc. are locked-in to the standard, so there is no alternative technology that can replace it.

By offering fair terms, the result is effectively the same as denial of use of an essential element.

This is an action that creates a

(5) Whether you unreasonably made it difficult for another business to do business

332 C. For the reasons discussed in paragraphs 1) and (4), the Respondents' refusal to license standard essential patents for the manufacture and sale of modem chipsets to competing modem chipset manufacturers, and their offering of only limited terms attached to unreasonable conditions, constitutes conduct that unreasonably restrains competing modem chipset manufacturers from doing business.268)

(6) Whether good cause exists

333 First, the granting of standard essential patent licenses to the Respondents to manufacture and sell modem chipsets does not prevent the Respondents from receiving just compensation for their investment.269) See supra Part II. As discussed in paragraphs 1) b) (3) (a) and (4), it is not clear that Respondents' essential patents are implemented only in the mobile phone stage, nor is it clear that licensing only in the mobile phone stage is the only way for Respondents to be fairly compensated for their patents. Furthermore, the Respondents are FRAND-committed patent holders who have declared FRAND commitments to standardization organizations, and licensing their FRAND-eligible patents on FRAND terms guarantees them the opportunity to receive physical samples. Therefore, licensing the FRAND-eligible patents to modem chipset manufacturers does not significantly impair the

Respondents' ability to receive fair compensation for their investment.

334 In addition, if a technology is selected as a standard, the holder of a standard-essential patent to implement that standard may claim that the selection of the standard significantly expands the base of users of that technology and that the

268) Because we have already discussed in detail the anticompetitive intent or effect of Respondents' conduct in entering into restrictive agreements without licensing to competitors, we do not reiterate the unfairness argument here.

269) In this regard, the Criteria provide that a restriction on the use of or access to an essential element may be justified if it significantly impairs the just reward for the investment of the undertaking providing the essential element, but a reduction in profits due to increased competition is not considered to be an impairment of just reward [Criterion IV. 3. c. (4). (a) of the Criteria for Abuse of Dominant Position].

You'll have the opportunity to license your patented technology to a wider range of parties, including optical
They benefit from a broader range of business opportunities. Therefore, the selection of a standard can be seen as a reward in itself for patent holders.

335 In addition, a decrease in profits resulting from increased competition is not considered a diminution of just compensation, such as a decrease in profits resulting from limited patent enforcement against a patentee as a result of the defendants' compliance with FRAND commitments, which is a consequence of increased competition.

336 Second, Respondents' licensing of the Standard Essential Patents to competing modem chipset manufacturers to manufacture and sell modem chipsets will not reduce supply to existing users.270) Respondents' Standard Essential Patents are intangible property rights, not tangible equipment that is limited in quantity. Therefore, there is no concern that licensing the Standard Essential Patents will reduce existing supply because licensing to a particular implementer does not restrict licensing to other implementers.

337 Third, the quality of the services already provided will not be materially impaired by the Respondents' licensing of the Standard Essential Patents to competing modem chipset manufacturers for the manufacture and sale of modem chipsets.271) The licensing of the Standard Essential Patents to new modem chipset manufacturers will not impair the quality of the services provided to existing modem chipset manufacturers because it will merely be a license to enforce the immovable property rights exclusively held by the Respondents.

338 Fourth, it is impossible for Respondents to license standard-essential patents to competing modem chipset manufacturers for the manufacture and sale of modem chipsets due to non-compliance

with technical standards.

270) In this regard, the Criteria provide that the impossibility of providing an essential element without significantly reducing the amount of provision to existing users may be a justifiable reason for restricting the use of or access to an essential element [Criterion IV. 3. c. (4). (b) of the Abuse of Dominant Position Test].

271) In this regard, the Criteria recognize that there are justifiable reasons for restricting the use of or access to an essential element if the provision of the essential element would significantly reduce the quality of services already provided [Criterion IV. 3. c. (4). (c) of the Criteria for Abuse of Dominant Position].

The essential patents for mobile telecommunications standards are published by standardization organizations in the Generation Mobile Telecommunications Table.
Since the patent implements the selected technology, there are no problems such as non-conformity with technical standards arising from the provision of licenses for standard-essential patents.

339 Fifth, Respondents' licensing of standard essential patents for the manufacture and sale of modem chipsets is not related to the life or physical safety of their customers.

(7) Sintering

340 Therefore, the Respondents' refusal to license or restrict the essential mobile communications standard patents to competing modem chipset manufacturers in 2. a. 4) above is illegal because it constitutes an act of unreasonably interfering with the business activities of other operators under Article 3 (2) (1) (3) of the Act and Article 5 (3) (3) of the Implementing Decree, which refuses or restricts the use of or access to elements essential to the production and sale of goods of other operators without justifiable reasons.

2) Act 2: Demanding a mobile phone manufacturer to enter into and fulfill a patent license agreement as a condition for supplying modem chipsets

a) Applicable laws and jurisprudence

(1) Applicable laws

Article 3-2 (Prohibition of Abuse of Market Dominance) ① A market dominant business shall not engage in any act falling under any of the following subparagraphs

(hereinafter referred to as "abuse").

3. unreasonably interfere with the business activities of other operators

Article 5 of the Enforcement Decree (Types or Criteria of Abusive Behavior) ③ Article 3(2) of the Act (Prohibition of Abuse of Dominant Position)

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272) In this regard, the Criteria provide that a justifiable reason for restricting access to or use of an essential element may be recognized if it is technically impossible to provide the essential element due to non-compliance with technical standards [Criterion IV. 3. c. (4). (d) of the Criteria for Abuse of Dominant Position].

Unreasonable interference with the business activities of another business pursuant to Paragraph 1, Item 3 shall be directly or
shall be deemed to have indirectly made it difficult for other business operators to conduct their business activities by engaging in acts falling under item 1 of the following subparagraphs.

1.~3. (omitted)

4. Acts notified by the Fair Trade Commission as acts that make it difficult for other businesses to conduct business activities in an unfair manner other than items 1 through 3.

Criteria IV. Detailed Types and Criteria for Abuse of Dominant Position

3. unfairly interfering with the business activities of another business (Article 3 (2) (1) (3) of the Act)

Directly or indirectly making it difficult for another business to conduct its business activities by engaging in any of the following acts (Article 5, Paragraph 3)

a. through c. (omitted)

d. Other acts that make it difficult for other business operators to conduct business (Article 5, Paragraph 3, Item 4), including the following acts

(1) to (2) (omitted)

(3) Unfairly coercing a counterparty into a transaction or behavior that disadvantages them.

(4) ~ (6) (omitted)

(2) Related jurisprudence

341 Article 3(2)(1)(3) of the Act prohibits market-dominant undertakings from unduly interfering with the business activities of other undertakings, and Article 5(3)(4) of the Enforcement Decree and Criterion IV. 3. of the Enforcement Decree specify that 'forcing a transaction or act that is unfairly disadvantageous to a counterparty' is an act that makes it difficult for another business to conduct business.

342 Thus, in order for the Respondents' conduct to constitute an abuse of dominance that unreasonably interferes with the business activities of other undertakings, the Respondents m u s t first be in a dominant position; second, the Respondents' conduct must amount to coercion of a transaction or conduct that unreasonably disadvantages a counterparty; and third, the Respondents' conduct must be

The above must constitute behavior that makes it difficult for other businesses to conduct business. Here, other
In determining the case of difficulty in the business activities of a business operator, the production, financial, and sales activities of other business operators shall be considered comprehensively, including cases where there is a possibility that business activities may become difficult.

343 Specifically, the meaning of "unfairness" of the imposition of disadvantages as an abuse of dominant position of a market-dominant undertaking is not only that a particular undertaking is individually disadvantaged by discrimination, but also that it has the intention or purpose of maintaining or strengthening its monopoly in the market, i.e., artificially affecting the market order by restricting free competition in the market, and that it is objectively likely to have the effect of restricting competition.[273)]

344 Where a dominant undertaking's unfavorable coercive conduct is proven to have had a realistic anticompetitive effect, there is a presumption that the dominant undertaking had the intention or purpose to cause anticompetitive effects at the time of the conduct. Otherwise, it is necessary to determine whether the conduct is likely to have the effect of restricting competition and whether there was an intention or purpose to do so by considering various circumstances, such as the history and motivation of the conduct, the circumstances of the conduct, the nature of the relevant market, the degree of disadvantage suffered by the counterparty due to the conduct, changes in prices and output in the relevant market, and whether innovation is stifled and diversity is reduced.

b) Whether you are a dominant player

273) Regarding the coercive act of disadvantage among the abuses of market dominance, the Seoul High Court stated, "The unfairness of the coercive act of disadvantage as an abuse of market dominance under Article 3(2)(1)(3) of the Act must be interpreted in accordance with the legislative purpose of 'promoting competition in an oligopolistic market', In all cases where a market-dominant undertaking engages in unfavorable coercive acts with unfair intent or purpose against a specific undertaking that is a counterparty to an individual transaction, or where the unfavorable coercive acts have caused a specific undertaking to suffer disadvantages, such as that the specific undertaking has suffered or is likely to suffer disadvantages in its business activities, the mere fact that the specific undertaking has suffered disadvantages is insufficient to recognize the unfairness, In particular, the unfairness may be recognized when an unfair coercive act is committed with the intention or purpose of maintaining or strengthening a monopoly in the market, that is, artificially affecting the market order by restricting free competition in the market, and can be objectively assessed as an act that may have the effect of restricting competition."(Seoul High Court, July 22, 2009, Decision 2008 No. 37543, confirmed by the High Court after remand)

345 C. 1) As discussed in (b)(2), Defendants have been accused of infringing CDMA, WCDMA and LTE, as well as the overall market for licensing the patented technology held by the defendants in each of the telecommunications standards, and the market for modem chipsets in each of the standards, CDMA, WCDMA, and LTE.

c) Whether it involves coercing a counterparty into a transaction or behavior that is disadvantageous to the counterparty.

346 A standard-essential patentee has a duty to negotiate in good faith with a person seeking to enter into a license agreement for a standard-essential patent once it has made a FRAND commitment to grant a license on fair, reasonable, and nondiscriminatory terms to a person seeking to practice the standard-essential patent.274)

347 Nevertheless, the Defendants did not license the essential mobile communication standard patents at the modem chipset stage, but at the mobile phone device stage, and thus established and operated a business policy of separating the supply of modem chipsets from the conclusion of patent license agreements with respect to the supply of their modem chipsets.275) Thus, a mobile phone manufacturer wishing to purchase a modem chipset had to enter into a patent license agreement with the Defendants in addition to entering into a modem chipset supply contract with the Defendants.276)

348 However, the Defendants did not just separate the modem chipsets and the patents, but implemented a business policy that linked them back together, i.e., (1) the Defendants did not supply modem chipsets to mobile phone manufacturers that did not enter into a patent license agreement with them in advance, and (2) the Defendants did not allow the mobile phone manufacturers to use the modem chipsets that they

supplied, so that they could purchase modem chipsets and use them in their own businesses.

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274) Aug. 24, 2012, Seoul Central District Court 2011 Ga-hap 39552

275) These Defendants' business policies differ from the business practices of competing modem chipset manufacturers such as *** and ***, which generally do not require the sale of modem chipsets to mobile phone manufacturers to enter into separate license agreements for their patents, or refuse to sell modem chipsets to mobile phone manufacturers because they do not enter into patent license agreements.

276) If the modem chipset manufacturer does not obtain a patent license from the accused, the mobile phone manufacturer that purchases and uses the modem chipset must enter into a patent license agreement with the accused to avoid patent infringement.

277) In their responses to the Commission, Respondents stated that "Qualcomm's policy is not to enter into component supply agreements with unlicensed mobile phone manufacturers" and that "Qualcomm does not sell mobile phone chips to mobile phone manufacturers that are not licensed by Qualcomm." (Exs. 32-34).

The manufacturer of the cell phone you intend to use must ensure that the subjects are informed about the
Patent License Agreement. Furthermore, the Defendants continued to link the patent license agreements and the supply of modem chipsets even after the patent license agreements were concluded by stipulating in the modem chipset supply agreements that the Defendants could terminate the modem chipset supply agreements or suspend or withhold the supply of modem chipsets even if the modem chipset manufacturer breached the patent license agreements. By entering into patent license agreements with mobile phone manufacturers under this tying structure, Defendants made it difficult for mobile phone manufacturers to negotiate fair license agreement terms on FRAND terms on an arm's length basis, and forced mobile phone manufacturers to accept unreasonable patent license terms278) by placing them at risk of having their supply of modem chipsets suspended for breach of the license agreement.

349 In the following, we'll take a look at the specific suns of disadvantaged transactions or behaviors and the process and practices for enforcing them.

(1) 'Linking modem chipset supply contracts to license agreements' and the resulting deprivation of fair negotiation opportunities in standard essential patent license agreements

350 Because the Respondents are FRAND-committed standard-essential patent holders, they must negotiate in good faith on FRAND terms when negotiating license agreements with mobile phone manufacturers for the standard-essential patents they hold. In other words, a mobile phone manufacturer that wishes to license a standard-essential patent is entitled to a fair opportunity to negotiate with the Respondents during the license negotiation process. Nevertheless, by using their own modem chipsets as a model279) and requiring license agreements to be signed prior to

the supply of modem chipsets, the Defendants have ensured that mobile phone manufacturers negotiate license agreements with the Defendants on FRAND terms.

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278) In practice, Defendants used the tying structure of modem chipset supply and license agreements to impose unfair patent license terms on mobile phone manufacturers, such as blanket license terms, self-determined royalty terms, and free cross-granting, without fair negotiation, as described in more detail in Part 3 of Conduct 3) below.

279) In this regard, *** attempted to implement the same business model as Defendants for its subsidiary in the modem chipset business in the mid-2000s, but unlike Defendants, *** did not have dominance in the modem chipset market and was ultimately unable to implement the business model.

We are depriving them of the opportunity to do so.

351 First of all, the Respondents are market dominant players in the CDMA, WCDMA and LTE modem chipset markets, especially in the Premium product segment, where the Respondents themselves recognize that there are no competitors280) and there are no substitutes for the Respondents' modem chipsets in that segment. Therefore, the ability to purchase modem chipsets from the Accused is an important prerequisite for a cell phone manufacturer to start a cell phone business or to launch a new cell phone. Under these circumstances, if the Respondents demand that the mobile phone manufacturers sign a license agreement before supplying the modem chipsets, it would be virtually impossible for the mobile phone manufacturers to refuse, and even if the Respondents demand unfair license terms that violate the FRAND covenant, the mobile phone manufacturers would be forced to accept the Respondents' demands for fear of disrupting the supply of modem chipsets, thereby depriving them of the opportunity to negotiate fairly and in good faith under the FRAND covenant. Even if the Respondents supplied the modem chipsets before entering into the patent license agreement, the mobile phone manufacturer would still be forced to enter into a patent license agreement with the Respondents for the use of the modem chipsets because of the provision in the modem chipset supply agreement that "the mobile phone manufacturer may not use the purchased modem chipsets or sell them in combination with other components without a patent license" and would be deprived of the opportunity to negotiate fairly and in good faith under the FRAND covenant.

352 This disadvantageous bargaining position is no different for mobile phone manufacturers who already have patent license agreements with the accused and source their modem chipsets from them. This is because when

new mobile communication standards emerge or license agreements expire, it is necessary to negotiate patent licenses with the accused companies in the process of amending or renewing existing agreements. Even in this case, due to the linkage clause in the modem chipset supply contract that allows the supply of modem chipsets to be suspended in the event of a breach of a patent license agreement not directly related to the supply of modem chipsets, mobile phone manufacturers are at a constant disadvantage in their license agreements with the defendants for standard-essential patents, as there is a risk that the supply of modem chipsets may be interrupted or delayed if they do not accept the defendants' unfavorable contract terms.

280) Exhibit 56, 2014 *March 2014 Internal data of the subjects

353 Thus, Defendants' conduct in tying the supply of modem chipsets to license agreements effectively deprives the other party to the license agreement, the handset manufacturer, of the opportunity to negotiate on a FRAND basis, and thus constitutes coercion of a counterparty into a transaction at a disadvantage.

(2) "Linkage of modem chipset supply contracts to license agreements" and the risk of disruption to the mobile phone business

354 Defendants' interconnection of modem chipset supply contracts and patent license agreements creates a risk to the entire mobile phone business, as mobile phone manufacturers are subject to disruption of modem chipset supply for breaches of license agreements not directly related to modem chipset supply.

355 In general, when a mobile phone manufacturer launches a new product, it takes a long period of time, such as two years, for the technical verification of the modem chipset's performance and compatibility with other components, and given the speed of development of mobile communication technology and the life cycle of mobile phone products, it is practically impossible to replace the modem chipset with another modem chipset once the modem chipset for the new product has been confirmed. Therefore, once a cell phone manufacturer has decided to use the Accused's modem chipsets and is in the development or production stage, if the cell phone manufacturer is unable to obtain modem chipsets from the Accused, the cell phone manufacturer will experience business difficulties that will force it to abandon or suspend production of the product.[282)]

356 Accordingly, Respondents' conduct in tying the supply of modem chipsets to a license agreement and setting the terms of the transaction so that only a breach of the license agreement would result in the cessation of the supply of modem chipsets constitutes coercion of a counterparty to its detriment.

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281) Moreover, Defendants' contracts with cell phone manufacturers to supply modem chipsets include the words [redacted] If not cured within ** days or ** days, the defendants may terminate the modem chipset supply agreement or suspend or withhold the supply of modem chipsets." The stipulation does not clearly define the criteria for determining whether a mobile phone manufacturer is in compliance with the patent license agreement, which may lead to the defendants arbitrarily determining that the mobile phone manufacturer is in breach of the license agreement.

282) In this regard, Mr. *** *** stated that "even if a substitute third-party chipset exists, the substitute would have to undergo inadequate technical verification and operator approval procedures, and until all such procedures are completed, it would be impossible to sell the product for a limited period of time, which would also cause significant disruption to our mobile phone business." (Exhibit 61, Declaration of Mr. *** ***).

(3) Enforcing a patent license agreement

(a) Case E

357 As shown by the admission in paragraph 2. a. 5) As shown in the admitted facts under b) (1), in the process of concluding a WCDMA patent license amendment agreement with E in 2003-2004, the Respondents had a dispute with E over the terms of the WCDMA-related "license agreement" and, based on the provisions of the modem chipset supply agreement linking the supply of modem chipsets to the patent license agreement, the Respondents terminated the memorandum of understanding regarding the modem chipset rebate and notified E to stop supplying modem chipsets, WHEREAS, E became concerned that the supply of modem chipsets would be interrupted or delayed due to the violation of the patent license agreement and that its mobile phone business would suffer as a result, and entered into an amended license agreement as requested by the Respondents.

358 WHEREAS, despite E's perception that the terms of the patent license agreement demanded by the Respondents were unfavorable to it, it was forced to enter into an amended license agreement with the Respondents less than two months after the Respondents threatened to stop supplying modem chipsets if it did not enter into the license agreement, because it feared that its business would be jeopardized by the Respondents' actual cessation of supply of modem chipsets, WHEREAS, in this process, E could not reasonably be expected to have been afforded the opportunity to negotiate in good faith and at arm's length under the FRAND covenant to which it is entitled in the licensing of standard essential patents.

(b) Case F

359 F entered into and maintained a CDMA patent license agreement with the Respondents in 1993, and in the early 2000s, F engaged in patent license negotiations with respect to the WCDMA standard. With respect to the *March* 2004 license negotiations between F and the Respondents, F*** stated to the Commission that "The Respondents' proposal is unacceptable to us.

283), but if license negotiations with the Defendants broke down, the Defendants' modem
It was difficult for us to maintain our hardline stance because we were concerned that we would not be able to obtain a smooth supply of chipsets. At the time, we were sourcing most of our modem chipsets from the Respondents, and there were no substitutes for the Respondents' modem chipsets, so any disruption in the supply of modem chipsets from the Respondents without a very long inventory period would result in a number of business disadvantages, including the possibility that we would not be able to launch our mobile phones immediately, and we would not be able to recover the cost of the new products we were developing....(emphasis added) It would take a considerable amount of time before the implementation fee could be reduced to a reasonable level, and the adverse business consequences would be even greater, so in view of these realities, we were forced to conclude that we should accept the Respondents' proposal and finalize the license agreement as soon as possible in order to obtain the modem chips quickly and smoothly."284) It can be seen that F relied on the Respondents' modem chipsets to maintain its mobile phone business and was forced to enter into a license agreement with the Respondents without a reasonable royalty adjustment process because it feared that its supply of modem chipsets would be interrupted due to the failure of the license negotiations, and was thus prejudiced by not having the opportunity to negotiate fairly and in good faith with the Respondents in accordance with the FRAND commitments in the licensing process of the Standard Essential Patents.

360 Furthermore, as shown in the admitted facts in paragraph 2. 5) b) (2) above, as shown in the admitted facts, around 2009, when F started a data modem business for PCs using the Respondents' modem chipsets, ***, an employee of the modem chipset division of Korea

Qualcomm, sent an e-mail to F on *March *, 2009, stating that the Respondents would stop supplying modem chipsets to F due to F's violation of the license agreement, WHEREAS, Respondents actually contacted the licensing division of Respondents' headquarters to inquire whether it would discontinue supplying modem chipsets to F, and Respondents and F subsequently entered into a patent license agreement covering the subject product on *, 2009. In light of these facts, the linkage between the Respondents' supply of modem chipsets and the license agreement creates a risk that F will stop supplying modem chipsets to the Respondents for breach of the license agreement.

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283) At the time, Respondents argued to F that there was no reason to reduce the existing royalty rate given that WCDMA is a wireless communication method no different from CDMA and that the 1993 Agreement was a comprehensive agreement covering all patents, including CDMA as well as WCDMA (Exhibit 61, Affidavit of F***).

284) Respondents and F then entered into a patent license agreement on or about the last day of *, 2004 for a *% royalty.

285) Exhibit 61, Affidavit of F***, F***'s Affidavit

Under the facts of the case, and in light of the defendants' reliance on the modem chipsets, it is reasonable to assume that F
In order to avoid this business risk, the defendants had no choice but to enter into the license agreement as demanded by the defendants.

<Judgment on the claims of the relevant defendants ⑧>

361 In this regard, Respondents argue that they have been entering into licensing agreements with large mobile phone vendors since they were only a small company, that their licensing terms were well known before the later WCDMA and LTE standards were standardized, that they have never stopped supplying actual modem chipsets even if there was a dispute over the patent licensing agreement, and that they have never used the issue of modem chipset supply as a means to gain an advantage in negotiations over the licensing agreement.

362 In sum, Respondents' arguments are unavailing in the following ways

363 First, the Defendants built their business model to leverage their supply of modem chipsets as leverage for patent license agreements, and in fact used their business model to leverage license agreements with mobile phone manufacturers. The Defendants' modem chipset supply agreements require handset manufacturers who wish to purchase modem chipsets to first enter into a license agreement, and stipulate in the modem chipset supply agreements that the Defendants may terminate the modem chipset supply agreement and stop supplying modem chipsets if the handset manufacturer fails to perform under the license agreement. As a result, mobile phone manufacturers have no choice but to accept the license terms demanded by the Defendants, even if they are unreasonable, in order to purchase the Defendants' modem chipsets and conduct business. Given that the Respondents are vertically integrated companies with dominant positions in both the modem chipset market and

the telecommunications standard essential patent market, the above coercive effect is in effect on all mobile phone manufacturers that rely on the Respondents' modem chipsets.

364 In addition, since the early 2000s, Defendants have been involved in modem chipset supply contracts with cell phone manufacturers, including

to the license agreement, and, after 2008, to competing modem chipsets. It does not license essential patents for mobile communication standards to manufacturers. Moreover, in 2012, the company separated its licensing business and modem chipset business into separate entities, creating a business policy to link modem chipsets and licenses by using its modem chipsets as leverage.

365 As a result, Respondents extended the terms of license agreements entered into in the 1990s, when their CDMA technology was selected as the standard, to the later selected WCDMA and LTE standards, despite the fact that Respondents' contribution to WCDMA and LTE was less than that of CDMA, because the license terms were based on the mobile This is being applied uniformly across phone manufacturers.

366 On the other hand, even if the Respondents have not actually stopped supplying modem chipsets, the fear that the supply of modem chipsets may be interrupted or delayed in the process of modifying the patent license agreement is sufficient to constitute a significant threat to the business, and even if the Respondents have not actually done so, the mere threat of interruption of the supply of modem chipsets by the Respondents is perceived by the handset manufacturers as a realistic threat, resulting in the handset manufacturers entering into the patent license agreement on a less than fair and mutually beneficial basis, thereby forcing the handset manufacturers to bargain at a disadvantage.

d) whether it unreasonably interferes with the business activities of other operators

(1) Intent and Purpose of Restraint of Competition

367 Considering Defendants' position as dominant vertically integrated players in the modem chipset market and the telecommunications standard-essential patent licensing market, Defendants' awareness of the need to comply with FRAND commitments, Defendants' business reports and internal documents, and Defendants' patent license negotiation process, Defendants unusually separated the modem chipset supply agreement from the patent license agreement by requiring that the patent license agreement be executed prior to the modem chipset supply agreement, thereby linking the two contracts, and then enforcing the modem chipset supply agreement even in the event of a breach of the license agreement.

The process of licensing telecommunications standard essential patents by enabling class discontinuation
that there was an intent or purpose to restrain competition in the marketplace.

(a) Anticompetitive incentives in the license market as a dominant vertically integrated operator

368 The Respondents are vertically integrated companies that are both manufacturers and sellers of modem chipsets and licensees of essential patents for mobile telecommunications standards, and as market dominant players in the modem chipset market, they have an incentive to use their dominance in the modem chipset market to their advantage in entering into patent licensing agreements with mobile phone manufacturers to increase their dominance in the mobile telecommunications standard essential patent licensing market by linking the supply of modem chipsets to license agreements, thereby increasing their revenues without going through a fair negotiation process.286)

369 Moreover, the defendants themselves were aware of these anticompetitive incentives, see supra.
1) b) (4) (a) As discussed above, this is also evidenced by emails between Defendants' executives circa *2013287) stating that "maintaining the competitiveness of modem chipsets is important to maintain revenue for the licensing business."

(b) intent to circumvent the obligation to negotiate in good faith under the FRAND commitment; or

370 FRAND-committed defendants are obligated to negotiate in good faith when entering into licensing agreements for standard-essential

patents, and the defendants themselves have stated on their websites that "FRAND principles require mutual good faith negotiation of license terms between the patent holder and the licensee."[288] Standard-essential patent holders that have made a FRAND-committed declaration have stated that

286) In addition, the Defendants have an incentive to maximize the effectiveness of patent hold-up and limit competition in the market for licensing essential patents for mobile communications standards by tying unfair contractual terms to the license agreements, and to use the unfairly obtained contractual terms, such as free cross-granting, to strengthen and maintain their dominance in the modem chipset market. This will be discussed in the next section, Act 3.

287) Emails between ***, the general manager of Defendants' modem chipset sales in China, and ***, the vice president of Qualcomm Asia, indicate that maintaining the competitiveness of Defendants' chipset division (QCT) is critical to maintaining a strong revenue stream for the licensing division (QTL) (Exhibit 50, 2013 * * *. Email between Defendants' executives; Exhibit 51, 2013 * * *. Email between Defendants' executives).

288) The original text reads "FRAND is a well-established principle that appropriately balances the interests of patent holders to obtain a fair return on their innovations and those of implementers to obtain a access to such innovations through good face bilateral negotiation of licensing terms

recognized that it was obligated to negotiate in good faith under the FRAND commitments.

371 Nonetheless, as discussed in c) above, by tying the supply of modem chipsets to license agreements and requiring the execution and performance of patent license agreements as a condition of supplying modem chipsets, Respondents have the intent and purpose of leveraging their dominance in the modem chipset market to circumvent the good faith negotiation process under the FRAND commitments to impose terms and conditions that are disadvantageous to handset manufacturers and favorable to themselves.

(c) Blocking a path for cell phone manufacturers to circumvent the accused's business model by using competing modem chipsets

372 Defendants did not license their standard-essential patents to competing modem chipset manufacturers, thereby denying mobile phone manufacturers the ability to use competing modem chipsets to circumvent Defendants' business model of tying license agreements to the supply of modem chipsets.

373 If competing modem chipset manufacturers could obtain from Defendants a full and exhaustive license to the standard-essential patents embodied in their modem chipsets pursuant to Defendants' FRAND commitments, cell phone manufacturers could eliminate the risk of patent infringement from Defendants by selecting competing modem chipsets or using this optionality in their negotiations rather than entering into unfavorable license negotiations to purchase Defendants' modem chipsets.289However, by not licensing competing modem chipset manufacturers and, even when entering into restrictive patent arrangements with some modem chipset manufacturers, restricting the sale of modem chipsets only to handset manufacturers with whom they had

entered into licensing agreements and requiring them to report the sales of competing modem chipset manufacturers and the number of units sold by each outlet, Defendants sought to cut off any avenue for handset manufacturers to purchase modem chipsets without first entering into a license agreement with them.

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and conditions."[http://www.qualcomm.com/invention/licensing, "LTE/Wimax PATENT LICENSING STATEMENT" (December 2008)]

289) Of course, it is clear that the Respondents' refusal to license at the modem chipset stage restricted competition in the modem chipset market, which allowed the Respondents to acquire, maintain, and consolidate their dominant position in the modem chipset market, and reduced the number of competing modem chipsets from which mobile phone manufacturers could choose.

374 Defendants' intentions are further evidenced by the 2013 email between Defendants' employees titled "Big Potential Risk with QTL's New Deal with ***," discussed above290). At the time, Defendants' 2013 agreement with *** lifted the distributor restrictions on *** and significantly relaxed its obligations to report sales information, including distributors, indicating that Defendants were concerned that losing the ability to monitor the sales of modem chipsets by competing modem chipset manufacturers would make it more difficult to enforce license agreements with handset manufacturers in the future.

(d) Using the threat of modem chipset supply disruption as a threat in actual license negotiations

375 As discussed in Section 2. a. 5) b) above, in the course of entering into the patent license agreements with E in 2003 and 2004 and with F in 2009, the Respondents alleged breaches of the license agreements by E and F, citing the linkage clause to the supply of modem chipsets and the cessation of the supply of modem chipsets. In this process, the defendants did not attempt to resolve the dispute over the licensing agreement of the standard-essential patents through good faith negotiations on FRAND terms, but instead used their dominance in the modem chipset market and the modem chipset supply contracts tied to the license agreements to impose their demands. These cases will demonstrate that the defendants' conduct in tying the supply of modem chipsets to license agreements was done with the intent to use this structure in their licensing arrangements with mobile phone manufacturers to benefit themselves and to restrict competition by interfering with their business activities.

(e) Unusual business practices that are different from competing modem

chipset manufacturers

290\ "[redacted]?" (Sogap No. 50, 2013. *. *. Email between employees of the defendants "Big potential risk from QTL new deal with ***")

376 Competing modem chipset manufacturers, such as *** and ***, do not require mobile phone manufacturers to enter into a license agreement for their patents when selling modem chipsets to mobile phone manufacturers, or refuse to supply modem chipsets because they do not enter into a patent license agreement. In addition, there is no example of a mobile phone manufacturer purchasing a modem chipset and entering into a separate patent license agreement with the Respondents. In other words, there is no example of a mobile phone manufacturer requiring the supply of a modem chipset and a separate license agreement for a standard essential patent in the mobile phone industry, which clearly demonstrates the Respondents' intent to restrict competition.291)

<Judgment of the claims of the relevant defendants ⑨>

377 In this regard, the Respondents argue that the use of their modem chipsets by mobile phone manufacturers that have not entered into a patent license agreement will inevitably infringe the Respondents' patents, and therefore the tying of the supply of modem chipsets to the license agreement was not intended to cooperate with infringement, but to protect their patent rights and not to restrict competition.

378 In sum, Respondents' arguments are without merit given that

379 First, it is difficult to accept Respondents' argument that cell phone manufacturers are automatically infringers simply because they purchase modem chipsets from Respondents to manufacture cell phones. First of all, in the case of patents embodied in modem chipsets, even if a mobile phone manufacturer that purchases a modem chipset infringes the patent, it is not the intention of the mobile phone manufacturer, but rather the

business policy of the Respondents to separate the patent from the modem chipset and not to license the patent at the modem chipset stage, unlike conventional modem chipsets. Secondly, in the case of patents not embodied in modem chipsets, the infringement of the patent is unrelated to the purchase and use of the modem chipset by the mobile phone manufacturer, so it is a legitimate patent protection difference to link the supply of modem chipsets to the license agreement to protect such patents.

291) In fact, Defendants spun off their modem chipset business and licensing business into separate entities in 2012 to separate the supply of modem chipsets from patent licensing agreements, and stated that the purpose of the spin-off was "to further protect and insulate our valuable patent portfolio." (Defendants' 2012 Annual Report).

It's even harder to see it as a circle.

380 Therefore, it is not necessary to enter into a patent license agreement with the defendants before supplying modem chipsets to protect the defendants' patents, and if there is a patent infringement problem, the defendants can always exercise their patent rights through contract negotiations or claim patent infringement through litigation.292)

381 Second, Defendants' refusal to supply modem chipsets as a means of protecting their patent rights, rather than negotiating license terms in good faith under FRAND commitments or pursuing legitimate judicial remedies, is hardly a reasonable way to protect their patent rights, as it is a self-help remedy that makes it difficult for mobile phone manufacturers to do business.

382 Third, as mentioned above in 1. Basic Facts, when a patent holder declares and discloses a patented technology that falls under a standard to a standardization organization, there is no separate procedure for the standardization organization to verify whether the patented technology actually falls under a standard-essential patent, and subsequently, in the process of concluding a license agreement, the parties verify whether the patented technology is actually a standard-essential patent, and it is known that there are not a few cases where the declared patents are found to be invalid or non-standard-essential patents. Therefore, it is likely that the accused seeking to receive royalties for implementing a standard-essential patent will have to prove which of the accused's patents the handset manufacturer is infringing and in what manner, and conversely, the handset manufacturer, as the patentee, should be guaranteed the opportunity to dispute the validity, essentiality, and extent to which the accused's standard-essential patent is

implemented in its handset.[293]

383 If this is to protect patent rights, a patent license agreement with the modem chipset supplier

292) In their brief to the Commission, Respondents responded that "even if a patent holder makes a patent claim, if the patentee disputes the validity of the patent claim, the patentee is not barred from practicing the patent until the patentee obtains a favorable court ruling." (Respondents' Brief at 187.)

293) The European Court of Justice (ECJ) also ruled on July 16, 2015, in a case between Huawei and ZTE, that a patentee must first inform the patentee of which patents are infringed and how before filing an anti-sale claim, and that the patentee must be given the opportunity to contest the essentiality and validity of the patents during license negotiations.

to utilize the self-help method of rejecting and discontinuing the supply of modem chipsets.
Rather, as a standard-essential patent holder that has declared a FRAND commitment, it would be appropriate to utilize license negotiations and agreements in good faith, or to pursue legitimate judicial remedies in the event of infringement. Nevertheless, the Respondents do not specifically disclose this or provide patent information in the process of signing patent license agreements with mobile phone manufacturers, and require the signing of license agreements as a condition of supplying modem chipsets, which is difficult to admit that this is for the purpose of protecting the Respondents' patent rights or preventing patent infringement.

(2) Anti-competitive effects

(a) avoid negotiating license terms in good faith in violation of the FRAND Declaration

Avoiding or circumventing the obligation to negotiate in good faith under FRAND commitments

384 A standard-essential patent holder that has made a FRAND commitment is obligated to negotiate in good faith with a person requesting to enter into a license agreement for a standard-essential patent, and since the Defendants have also made a FRAND commitment for their standard-essential patents during the standard selection process, they must negotiate in good faith for FRAND license terms when negotiating a license agreement for a standard-essential patent with a mobile phone manufacturer. However, it is clear from the foregoing that the defendants, who are obligated to negotiate in good faith, violate the FRAND commitments by tying the license agreement to the supply of modem chipsets and using their dominance over the modem chipsets to force the signing of the license agreement.

Competition Restraining Effects of FRAND Covenant Violations

385 Given the fundamental nature of standardization, which involves the adoption of a particular technology as a standard and the exclusion of competing technologies, and the role of FRAND covenants in curbing the abuse of dominance by standard-essential patent holders by displacing competing technologies after standardization, breach of FRAND covenants can be a significant threat to competition resulting from standardization.

This has the effect of removing the only means of preventing one behavior. Furthermore, if a particular standard is set to
Where no substitute standard exists, or where the competitive structure of the sub-markets in which the standard is applied has no way of controlling the abuse of monopoly power by the standard-essential patent holder, the breach of FRAND commitments by the standard-essential patent holder is more likely to result in direct anti-competitive harms, such as higher manufacturing costs, reduced product variety, limited technological innovation, and reduced consumer benefits.

386 However, the Respondents are monopolistic, vertically integrated firms with dual dominance as the holders of telecommunications standard-essential patents and as dominant players in the market for modem chipsets that implement these standard-essential patents. Accordingly, Respondents' violations of the FRAND commitments have anticompetitive effects in the market for licensing telecommunications standard-essential patents, the market for modem chipsets, and the market for innovation.

③ Competition Restrictiveness of Anti-Sale Claims by Standard Essential Patent Holders

387 It is a violation of the FRAND principle and likely a violation of the law for a standard-essential patent holder that has made a FRAND declaration to seek to enjoin sales against a potential licensee that is negotiating in good faith to obtain a license.

388 (i) Article 3(2)(1)(3) of the Act prohibits a dominant undertaking from unduly interfering with the business activities of other undertakings, and Article 5(3)(4) of the Act and Article IV. 3. l. (6) specifies that the prohibited type of abuse is the act of a dominant business entity "improperly using patent infringement lawsuits, patent invalidation judgments, and other

judicial and administrative procedures related to intellectual property rights to make it difficult for other business entities to conduct business. In addition, the "Guidelines for Examination of Unfair Exercise of Intellectual Property Rights" recognizes legal proceedings such as patent infringement lawsuits as an important means of ensuring the rights of patent holders, but stipulates that acts that abuse legal and administrative proceedings such as patent infringement lawsuits may be judged to be outside the legitimate scope of patent rights.

389 (ii) the European Commission, unlike a typical patentee's claim for non-infringement,

In the case of Samsung Electronics' injunction claim against Apple, the European Court of Justice (ECJ) decided that the act of a standard-essential patent holder who has declared FRAND to file an injunction claim against a willing implementer may constitute an abuse of dominant position294) and agreed that Samsung Electronics would not file a patent infringement claim in Europe based on Samsung Electronics' misrepresentation of standard-essential features against a potential implementer that complies with the FRAND terms295) and follows the determination system for determining the terms of implementation295).

390 (iii) The European Court of Justice has held that the filing of an injunction claim by a standard-essential patent holder with a FRAND commitment may constitute an abuse of a dominant position, provided that ⓐ the patent holder has warned the infringer of the infringement before filing the action, ⓑ the infringer is willing to license on FRAND terms, and ⓑ the patent holder has made a specific written proposal on the licensing terms, in particular on the royalties and the way in which the royalties are calculated, The court held that an injunction claim may not constitute an abuse of a dominant position if the infringer continues to exploit the patent and fails to respond in good faith to the patent holder's proposal, particularly with respect to royalties and the manner in which royalties are calculated.297)

391 (iv) The U.S. Federal Trade Commission adopted a consent decree prohibiting injunctive actions against operators seeking to enter into license agreements for standard-essential patents. Specifically, the consent order requires that ⓐ a minimum six-month negotiation period be held; ⓑ a binding offer of a license agreement be made at the licensee's request within 60 days of the expiration of the six-month period; ⓑ the licensee may request that the FRAND terms be determined by court

adjudication or arbitration; and ⓓ an anti-infringement claim may only be brought if the licensee has made it clear that it is unwilling to pay a royalty for the standard-essential patent or if the licensee is located outside the jurisdiction of a U.S. court.[298)]

294) EU Commission, Case AT.39985 - Motorola - Enforcement of GPRS Standard essential patents, Apr. 29, 2014

295) In this context, the European Commission's system for determining the FRAND-compliant implementation terms means (a) a negotiation period of up to 12 months, and (b) if no agreement is reached by the end of the negotiation period, the FRAND terms are determined by a redetermination or court ruling.

296) EU Commission, Case AT.39939 - Samsung - Enforcement of UMTS Standard essential patents, Apr. 29, 2014

297) July 16, 2015, European Court of Justice (ECJ) Case C-170/13

298) Federal Trade Commission, In the Matter of MOTOROLA MBILITY LLC and

392 (v) In an amicus brief filed with the Federal Circuit Court of Appeals in a patent dispute between Apple and Motorola, the U.S. Federal Trade Commission stated that "license fee negotiations conducted under the threat of an injunction may result in undue consideration of the patentee's interests in a manner that conflicts with the RAND commitment. Because the licensee is locked into the standard technology, the patentee will be able to adopt unreasonable terms despite the RAND commitments. In particular, if an injunction claim is based on a patent covering only a few components of a complex multicomponent system, as is often the case in the telecommunications industry, it could enable the standard-essential patent holder to realize implementation fees above the value of the patented technology and lead to price increases for consumers. "[299)]

393 (vi) The U.S. Department of Justice supported the standardization organization IEEE's change in policy to limit the threat of exclusion from the marketplace for patent holders with RAND commitments, arguing that limiting the threat of exclusion from the marketplace would reduce the likelihood of patent enforcement using standard-essential patents and give licensees peace of mind to continue product development.[300)]

④ Stronger restraint of competition than non-sale claims

394 As discussed in Sections 1 through 3 above, a FRAND-committed standard-essential patent holder may not, in negotiating the license of a standard-essential patent, make a non-solicitation claim in violation of the FRAND commitment, or conduct that has the effect of a de facto non-solicitation claim.

395 Nonetheless, as a substitute for the antitrust claims, the Defendants arranged for the parties to enter into and execute patent license

agreements with them in advance as a condition of supplying the modem chipsets. In doing so, Defendants circumvented the restrictions imposed by the FRAND commitments and imposed their own unfair patent license agreements on handset manufacturers who had no choice but to purchase their modem chipsets. This is a stronger and

GOOGLE INC. Docket No.C-4410

299) Dec. 4, 2012. Federal Trade Commission, BRIEF OF AMICUS CURIAE FEDERAL TRADE COMMISSION SUPPORTING NEITHER PARTY Nos. 2012-1548, 2012-1549

300) Amicus Brief 86, Position on U.S. Department of Justice's Changes to IEEE Intellectual Property Policy

It unfairly forces phone manufacturers to enter into patent license agreements.

396 (i) An injunction claim is usually filed by the patent holder in a court of competent jurisdiction, where the court considers the relevant laws and contractual terms to decide whether to grant the injunction, and both parties have the opportunity to fully present their positions. On the other hand, the defendants' actions in connection with the modem chipset supply contract and the license agreement (Act 2) are self-help remedies in which the defendants determine whether the mobile phone manufacturer is in violation of the patent license and, if so, refuse or discontinue the supply of the modem chipset. In this process, the mobile phone manufacturer is forced to accept the defendants' unilateral demands without being able to fully present its position or negotiate fairly on an equal footing with the defendants.

397 (ii) Even if an injunction is filed, it usually takes several years for the technical analysis of patent infringement, etc. to be completed, so there is little risk that the injunction will immediately halt the mobile phone manufacturer's business activities. On the other hand, the Defendants' conduct 2 may result in an immediate and total suspension of the supply of modem chipsets at the Defendants' discretion, and it is difficult for mobile phone manufacturers to predict whether such suspension will occur.

398 (iii) The effect of an injunction is usually limited to the jurisdiction of the court in which the claimant sought the injunction, and the mobile phone manufacturer can continue its business activities outside the jurisdiction. On the other hand, if the defendants refuse or stop supplying the modem chipsets, the cell phone manufacturer has no choice but to stop producing all cell phones equipped with the defendants' modem chipsets, regardless of the geographical scope.

399 Ultimately, Defendants' conduct 2 violates the FRAND Undertakings because, as shown in Table 52 below, Defendants self-imposed, virtually instantaneously, a worldwide ban on the business of mobile phone manufacturers.

<Table 52 Comparison of the injunction claim and the defendants' conduct 2

	Injunction	Behavior 2: Refuse to supply the modem chipset - discontinued
Judgment Subject	Neutral authorities, such as courts	Subjects' self-help
Judgment Criteria	Based on applicable laws, contractual terms, etc.	Subjects' arbitrary judgment
When the effect occurs	After final judgment, etc.	Immediate
Scope of impact	Jurisdiction-specific	Entire business scope

<Judgment on the claims of the relevant defendants ⑩>

400 In this regard, Defendants argue that while an injunction would completely prohibit mobile phone manufacturers that refuse to license from Defendants from manufacturing and supplying mobile phones, it would not prevent mobile phone manufacturers from manufacturing mobile phones because they could purchase modem chipsets from other modem chipset manufacturers even if they refused to supply modem chipsets in violation of their patent licenses.

401 In sum, Respondents' arguments are without merit given that

402 First, as discussed above, Defendants' refusal to supply modem chipsets poses significantly greater business uncertainty and risk to Hewlett-Packard because of the arbitrariness of the decision, the immediacy of the effect, and the breadth of the impact, compared to a sales ban.

403 Second, the Respondents are market leaders in the modem chipset

markets for the CDMA, WCDMA, and LTE standards, with virtually no competition, particularly in the CDMA modem chipset, CDMA backward compatible multimode modem chipset, and high-end LTE-premium modem chipset product lines, and the Respondents impose restrictions on competing modem chipset manufacturers that require them to sell their modem chipsets only to the Respondents' licensees, making it difficult to obtain modem chipsets from competing modem chipset manufacturers.

(b) transferring dominance in the modem chipset market to the standard essential patent licensing market

404 Moreover, the Respondents are the dominant players in the CDMA, WCDMA and LTE modem chipset markets, especially in the Premium product segment, where the Respondents themselves recognize that there are no competing products that can replace their modem chipsets in that segment. Therefore, cell phone manufacturers have no choice but to rely on the supply of modem chipsets from the Respondents for the continued and stable operation of their business.

405 However, the defendants used their dominance in the modem chipset market to create the

In the process of entering into the E patent license agreement with the Respondents from 2003 to 2004 and the F patent license agreement with the Respondents around 2009, the Respondents entered into patent license agreements with mobile phone manufacturers without conducting good faith negotiations in accordance with the FRAND commitments by threatening to stop supplying modem chipsets.

406 Furthermore, *** stated to Respondents in 2013 that "Qualcomm is able to collect *% royalties because Qualcomm chipsets are essential" and that "*** would not have entered into the license agreement if it did not rely on Qualcomm's modem chipsets", Given that Defendants admittedly believed that maintaining the competitiveness of their modem chipset segment was critical to maintaining strong revenues in the licensing segment301), Defendants abused their dominance by leveraging their modem chipset dominance to violate the FRAND commitments they were bound by as licensees in the standard-essential patent licensing market.302)

301) Smallpox No. 51, between *** and ***, 2013. *. *. Mail
302) Even if the defendants have dominance in the modem chipset market, there is not much

concern that their dominance will be transferred to the patent license market if they do not link the supply of modem chipsets to patent license agreements. This is because the defendants are constrained by the FRAND declaration like other standard-essential patent holders, i.e., even if the defendants offer unfair license terms, mobile phone manufacturers can enter into a license agreement under FRAND terms through arbitration, litigation, etc. However, by blockading the supply of modem chipsets and the mobile phone business itself until the mobile phone manufacturers enter into a license agreement, the Defendants circumvented their obligation to negotiate in good faith with the mobile phone manufacturers and forced them to accept their proposed license terms.

(c) function as a means of patent holdup due to linkage

407 Defendants' conduct in linking the supply of modem chipsets to the execution of patent license agreements provides a means of enabling patent holdup in the patent license process. Using the supply of modem chipsets as leverage, Defendants have coerced and forced mobile phone manufacturers to enter into patent license agreements with unfair terms, such as blanket licenses and unilateral royalty terms, and to receive gratuitous cross-granting of limited patents in exchange for patent licenses. The unfavorable license terms obtained by the Defendants, i.e., royalties based on the full price of the mobile phone and free cross-granting, lead to exclusionary effects in the modem chipset market by increasing the indemnification costs of competing modem chipset manufacturers or causing cost differentials between themselves and competing modem chipset manufacturers, and the unfavorable license terms stifle innovation in mobile communications.

408 The detailed facts of the conduct and its anticompetitive effects are discussed in the following section 3), Conduct 3.

<Judgment on the allegations of the relevant defendants ⑪>.

409 On the other hand, the Respondents argue that even if they tied the supply of modem chipsets to license agreements and forced mobile phone manufacturers to enter into disadvantageous license agreements, this was only to the disadvantage of mobile phone manufacturers and had no anticompetitive effect as an exclusionary effect.

410 In sum, Respondents' arguments are without merit, given that

411 First, the Respondents' ability to impose disadvantageous license terms on mobile phone manufacturers is not a matter of mere tying or exploitative behavior because the Respondents used their control of the modem chipsets to avoid the FRAND commitments required of them as standard-essential patent holders. The defendants, as well as the cell phone manufacturers, were able to use the defendants' modem chip

3 The perception that dominance is a factor in entering into unfavorable licensing agreements is a function of

In light of the admissions and evidence discussed in (b) above, it is sufficiently admitted.

412 Second, the Respondents' use of their dominance in modem chipsets to violate the FRAND commitments and withhold patents is not merely exploitative. By violating the FRAND agreement, which is itself a means of controlling the restrictive effects of standardization on competition, and by using the unfavorable terms and conditions obtained in the licensing process to their own advantage, such as unilateral royalties based on the total price of the mobile phone and free cross-granting, as described in (c) above, they restrict competition in the modem chipset market by increasing the costs of competing modem chipset manufacturers, and inhibit innovation in the mobile communications industry by causing an imbalance in the input and reward of technology development.

m) Sintered

413 Therefore, the Respondents' act 2. a. 5) above (requiring mobile phone manufacturers to enter into and perform a patent license agreement as a condition of supplying modem chipsets) is illegal because it constitutes an act of making it difficult for other businesses to conduct their business activities by other unreasonable means under Article 3 (2) (1) (3) of the Act and Article 5 (3) (4) of the Enforcement Decree, and is an act of forcing a transaction or act that is unfairly disadvantageous to a counterparty.

3) Act 3: When concluding a patent license agreement with a mobile phone manufacturer, the act of presenting contract terms including a

comprehensive license term, a license fee term set by the company, and a free cross-grant term

a) Applicable laws and jurisprudence

414 The relevant statutes and legal principles for Act 3 are the same as those for Act 2 in 2. c. above. 2) The relevant statutes and legal principles are the same as those for Act 2 in a) above.

b) Whether you are a dominant player

415 As discussed in Section 2. c. above. 2) As discussed in b), the Respondents are recognized as dominant players in the overall market for patent licenses held by the Respondents for each of the telecommunications standards, CDMA, WCDMA, and LTE, and in the market for modem chipsets for each of the standards, CDMA, WCDMA, and LTE.

c) Does it coerce a counterparty into a transaction or behavior that is disadvantageous to the counterparty?

416 As discussed above, defendants who have made FRAND commitments to standardization organizations such as ETSI and TIA for their essential patents are obligated to negotiate in good faith on FRAND terms when negotiating license agreements for their essential patents with mobile phone manufacturers who wish to license them for a fair consideration.

417 In a typical patent license agreement negotiation process, the parties identify the patents to be licensed, estimate the scope and value of the implementation of each patent, decide whether to license the patent, and determine the level of the royalty, i.e., the patentee provides the licensee with (i) a list of the patents and a claim chart so that the licensee can understand the scope of the patents and the extent of the patent's rights before entering into the agreement303),
(ii) give the licensee an opportunity to verify the validity of the patent being licensed by checking its expiration date, relationship to prior art, etc.

418 In addition, if infringement is alleged, the FRAND-certified patentee must warn the licensee of the alleged infringement by notifying the licensee of which patents have been infringed and how, and, if the

infringer wishes to license on FRAND terms, offer a specific written proposal for the terms of the license, including royalties and the manner in which the royalties will be calculated.[304]

303) Since a registered patent determines the scope of rights and determines whether the patent is infringed, claim interpretation is necessary for the licensee to determine whether the patent is infringed.

304) CJEU, Case C-170/13, Huawei Technology Co. LTD v. ZTE Corp.

419 Nonetheless, as shown in Acts 1 and 2 above, Defendants used the modem

By refusing to license its standard-essential patents to competing modem chipset manufacturers at the chipset stage, creating a structure that requires mobile phone manufacturers to enter into licenses at the handset stage (Conduct 1), and then using its dominance in the modem chipset market to tie license agreements to the supply of modem chipsets, forcing mobile phone manufacturers to first enter into patent license agreements with the accused (Conduct 2), and, as discussed below, (i) precluding mobile phone manufacturers' opportunity to negotiate FRAND terms for standard-essential patents by imposing blanket license terms without providing them with adequate and necessary information to negotiate patent license agreements; and

(ii) separate licenses for telecommunications standards or other patents that you can choose from

(iii) forcing handset makers to accept his royalty terms without giving them the opportunity to do so; and (iv) forcing handset makers to cross-grant his patents at no cost, thereby imposing terms and conditions that were favorable to him and that the handset makers would not have accepted in a fair bargaining process on FRAND terms.

(1) Forcing favorable contract terms by eliminating the opportunity for FRAND negotiations and reasonable licensing choices

420 At the level of standardization organizations, there is no separate verification procedure for the validity of standard-essential patents, such as whether the patent holder has disclosed and registered his patent as a standard-essential patent as defined by the standardization organization. Therefore, such verification and evaluation of validity is usually carried out at the time of the actual conclusion of the license agreement. Therefore, if a standard-essential patent holder wishes to enter into a paid license agreement, it must reasonably explain that its patent qualifies as a

standard-essential patent and that the other party is infringing its standard-essential patent305), and this is no different for large standard-essential patent holders.

421 However, when entering into patent license agreements, the defe[redacted]t wi[redacted]

305)

.(Sgt. 69, *** Negotiated in 2003-2004.

Status internal reporting)

Any intellectual property rights of any person (including QUALCOMM's Intellectual Property or QUALCOMM's
Licensed Patent Claims)'[306]), which includes both patents implemented in modem chipsets and patents implemented outside of modem chipsets, standard-essential and other patents, and standard-essential patents for each CDMA, WCDMA, and LTE standard.

422 Nevertheless, as shown in the admitted facts in 2. a. 6) b) above, during the conclusion of the Patent License Agreement, the Respondents did not specify which of their patents were infringed by the mobile phone manufacturers, nor did they provide the mobile phone manufacturers with the basis for license negotiations, such as a list of the patents to be licensed and a claim analysis, despite the requests of the mobile phone manufacturers, including ***, ***, ***, ***.

423 As a result, mobile phone manufacturers enter into patent license agreements with the Respondents without knowing even the basic information necessary to negotiate a patent license, such as the list, number, validity, and contributory value of the patents included in the Respondents' IPRs, entering into blanket license agreements against their will, without having adequately reviewed whether patents that are unnecessary to them307 or substitutable, such as non-standard essential patents (Non-SEPs) or non-cellular SEPs, are included in the accused's IPR estate.[308])

424 In addition, as the generations of mobile communication standards such as CDMA, WCDMA, and LTE change

306) "[redacted]." (Fingerprints 9 and 10)

307) *** stated in its reply brief to the Commission that "Qualcomm argues that 'technologically necessary IPRs' and 'commercially
Necessary IPRs (non-standard-essential patents)," when in fact *** did not need these non-standard-essential patents (Ex. 79), and *** nevertheless entered into patent license agreements with Respondents that included non-standard-essential patents.

308) In this regard, the National Development and Reform Commission of China decided on February 9, 2015 as follows.

"The evidence submitted by a number of licensees shows that Respondents' assertion that they always had the option to license only the essential telecommunications patents is not true. The Commission's investigation shows that, although some licensees voluntarily sought to license patent portfolios from the Respondents, others were required to accept licenses to the Respondents' non-standard-essential patents in order to license the Respondents' telecommunications standard-essential patents. (Emphasis added.) Defendants have refused to provide licensees with patent lists and have not generally offered licensees licenses covering only their telecommunications standard-essential patents."

The proportion of the subject's standard-essential patents in total standard-essential patents by mobile phone generation is

Despite the significant decrease309), as shown in the acknowledged facts in paragraph 2. a. 6) b) above, Respondents' representatives Steve Altman and Paul Jacobs repeatedly publicly stated that mobile phone manufacturers should pay the same royalty even if they use only one of their patents, and proposed uniform licensing terms, such as *% of the price of the mobile phone, as a blanket license. In response, even when mobile phone manufacturers such as ***, ***, ***, and *** requested to enter into patent license agreements separately for each generation of mobile communication standards or to provide patent information, the Respondents refused to do so310) and unilaterally determined the royalty rate based on the total price of the mobile phone regardless of the mobile phone manufacturer's intentions, such as refusing to cross-grant or adjusting the royalty rate to reflect the value of the cross-granted patents. On the other hand, other standard-essential patent holders, such as *** and ***, have, for their standard-essential patents, separated standard-essential patents from non-standard-essential patents at the request of licensees, set actual fees separately for each mobile communication standard, or reasonably adjusted the level of fees to reflect the value of the other party's patents. In the end, unlike other standard-essential patent holders, Defendants have unusually deprived mobile phone researchers of a fair opportunity to negotiate or to choose as an enforcer of non-standard-essential patents or other patents, and have presented them with license agreement terms favorable to themselves, including blanket license terms and license fee terms set by them, and forced them to enter into the agreement.

425 As a result, Defendants' blanket licenses and unilateral royalty terms deprive mobile phone manufacturers of the opportunity to negotiate on a FRAND basis in their standard-essential patent licensing agreements

with Defendants, while at the same time depriving them of the ability to negotiate on a FRAND basis as licensees of non-standard-essential patents and other patents.

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309) Based on ETSI-registered standard-essential patents, the share of standard-essential patents held by the accused in the total number of standard-essential patents was 90% in the CDMA standard, but decreased significantly to 27.3% in the WCDMA standard and 16% in the LTE standard.

310) [redacted] (Exhibit 73, email from Defendants' license negotiator, dated 2004.*.*.)

They'll say you've been blocked from choosing.

(2) Setting the terms of a cross-grant for free without providing fair consideration

426 In addition to (1) above, by including in their patent license agreements with mobile phone manufacturers the terms of a free cross-grant of the mobile phone manufacturers' patents, the mobile phone manufacturers not only receive no payment from the Respondents for their patents, but also lose the opportunity to receive fair consideration, including physical samples, from customers who purchase the Respondents' modem chipsets.

(3) Enforceability cases

427 As discussed in the previous section on Conduct 2, these comprehensive, unilateral, royalty-bearing patent license agreements between the Defendants and mobile phone manufacturers, including gratuitous cross-granting, were entered into consistently311) without regard to the willingness of the mobile phone manufacturers to cooperate with the Defendants in order to leverage the Defendants' dominance in the modem chipset market under the modem chipset supply and license agreement nexus structure established by the Defendants, as illustrated by examples (a) through (c) below.

(a) *** case

428 When *** entered into a WCDMA-related patent license agreement with Respondents in 2003, *** pointed out that Respondents were using ***'s patents without paying any royalties and sought compensation for

this312). However, the Respondents pointed out that ***'s previous CDMA-related patent license agreement was on a "royalty-free" basis, and that the Respondents were not required to pay any

311) Defendants have entered into substantially identical agreements (blanket licenses, unilateral royalty terms, and free cross-granting) with *** mobile phone manufacturers worldwide under their patent license standard agreements.

312) At that time, *** conveyed to Respondents its position that "the importance of Respondents' IPRs to the WCDMA standard has diminished" and that "at the time of the original (CDMA) agreement, *** had no patents, but now has a significant number of patents, and the royalty rate should be adjusted to reflect the value." (Exhibit 69 to ***'s internal report on the status of negotiations in 2003-2004).

No royalties have been paid to any other licensee, such as ** or **, and no royalty adjustments have been made to the 'p

It refused to pay for the cross-grant on the grounds that it was "difficult because it would affect their entire licensing program."[313]) This demonstrates that the defendants have been accepting cross-grant at no cost in order to maintain their usual standard contractual arrangement of a blanket license, even when mobile phone manufacturers request just compensation for their standard essential patents.

(b) *** case

429 In its response to the Commission, *** stated that ***'s acceptance of the terms of the patent license agreements with the Respondents was due in large part to the way the Respondents tied the license agreements to the supply of modem chipsets, and that unlike the license agreements with the Respondents, which are dominant in the modem chipset market, ***'s license agreements with other patentees are bilateral and it has not been forced to cross-grant or pay high royalties to other patentees.314)

(c) ***、*** cases

430 On the other hand, Respondents did not impose unilateral royalty terms on ***, which did not rely on modem chipsets, or secure a cross-grant of ***'s patents315), and Respondents list *** as the only patentee from whom they did not secure a cross-grant. In addition, *** told the Commission that "*** trades with Qualcomm on normal terms for non-cellular components (Bluetooth, Wi-Fi) where Qualcomm does not have dominance and

313) Internal report on the status of the 2003-2004 negotiation of the Sogetsu 69 certificate ***.

314) *** believes that "Qualcomm's practice of supplying baseband chipsets only contingent upon the execution of a license agreement and the absence of a challenge to its terms plays a significant role in the terms of the license agreement" and that "*** has entered into

license agreements with other patent holders on a bilateral basis. Unlike its agreements with Qualcomm, which has a dominant position in the market for modem chipsets, *** has not been forced to grant free cross-licenses or pay high royalties in its agreements with other patent holders." (Exhibit 79, ***'s Response to the Commission's Request for Production).

315) The subjects' typical practice rate is *%, [redacted]

316) [redacted]

Defendants do not require a separate patent license for these components, but rather use the
It sells at a loss and guarantees supply. Qualcomm's relationship with *** regarding modem chipsets is unusual."317)

431 These cases demonstrate that the terms of the patent license agreements, including the free cross-grant, that Defendants secured from mobile phone manufacturers were obtained by leveraging their dominance of modem chipsets.

432 Therefore, if the Respondents did not have dominance in the modem chipset market and the mobile telecommunications standard essential patent licensing market, or even if they did, it is difficult to imagine that a mobile phone manufacturer would have entered into a patent license agreement with the Respondents that included such terms as a blanket license, unilateral royalties, and free cross-grant without being provided with specific information about the patents to be licensed by the Respondents, without an analysis of the value of the patents, and without substantive negotiation of the terms of the license agreement. In other words, the fact that the defendants entered into a patent license agreement with a mobile phone manufacturer under terms that were favorable to the defendants and that the mobile phone manufacturer would not have accepted if it had had a fair opportunity to negotiate, indicates that the defendants unilaterally forced the mobile phone manufacturer to accept the terms of the agreement without going through a fair negotiation process on FRAND terms.

<Judgment on the allegations of the relevant defendants ⑫>.

433 In this regard, Respondents argue that the cross-grants they received from handset manufacturers were not gratuitous, and that they compared the value of the mutually held patents and factored them into

the calculation of the total license consideration they were receiving from the handset manufacturers.

434 In particular, (i) Defendants themselves represented that most cell phone manufacturers did not pay cross-grant consideration because they either had no patents or the value of the patents they did have was insignificant; and

317) Aug. 17, 2016, 2nd deliberation of the KFTC's plenary session

318); and (ii) the terms of the patent license agreements that Defendants entered into with mobile phone manufacturers.

If the cross-grant consideration was factored into the calculation of the overall license fee consideration, as Respondents contend, there should be a difference in the license fee based on the proportion of the value of the patents owned by the mobile phone manufacturers, yet there is no difference in the license fee terms even among mobile phone manufacturers with significant differences in the value of their patents319), (iii) that foreign competition authorities, including China320) and Japan321), have already prohibited licensees from cross-granting their patents to themselves and their customers without paying reasonable consideration to the Respondents, the Respondents' assertion that the license consideration is calculated by comparing the value of the patents is without merit.

d) whether the conduct unreasonably interferes with the activities of other operators

(1) Intent and Purpose of Restraint of Competition

(a) Intent and purpose to strengthen its dominance in the market for licensing telecommunications standard essential patents

435 As discussed above, Defendants (i) failed and refused to negotiate in good faith with cell phone manufacturers in violation of the FRAND covenant, (ii) held standard essential patents

318) During the course of the investigation, the examiner repeatedly asked the respondents to explain the basis on which they calculated the value of the cross-grant, if any, but the respondents were unable to provide any evidence.

319) [redacted]

320) In this regard, the National Development and Reform Commission of China decided on February 9, 2015 as follows.

"Respondents have failed to submit competent evidence to establish that Respondents paid relevant consideration for the patents cross-licensed by certain licensees in licensing their patents, and Respondents' assertion that gratuitous cross-licenses are part of an overall quid pro quo between Respondents and the licensees lacks facts and evidence to support it. (emphasis added) The Commission finds that Respondents used their dominant position in the market for licensing essential patents for telecommunications standards to coerce Licensees to provide patent cross-licenses to Respondents for free (without the value of the patents cross-licensed by Licensees being deducted from the fees paid to Respondents or the expected consideration being paid)."

321) On September 28, 2009, the Japan Fair Acquisition Commission decided the following.

"Qualcomm shall terminate any contractual provisions that purport to grant to Qualcomm free of charge the right to practice intellectual property rights held or to be held by a domestic terminal manufacturer for the manufacture and sale of CDMA chips, that purport to undertake not to assert rights against Qualcomm or Qualcomm's customers, and that purport to undertake not to assert rights against Qualcomm's licensees." (Japan Fair Trial Commission No. 22, 2009)

a dominant player in the modem chipset market, supplying modem chipsets and licensing patents
It was the only business that did not supply modem chipsets prior to entering into the patent license agreements by separating and linking the agreements; (iii) in entering into patent license agreements with mobile phone manufacturers under the above structure, it did not specify the patents to be licensed or provide any data to evaluate the value of the patents, but imposed unilateral license terms under a standard contractual arrangement called a blanket license; and (iv) whereas, ***, which did not rely on Respondents' modem chipsets, did not accept unilateral royalty terms or secure a cross-grant of ***'s patents, failed to impose unilateral royalty terms on ***, which was not dependent on Respondents' modem chipsets, or to secure a cross-grant of ***'s patents, (v) the blanket license terms appear to have been intended to extend Respondents' overwhelmingly dominant share of the CDMA source technology to the WCDMA and LTE standards, in which they have a lesser share, and to extend the dominance of their standard-essential patents to non-standard-essential patents, with the intent or purpose of consolidating their dominance in the telecommunications standard-essential patent licensing market.

(b) intent and purpose to exclude competitors from the market for modem chipsets

436 By refusing to license their standard-essential patents to competing modem chipset manufacturers, while tying the supply of their modem chipsets to patent license agreements with mobile phone manufacturers, Respondents created a "patent umbrella" in which they cross-granted patents owned by mobile phone manufacturers without paying fair consideration to the mobile phone manufacturers.

437 Defendants' conduct is consistent with the fact that (i) the risk of

patent disputes and the availability of indemnification is an important factor in a mobile phone manufacturer's decision to purchase a modem chip set; (ii) Defendants have made it clear in their sales pitches and business reports that only by purchasing their modem chipsets with the knowledge that they will be protected from Defendants' and third parties' patents, white papers, and other publications since 2004; (iii) Paul Jacobs, Defendants' CEO, stated at a London investor conference that purchasing their modem chips would not require a separate enforcement fee to be paid to other patent holders, whereas purchasing a competing modem chip set would require a separate fee.

In light of its emphasis on the need for a more efficient and cost-effective modem chipset322), it is not surprising that the company has positioned itself as a more attractive alternative to competing modem chipsets.
The intent and purpose of the conspiracy was to limit competition in the modem chipset market.

(2) Anti-competitive effects

(a) Strengthening monopoly power in the telecommunications standard essential patent license market

Gain control over their entire patent estate with comprehensive license terms

438 By imposing blanket licenses that make it difficult to distinguish between mobile communications standard-essential patents, other patents, and the generational performance requirements of mobile communications standards, Defendants have extended their control over standard-essential patents to other patents and extended their control over CDMA standard-essential patents to later standards such as WCDMA and LTE, thereby limiting the patent choices of mobile phone manufacturers and increasing their control over the entirety of their patents.

439 Moreover, the blanket license terms deepened the dependency on the entirety of Defendants' patents. Mobile phone manufacturers have already paid for all patents in existence or developed in the future that are covered by Defendants' patents, significantly reducing their incentive to enter into patent license agreements even if circumvention designs are possible or alternative technologies exist.

Avoiding good faith negotiation of license terms in violation of FRAND commitments.

440 Despite the FRAND commitments, the defendants refused to supply modem chipsets to the

322) Paul Jacobs said at the time "Over 100 companies have provided us with several bundles of IP transfer rights, which means that if I sell a modem chip and software to a company, they have access to Qualcomm IP, and as a result, they have access to IP held by over 100 companies. This prevents the accumulation of a large amount of potential royalties, because otherwise, if they got the chips from another company, or if they didn't get the IP rights, they would have to negotiate with each of the other companies individually, and possibly pay royalties to each of the other companies." (Exhibit 82, Exhibit 5, 2005 Investor Meeting).

and forced them to sign patent license agreements in advance,323) and, in the process, to take advantage of the standard-essential patent
and other patents, and the licensing terms of mobile telecommunications generation-specific standards in a manner that renders them indistinguishable, thereby circumventing their obligation to negotiate in good faith under the FRAND commitments. As a result, the benefits of standardization are lost and the harms of monopolization remain, and, moreover, given Defendants' position as vertically integrated players in the mobile telecommunications semi-essential patent licensing market and the modem chipset market, they directly raise the costs of competing modem chipset manufacturers, thereby excluding competitors, and strengthen their dominance in the mobile telecommunications standard essential patent licensing market, causing anticompetitive harms in the form of reduced product diversity, restricted technological innovation, and reduced consumer benefits.

(b) Patent hold effects

441 By unilaterally imposing royalty terms that apply a percentage of the total selling price of a handset without providing a specific basis for calculating the royalty, Defendants force handset manufacturers to (i) pay royalties to Defendants for any increase in the value of the handset due to the handset manufacturer's own innovations that are unrelated to Defendants' patents.324); and (ii) by establishing long-term and perpetual contracts, rather than licensing on a per-carrier standard basis, Defendants will be able to maintain similar royalty rates regardless of changes in the value of their patents, such as a decrease in the value of their standard-fill patents as carrier standards change.325)

442 In addition, the Defendants' imposition of unfair licensing terms on handset manufacturers, including blanket licenses, arbitrary royalty terms, and free cross-granting, not only excludes them

from the mobile communications standard-essential patent licensing market, but also from the modem chipset market and the innovation market, as follows

323) In light of the foregoing, and the fact that the standard-essential patent holder's exclusionary claims are in violation of the FRAND covenant, Respondents' conduct has a stronger anticompetitive effect than exclusionary claims in the standard-essential patent licensing market.

324) The result is that the value of the various technologies and values that make up a smartphone, such as components such as memory and cameras, operating systems, voice recognition, fingerprinting, mobile payment systems, and other technologies and values that are not related to the accused's essential patents for mobile communications standards, are also included in the implementation fees for the accused.

325) 더욱이 피심인들은 ***、***、***、*** 등과 체결한 특허 라이선스 계약서에서는 계약이 종료되더라도 피심인들이 제공받은 크로스 그랜트는 유지되도록 규정되어 있다.

(c) Exclusion of competitors in the modem chipset market

443 The "patent umbrella" created by Defendants' free cross-grant from 195 mobile phone manufacturers had the effect of (i) discriminating against their modem chipset customers by unfairly favoring their implementation fees; (ii) raising competitors' costs due to the cost of patent infringement indemnification; (iii) the prolonged and anticompetitive effects of perpetual contractual agreements; and (iv) the anticompetitive effects of the integration of modem chipsets and application processors, (iii) the long-term and perpetuation of the anticompetitive effects of long-term and perpetual contractual terms, and (iv) the spillover effect of the anticompetitive effects of the integration of modem chipsets and application processors into adjacent markets, resulting in the exclusion of competing modem chipset manufacturers from the modem chipset market. A more detailed description of the anticompetitive effects has already been provided in the section on the anticompetitive effects of Act 1 (refusal to license or restriction of essential patents for telecommunications standards to competing modem chip set manufacturers) in b) (4) above.

(d) impede technological innovation by limiting the incentives of other patent holders to develop technologies

444 (i) the cell phone manufacturers are deprived of the opportunity to be fairly compensated for their patented inventions because of the Respondents' coercion of the free cross-grants, and (ii) the duration of the cell phone manufacturers' free cross-grants to the Respondents is prolonged, Given that the free cross-grant provided by the mobile phone manufacturers to the Defendants has not been terminated but perpetuated even after the

termination of the licensing agreement with the Defendants, and that the mobile phone manufacturers have stated that the free cross-grant to the Defendants makes it difficult for them to continue innovative research and development on the patents327), the incentive for mobile phone manufacturers to invest in technology development is significantly reduced.

326) This anticompetitive effect is magnified by (i) the fact that Defendants have required cell phone manufacturers to cross-grant not only standards-essential patents, but also non-standard-essential patents; (ii) the fact that Defendants' large share of the modem chipset market means that they have little chance of receiving fair compensation from cell phone manufacturers who do not purchase Defendants' modem chipsets; and (iii) the fact that 195 cell phone manufacturers worldwide provide free cross-granting to Defendants.
With respect to the cross-granting of dual non-standard-essential patents, the recently revised IEEE Intellectual Property Policy permits a standard-essential patent holder to require a licensee to cross-license a standard-essential patent in exchange for a license to the standard-essential patent, but prohibits the standard-essential patent holder from requiring the licensee to license the non-standard-essential patent in exchange for a license to the standard-essential patent. An important consideration in this decision was that mandatory cross-licensing requirements can be a disincentive to innovation. [Business Review Letter from the U.S. Department of Justice to IEEE regarding revisions to IEEE's intellectual property policy].

327) In its response to the Commission, ***, a cell phone manufacturer, stated that "***'s current and newly developed technologies were provided to Qualcomm and Qualcomm's customers without any consideration to ***. As a result, *** has no interest in the

and stifle technological innovation.[328)]

445 (ii) Respondents' imposition of a one-size-fits-all royalty based on the price of a handset prevents handset manufacturers from fully recovering the increased value of their handsets as a result of the technology, while Respondents are free-riding on the value of other operators' innovations, which in turn reduces the incentive for other operators in the telecommunications industry to develop new technologies.[329)]

446 (iii) In addition, as Defendants impose blanket licenses for standard essential patents and other patents, mobile phone manufacturers will have no incentive to use other patents of other operators through evasive and circumventive designs, and the incentive for other operators to develop technologies will be reduced.

447 The above imbalance in incentives to develop technologies between the Respondents and other operators restricts innovation competition in the telecommunications industry as a whole, leading to a "technology development → patent acquisition → standardization contribution
→ compensation through standard-essential patent licensing," and ultimately may increase the influence of the subjects in the future standardization process, thereby consolidating the position of the subjects.

(d) Hindering end-consumer benefits

448 Defendants' imposition of unfair patent licensing terms on cell phone manufacturers

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Without reasonable compensation, there is no way to continue innovative research and

development." (***'s Response to the Commission's Exhibit 95).

328) In fact, in the patent litigation between Samsung Electronics and Apple from 2012 to 2014, the Dutch court held that Samsung Electronics' patents were exhausted because Samsung Electronics made sub-licensing agreements with customers of the defendants' modem chipsets and Apple used the defendants' modem chipsets. For reference, in the WCDMA standard, the defendants hold 1,828 standard-essential patents and Samsung holds 260 standard-essential patents, and in the LTE standard, the defendants hold 1,647 standard-essential patents and Samsung holds 1,039 standard-essential patents.

329) In this regard, Dr. *** (Professor of Economics, *** University), who participated as an expert for the interested parties in the Commission's deliberations, argued that (i) by collecting excessive implementation fees from cell phone manufacturers, the Respondents' investment incentives are unfairly increased and the investment incentives of their competitors are unfairly reduced, thus giving the defendants an unfair advantage in the competition for the development of next-generation products, and (2) the defendants' exploitative effects lead to exclusionary effects, as the defendants discriminatorily grant c r o s s - g r a n t of mobile phone manufacturers' patents obtained through license agreements only to their m o d e m chipsets, which has the effect of increasing the costs of c o m p e t i t o r s . (August 17, 2016, 2nd plenary meeting of the Fair Trade Commission)

If the phone manufacturer's costs increase, this will be reflected in the price of the phone, which will ultimately be reflected in the
The burden is passed on to consumers. In addition, the exclusion of competitors reduces product diversity and limits technological innovation, which in turn leads to the harmful effects of competition restrictions in terms of reduced consumer welfare.

<Judgment on the allegations of the relevant defendants ⑬>.

449 Defendants argue that the terms of the blanket license, the terms of the self-determined royalty, and the terms of the free cross-grant are distinct acts, and that it would be a shifting of the burden of proof to find them unlawful as a group.

450 (i) the defendants themselves have described the blanket license as a "bundle of royalties based on the price of the handset and a free cross-grant"; (ii) all of the above practices are common to the defendants' standard agreements, and the defendants have consistently provided for them in their patent license agreements with handset manufacturers, (iii) the above conduct is a means for the Respondents to circumvent FRAND commitments and implement patent restraint, and it is therefore appropriate to determine the illegality of the conduct as a whole; and (iv) in terms of anticompetitive effects, the conduct is all linked to the exclusionary effect in the modem chipset market and the standard essential patent license market, and it is therefore appropriate to view the conduct as a single entity of patent restraint in terms of mode of conduct and effect, and the Respondents' argument is without merit.

m) Sintered

451 Therefore, the Respondents' act 3 in 2. a. 6) above (imposing

contractual terms including comprehensive license terms, self-determined royalty terms, and free cross-grant terms when entering into a patent license agreement with a mobile phone manufacturer) is illegal because it constitutes the act of imposing transactions or acts that are unfairly disadvantageous to a counterparty by making it difficult for another business to conduct its business activities by other unfair means under Article 3(2)(1) of the Act and Article 5(3)(4) of the Enforcement Decree.

4) Linkage between conduct 1, 2, and 3 and the resulting anticompetitive effects

452 As discussed above under 2. c. 1) through 3) above, Acts 1, 2, and 3 individually have anticompetitive effects and constitute violations of the Act. Moreover, as discussed in 2. a. 3) above, Acts 1, 2, and 3 were implemented under the Respondents' business policies and were further consolidated by becoming mutual prerequisites or means of each other and circulated in an organic linkage, thereby increasing their anticompetitive effects.

453 First, Respondents' refusal to license and restriction of licenses to competing modem chipset manufacturers allows Respondents to use the supply of modem chipsets to force mobile phone manufacturers to enter into patent license agreements. This is because in order for the Defendants to require mobile phone manufacturers to enter into patent license agreements, they must prevent the exhaustion of licenses at the modem chipset stage.

454 Second, Defendants are able to use the structure to force handset manufacturers to enter into patent licensing agreements under the guise of supplying modem chipsets, which allows them to impose patent licensing terms on handset manufacturers that are unfavorable to handset manufacturers, including blanket licenses, arbitrary royalty terms, and free cross-granting.

455 Third, by refusing and restricting licenses to competing modem chipset manufacturers, exposing competing modem chipset manufacturers to the risk of patent attacks by Defendants, and by imposing unfavorable licensing terms, particularly free cross-granting, on mobile

phone manufacturers, Defendants created a "patent umbrella" for their modem chipsets. As a result, Defendants consolidated their dominance in the modem chipset market and the market for licensing essential patents for telecommunications standards, and created a "tilted playing field," a gap in patent protection for modem chipset customers.

5) Behavior 2 and 3: Taking unfair advantage of your trading position to trade with another party

a) Applicable laws and jurisprudence

(1) Applicable laws

Article 23 (Prohibition of Unfair Trade Practices) ① A business shall not engage in any of the following acts that may impede fair trade (hereinafter referred to as "unfair trade practices"), or cause its affiliates or other business entities to engage in such acts.

4. take unfair advantage of one's trading position to trade with another party

Article 36 of the Enforcement Decree (Designation of Unfair Trade Practices) ① The types or criteria of unfair trade practices pursuant to Article 23 (Prohibition of Unfair Trade Practices), Paragraph 3 of the Act are as shown in Tables 1 and 2.

[Exhibit 1 of 2] Types and criteria of unfair trade practices (related to Article 36(1))

6. abuse of trader status

In Article 23 (Prohibition of Unfair Trade Practices), Paragraph 1, Item 4 of the Act, "the act of trading with another party by unfairly using one's trading position" means the act falling under any of the following items.

a. ~ c.

(omitted) d.

(omitted)

Provide

penalties

Setting or changing the terms and conditions of a transaction to disadvantage the counterparty or disadvantaging the counterparty in the process of implementing the transaction, other than acts falling under any of the preceding items.

(2) Related jurisprudence

456 Whether a business entity has a dominant position over a counterparty

shall be determined by comprehensively considering the existence of an ongoing business relationship, the degree of transaction dependence, the ease of securing alternative business lines, the business entity's business control and supervision, market conditions, and the nature of the goods or services involved.

457 In addition, dominant position is recognized when one of the parties has a relatively superior position, or at least a position that can significantly influence the other party's trading activities.

It will be said that it is sufficient to do so, which is based on the market conditions faced by the parties, the overall level of business ability
It should be judged by considering the characteristics of the vehicle and the goods to be traded.[330)]

458 Whether a person has used its dominant position to provide unfair disadvantages shall be determined by determining whether it is recognized that it has provided disadvantages in the process of setting or changing the terms and conditions of the transaction or in its implementation, and whether it is likely to impede fair trade as a departure from normal trade practices in light of the specific circumstances of the transaction, the nature of the product, the extent of the dominant position in the market, and the content and extent of the disadvantages suffered by the other party, such as the intention and purpose, effect and impact of the act, etc.)

b) Act 2: Requiring mobile phone manufacturers to enter into and implement patent license agreements as a condition of supplying modem chipsets

(1) Whether you have trader status

459 In light of the foregoing and the following considerations, the Respondents are in a position of relative dominance over their counterparty, the cell phone manufacturer, or at least in a position to significantly influence its trading activities with the counterparty.

460 First, the Respondents hold a 90% share of the global CDMA modem chipset market, a 50% share of the WCDMA modem chipset market,

and a 60-90% share of the LTE modem chipset market, and as mobile phone manufacturers rely on the Respondents for a significant portion of their modem chipset supply, maintaining stable business relationships with the Respondents is critical to their business activities. Moreover, domestic mobile phone manufacturers are even more dependent on the Defendants, purchasing more than 95% of their LTE standard modem chipsets from the Defendants, making it virtually impossible for mobile phone manufacturers to reject the terms of trade offered by the Defendants.

330) See Supreme Court decision of June 29, 2006, 2003 No. 1646

331) See Supreme Court decision of Sept. 8, 2006, No. 2003d 7859

461 Second, cell phone manufacturers are at risk of going out of business if their business with the defendants is disrupted because they will not be able to replace them with other modem chipsets in the short term. This is because mobile phone manufacturers develop and launch new mobile phones and decide on the modem to be used in those phones through a lengthy technical verification and customer approval process for nearly two years, including the performance of the modem chipset and its compatibility with other components, before launching a new product, and once they decide on the modem chipset to be used in a particular model of mobile phone, it is virtually impossible to replace it with another modem chipset.

462 In these circumstances, the Defendants are in a position to significantly influence the business activities of the cell phone manufacturers because if the Defendants were to cease supplying modem chipsets, the cell phone manufacturers would be forced to abandon and cease production of cell phone products that use the Defendants' modem chipsets.

(2) Whether the behavior is unfairly disadvantageous (a) whether a disadvantage has occurred

463 As discussed in part 2. c) above. 2) As discussed in part 2 of this section, the Respondents' practice of requiring mobile phone manufacturers to enter into and perform patent license agreements as a condition of supplying modem chipsets (Practice 2) deprived the

mobile phone manufacturers of the opportunity to negotiate the terms of the agreements on FRAND terms and subjected them to the risk that a breach of the license agreements would result in the cessation of the supply of modem chipsets and thus the cessation of their entire business, and thus constituted an unfair term of trade for the counterparty, Mobile Phone Research.

(b) whether you used your dominant position to unfairly disadvantage them.

464 The Respondents' conduct in requiring cell phone manufacturers to enter into and perform patent license agreements as a condition of supplying modem chipsets (Act 2), which collectively

in light of the fact that it abused its dominant position to improperly demand that its counterparty, a cell phone manufacturer, pay the
You've penalized them.

465 First, as discussed in Section 2. d. above. 2) above, Respondents intentionally designed business policies that allowed them to transfer their dominance in the modem chipset market to the patent licensing market and used it to negotiate patent licenses with mobile phone manufacturers in order to circumvent the constraints imposed by the FRAND commitments and to impose contractual terms favorable to themselves.

466 Second, the Respondents' conduct in requiring the Respondents to enter into and perform a patent license agreement as a condition of supplying the modem chipsets (conduct 2) constituted an unfair term of trade contrary to normal trade practices, as follows

467 (i) no competing modem chipset manufacturer has ever required a mobile phone manufacturer to enter into a patent license agreement before selling a modem chipset to a mobile phone manufacturer or refused to sell a modem chipset to a mobile phone manufacturer for failure to enter into a patent license agreement; (ii) the Respondents separated the patents from their modem chipsets and tied the modem chipsets to the patent license agreement, forcing the mobile phone manufacturer to enter into a license agreement with them before purchasing the modem chipsets. (iii) Despite the fact that a standard-essential patent holder with a FRAND commitment is not permitted to use threats, such as injunctions, to unilaterally force a license agreement, the Defendants circumvented the process of good faith licensing negotiations by supplying their modem chipsets in violation of the FRAND commitment and forced mobile phone manufacturers to enter

into license agreements with them.

468 Third, it creates a risk of business interruption for handset manufacturers for unforeseeable reasons. According to the modem chipset supply agreements between the Defendants and handset manufacturers, the Defendants may unilaterally stop supplying modem chipsets to a handset manufacturer if the Defendants believe that the handset manufacturer has breached the license agreement. Moreover, given the mobile phone manufacturers' heavy reliance on the Defendants' modem chipsets, the risk of such a modem chipset supply disruption is the risk of a de facto shutdown of the entire mobile phone business.

It can get messy.

469 Fourth, the tying of Defendants' supply of modem chipsets to license agreements serves as a means of uniformly imposing unfair licensing terms on handset manufacturers, i.e., Defendants have used this tying to ensure that handset manufacturers have no choice but to enter into license agreements with Defendants, and have entered into license agreements with unfair terms for virtually all handset manufacturers.

(3) Sintering

470 Therefore, the Respondents' conduct 2. a. 5) above is illegal as it falls under Article 23(1)(4) of the Act.

C) Act 3: When entering into a patent license agreement with a mobile phone manufacturer, presenting the terms of the agreement, including the terms of a comprehensive license, the terms of a license fee set by the company, and the terms of a free cross-grant.

(1) Whether you have trading status

471 The Respondents have a dominant position over the mobile phone manufacturers for the reasons set out in paragraphs 2. c) (1) above in relation to the act (act 3) of imposing contractual terms, including blanket license terms, self-determined royalty terms, and free cross-grant terms, when entering into patent license agreements with the mobile phone manufacturers. 5) b) For the reasons set out in (1) above, they have a dominant position over the cell phone manufacturers.

(2) Whether the behavior is

unfairly disadvantageous (a)

whether a disadvantage has

occurred

472 First, by circumventing the FRAND constraints imposed on them as standard-essential patent holders and offering blanket licensing terms, Defendants have allowed cell phone manufacturers to combine standard-essential patents and

It has become impossible to distinguish between the license terms of different patents, and unnecessary or
The licensee's right to choose was compromised by being forced to license alternative technologies.

473 Second, the Respondents failed to provide the handset manufacturers with a basis upon which they could fairly assess the value of the patents, which put the handset manufacturers at a disadvantage by making it impossible for them to determine whether the terms of the license agreements were reasonable, and thus to accept the Respondents' proposed terms of performance.

474 Third, the defendants imposed uniform terms and conditions on cell phone manufacturers based on the price of the cell phone. As a result, the cell phone manufacturers were forced to pay the same terms and conditions, despite differences in product features and components, the degree of infringement, the functional value of the patents, and the proportion of the patents in the overall product.

475 Fourth, the Respondents have accepted cross grants from cell phone manufacturers for their modem chipset business without paying fair consideration. As a result, cell phone researchers have been deprived of their right to receive royalties without fair consideration and have been unable to enforce their patent rights in the event of infringement.

(b) whether you used your dominant position to unfairly disadvantage them.

476 Although the holder of a standard-essential patent with a FRAND commitment is obligated to negotiate license terms in good faith and in accordance with FRAND terms, the Respondents unilaterally imposed their

own terms and conditions on the license agreement. In addition, if a license agreement for a standard-essential patent is concluded through normal trade practices and negotiations between equal parties, it would be difficult to expect that one of the parties would unilaterally accept contractual terms that are unfavorable to it by giving up its rights or interests under FRAND terms without any consideration, so the Respondents' conduct constitutes unfair trade practices by taking advantage of their trading position.

You've penalized your opponent.

(3) Sintering

477 Therefore, the Respondents' conduct 3 in 2. a. 6) above is illegal as it falls under Article 23(1)(4) of the Act.

3. disposition

A. Corrective actions and penalties

478 Of the acts of the Respondents in this case, the acts in 2. a. 4) above violate Article 3(2)(1)(3) of the Act332), the acts in 2. a. 5) above violate Article 3(2)(1)(3) and Article 23(1)(4) of the Act, respectively, and the acts in 2. a. 6) above violate Article 3(2)(1)(3) of the Act and Article 23(1)(4) of the Act. On the other hand, since the legislative objectives and legal interests of prohibiting abuse of dominant position under Article 3(2) of the Act and unfair trade practices under Article 23 of the Act are different, and since the acts of unfair trade practices cannot all be included in the acts of abuse of dominant position, these two provisions can be applied in principle to the acts in 2. a. 5) and 6) above.

479 The acts of the Respondents in 2. a. 4) through 6) above constitute very serious violations that have a significant effect on the order of fair and free competition and have a significant impact on consumers, and therefore, in the case of abuse of dominant market position, a corrective order is imposed pursuant to Article 5 of the Act, and a fine is imposed pursuant to Articles 6 and 55(3) of the Act, Articles 9 and 61 of the Enforcement Decree of the Act, [Appendix 2], and the Notice on Detailed Standards for Imposition of Fines (2015. 10. 7. 2015. 7. KFTC Notice No. 2015-14, hereinafter referred to as the 'Notice of Penalty'), and in the

case of unfair trade practices, impose a corrective order in accordance with Article 24 of the Act, and impose a penalty in accordance with Articles 24(2) and 55(3) of the Act, Articles 9, 61 and [Appendix 2] of the Enforcement Decree, and

332) The acts in 2. a. 4) are violations of Article 5 (3) (4) of the Enforcement Decree and Article 5 (3) (3) of the Enforcement Decree, respectively. However, since the facts underlying the violations are the same, and the period of violation and the amount of related sales are the same, it is decided not to impose separate fines for each violation.

Enforce penalties in accordance with the penalty notice.

<Judgment on the merits of the relevant defendants' claims ⑭>.

480 In this regard, the Respondents argue that, as a matter of international decorum, it is necessary to refrain from issuing a cease-and-desist order for patents registered in foreign territories and foreign countries, as it may infringe on the sovereignty of other countries, and that the scope of the cease-and-desist order should be limited to patents registered in Korea.

481 On the face of it, the Respondents operate the same licensing policy and modem chip set supply policy in the global market. Accordingly, the Respondents' violations in this case do not occur only in relation to patents registered in Korea for Korean operators in one geographical area, but are simultaneously carried out globally, both domestically and internationally, and the effects of the violations affect not only the domestic market but also the international market. Therefore, with regard to the content of the remedial measures and whether to extend the scope of their application to foreign countries, the scope of the remedial measures must be examined in terms of the principle of proportionality in accordance with the content and extent of the violation in order to effectively correct the violation and eliminate the restrictive effects of competition in the market. The violations of the Respondents in this case are structurally related to the maintenance and enhancement of an unfair trading order on a global scale and have caused anticompetitive effects, and the anticompetitive effects in the global market are also having a negative impact on the domestic market and domestic consumers, WHEREAS, the Respondents' business policies and transactions are interconnected, and the effects of the violations are interconnected, and therefore it is difficult and ineffective to distinguish between domestic and foreign remedies to remedy the anticompetitive

effects of the violations, and it is appropriate not to limit the content and scope of the remedial measures to the Korean region and Korean patents in order to effectively remedy the anticompetitive effects on the domestic market.

482 Regarding the scope of the remedial measures, we decided to limit them to modem chipset manufacturers and cell phone manufacturers that are deemed to have a direct, substantial and reasonably foreseeable impact on the domestic market, as described in paragraph 9 of the remedial order.

483 The issue of international comity with respect to the enforcement of laws of other countries is an issue that will be considered if the Commission's cease-and-desist order in this case conflicts with the enforcement of foreign laws, and does not arise simply because the cease-and-desist order covers conduct that occurred abroad. At this time, the Commission's Cease-and-Desist Order does not specifically conflict with existing foreign law enforcement, and as Paragraph 10 of the Cease-and-Desist Order provides that "if any binding final judgment or action or order of a foreign court or competent authority rendered after the date of this Cease-and-Desist Order conflicts with this Cease-and-Desist Order and makes it impossible to comply simultaneously, Respondents may request that the Commission reconsider this Cease-and-Desist Order," we believe that conflicts with the law enforcement of other countries may be resolved in the future.

B. Calculation of penalties for violations of the prohibition of abuse of a dominant position (Article 3(2) of the Act)

484 In view of the fact that the violations in this case are carried out by the Respondents in coordination with each other and in accordance with a single business policy, so that the anticompetitive effects of the violations are multiplied, and the effects of each violation are related to each other and affect the organic circulation structure, the penalty for each violation in this case shall be imposed collectively as a whole.

1) Calculation Basis

a) Calculate the relevant sales amount

485 The duration of a violation generally means the period from the

beginning of the violation to the end of the violation, unless the violation has ended by the date of the deliberations of the committee imposing the penalty, in which case the end of the violation is the date of the committee's deliberations on the case, or the last deliberation if the deliberations continued for more than two days. Therefore, the end date for each of the conduct in this case is December 21, 2016, the date of the Commission's last deliberation in this case.

486 On the other hand, for the same reasons as above with respect to the date of commencement of the violation, the penalty may be assessed at

We have grouped the acts together and imposed them collectively to align the time periods for the imposition of the penalties, using Act 1 as the basis for the timing of the violations because each of the Respondents' acts was based on and occurred in connection with the refusal and restriction of licenses to competing modem chipset manufacturers (Act 1), The time of the violations in this case is defined as the point in time when, among the many instances of refusal and restriction in Conduct 1, Defendants explicitly refused to respond to a competing modem chipset manufacturer's request to provide a license, and the intent or purpose of Defendants' conduct and the wrongfulness of that conduct were clearly and conclusively established by the evidence. Accordingly, the date for calculating the penalty for the violations in this case shall be *March *, 2009, when Respondents rejected A's request to enter into a license agreement and entered into a restrictive side agreement with A.

487 On the other hand, the relevant turnover, which is the basis for the calculation of the penalty, means the turnover or equivalent amount of the relevant products sold by the Respondents during the period of violation, and the relevant products for the calculation of the penalty in this case are (i) modem chipsets related to CDMA, WCDMA and LTE standards, and (ii) patent licenses related to CDMA, WCDMA and LTE standards. Accordingly, the relevant revenues for the purposes of calculating the penalty shall be the following revenues of the Respondents that are deemed to have a direct and substantial effect on the domestic market during the period of violation

488 (1) With respect to modem chipsets related to CDMA, WCDMA, and LTE standards among the Relevant Products, the sales of modem chipsets related to CDMA, WCDMA, and LTE standards sold by the Defendants to domestic mobile phone manufacturers during

the Violation Period, and the sales of modem chipsets related to CDMA, WCDMA, and LTE standards sold by the Defendants to foreign mobile phone manufacturers that are installed in mobile phones sold in Korea

489 With respect to patent licenses related to CDMA, WCDMA, and LTE standards among related products, the patents for the domestic sales portion of the Defendants' revenues from licenses related to CDMA, WCDMA, and LTE standards to domestic mobile phone manufacturers and the domestic sales portion of the Defendants' revenues from licenses related to CDMA, WCDMA, and LTE standards to foreign mobile phone manufacturers during the period of infringement

License revenue

490 The related revenues totaled $18,734,319,351 for modem chipsets related to the CDMA, WCDMA and LTE standards333) and $13,326,302,653 for patent licenses related to the CDMA, WCDMA and LTE standards334).

491 However, the sales related to modem chipsets are all sales of QCTAP, and the sales related to patent licenses are all sales of QI. Therefore, the relevant sales by subject are the sales of QCTAP. 18,734,319,351 and QI is $13,326,302,653.

<Judgment on the allegations of the relevant defendants ⑮>.

492 The Respondents argue that the revenue from the Respondents' licenses and sales of modem chipsets to domestic mobile phone manufacturers should be limited to the revenue from sales and licenses of modem chipsets installed in mobile phones sold for domestic use.

493 However, the relevant goods must be determined individually and specifically by considering the type and nature of the goods directly or indirectly affected by the violation, the trading area, the counterparty, the stage of the transaction, etc.

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333) The Respondents only submitted relevant sales data through September 30, 2016, and therefore, for the entire violation period, only October 1, 2016, and October 1, 2016, respectively.
~ The relevant sales for the period from December 21, 2016 to December 21, 2017 were estimated on a pro rata basis based on the relevant sales of the Respondents for the previous year.

334) However, the Respondents have stated that it is not possible to calculate the sales of modem chipsets sold by the Respondents to cell phone manufacturers headquartered outside of Korea that are installed in cell phones imported into Korea, The relevant revenues for this portion are estimated based on the number of cell phones sold in Korea equipped with Respondents' modem chipsets by ***, a cell phone manufacturer headquartered outside of Korea that has consistently maintained a meaningful share of the Korean cell phone market since the beginning of the infringement period, and the average unit price of modem chipsets purchased from Respondents. The specific calculation is as follows

λ [redacted]

B [redacted]

While both chipsets and licenses can be viewed as related products, the degree of impact, geography, and geography of the
There is no merit to the Respondents' argument, as the relevant sales amount includes all the goods sold by the Respondents to domestic operators in consideration of the counterparty, but excludes the goods sold to overseas operators that were not sold in Korea.

494 The Respondents argue that the Commission has already imposed penalties on the Respondents for discriminatory charging of implementation fees and provision of conditional rebates in connection with the sale of modem chipsets335) and that sales up to the date of the decision in the present case (December 30, 2009) should be excluded from the relevant sales, but the end of the period of violation for the purpose of calculating the penalty in the present case is July 15, 2009, and the period of violation relevant to the imposition of the penalty in the present case is after that date, 2009. *. *Therefore, the Respondents' argument is without merit.

495 ③ The Respondents argue that revenues corresponding to periods in which the Respondents' share of the modem chipset market for the WCDMA and LTE standards is less than 50% should be excluded from the relevant revenues because the Respondents' dominant market position cannot be recognized during such periods, but this argument is not supported by the findings in 2. c. above. As discussed in paragraphs 1) and (2) above, the Respondents' argument is without merit as the Respondents are recognized as having a dominant market position in the relevant market.

496 ④ The Respondents argue that only license revenues should be included in the relevant revenues where there is actual proof of the imposition of license agreement terms at the same time as the mobile phone manufacturer purchases the Respondents' modem chipsets, but (i) the

Respondents' conduct in this case was not limited to a few specific operators, but was carried out consistently and continuously in virtually all contracts in accordance with the Respondents' business policy; and (ii) the Respondents' structure, even from the perspective of an individual mobile phone manufacturer, was already created by the Respondents, The contracts were entered into under a structure that Defendants had already created, namely, refusing to license modem chipset manufacturers, tying license agreements to the supply of modem chipsets to handset manufacturers, and using their dominance over modem chipsets to dictate the terms of the licensing agreements; (iii) Defendants themselves have stated that "almost all of the hundreds of license agreements they have entered into with handset manufacturers around the world are substantially similar in structure and terms"; and (iv) until now, the

335) FTC Resolution No. 2009-281 (Dec. 30, 2009)

The terms of your contract with the phone manufacturer (including the terms of the blanket license, your own implementation
fee terms, and free cross-grant terms) were favorable to the Respondents and that the handset manufacturers could have negotiated different terms that would not have disadvantaged them if they had been given a fair opportunity to choose; and (v) there is no merit to the Respondents' argument that the relevant market affected by the Respondents' conduct is the entire licensing market, given that the relevant sales should be limited to only those licenses that are actually proven.

b) Charging base rate

497 The Respondents' conduct in this case is not only harmful in that (i) the Respondents have been engaging in the conduct in question for a long period of time since the beginning of their business; (ii) the Respondents' conduct in question impairs the competitive conditions for efficient competition on the basis of merit and restricts competition in the modem chipset market and the standard essential patent license market, the fair competition in the market for technological innovation is distorted; (3) the violations are global in scope and the affected markets are also global in scope; and (4) the relevant turnover of the violations significantly exceeds KRW 300 billion, which constitutes a very serious violation.7% for very serious offenses.

C) Calculation Criteria

498 Multiplying the above relevant sales per subject by the base rate of 2.7%, the base per subject is $505,826,622 for QCTAP and $359,810,171 for QI.

2) Primary and secondary adjustments

499 Since the defendants do not have a primary or secondary reason for the adjustment, the primary and secondary adjustment calculation is the same as above.

3) Determining penalties

500 As there are no additional reasons for adjustment for the Respondents, the above secondary adjustment calculation criteria remain unchanged, and the amount of the penalty for each Respondent is KRW 602,540,000,000 for QCTAP and KRW 428,605,000,000 for QI by applying the trading reference rate (1USD=1,191.2 KRW) first announced by Hana Bank336) on the date of the Commission's agreement (December 21, 2016) and discarding the amount less than KRW 1 million.

C. Calculating penalties for violations of the Unfair Trade Practices Prohibition (Article 23 of the Act)

501 The Respondents' Acts 2 and 3 constitute abuses of a dominant position and unfair trade practices. For the reasons set forth in paragraph 3. b. above, the imposition of penalties for each of Acts 2 and 3 in this case is based on the aggregate penalties imposed for each act as a whole.

1) Calculation Basis

a) Calculate the relevant sales amount

502 For the same reasons discussed in 3. b. 1) a) above, the period of violation is from *Month* 2009 through December 21, 2016, and the relevant sales per Respondent337)338) are as follows

336) The penalty notice stipulates that 'the penalty shall be determined by applying the trading reference rate first announced by the Korea Exchange Bank on the date of settlement by the Fair Trade Commission and converting it into Korean won.' The Korea Exchange Bank merged with Hana Bank on September 1, 2015, to become Hana Bank.

337) The Respondents only submitted relevant sales data through September 30, 2016, and therefore, for the entire violation period, only October 1, 2016, and October 1, 2016, respectively.
~ The relevant sales for the period from December 21, 2016 to December 21, 2017 were estimated on a pro rata basis based on the relevant sales of the Respondents for the previous year.

338) However, the Respondents have stated that it is not possible to calculate the sales of modem chipsets sold by the Respondents to cell phone manufacturers headquartered outside of Korea that are installed in cell phones imported into Korea,

The relevant revenues for this portion are estimated based on the number of cell phones sold in Korea equipped with Respondents' modem chipsets by ***, a cell phone manufacturer headquartered outside of Korea that has consistently maintained a meaningful share of the Korean cell phone market since the beginning of the infringement period, and the average unit price of modem chipsets purchased from Respondents. The specific calculation is as follows

λ [redacted]

B [redacted]

18,734,319,351 and QI is $13,326,302,653.

b) Charging base rate

503 For the same reasons discussed in 3. b. 1) b) above, we apply 1.8% for very serious offenses.

C) Calculation Criteria

504 The per subject calculation based on the above per subject related sales multiplied by the 1.8% base rate is $337,217,748 for QCTAP and $239,873,447 for QI.

2) Primary and secondary adjustments

505 Since the defendants do not have a primary or secondary reason for the adjustment, the primary and secondary adjustment calculation is the same as above.

3) Determining penalties

506 As there are no additional reasons for adjustment for the Respondents, the above secondary adjustment calculation criteria shall be maintained, and the amount of the penalty for each Respondent shall be KRW 401,693,000,000 for QCTAP and KRW 285,737,000,000 for QI by applying the trading reference rate (1USD=1,191.2 KRW) first announced by Hana Bank on the date of the Commission's agreement (December 21, 2016) and discarding amounts less than KRW 1 million.

D. Determining the final penalty amount

507 Act 2 and Act 3 in 2. a. 5) and 6) above are related to the prohibition of abuse of a dominant position and the

[redacted]

The prohibition against unfair trade practices may apply in conflict, but only if the facts underlying the violation of the law are the same.
Given that there is only one thread, we will only impose penalties for violations of the prohibition on abuse of dominant position, which has a higher statutory penalty threshold.

508 Accordingly, the final penalties imposed on the Respondents for acts 1 through 3 of 2. a. 4) through 6) above are KRW 602,540,000,000 for QCTAP and KRW 428,605,000,000 for QI, based on the penalty calculation under the prohibition of abuse of dominant position.

4. conclusion

509 The conduct of the Respondents in 2. a. 4) above is in violation of Article 3(2)(1) of the Act, and the 2. a. 5) is in violation of Article 3 (2) (1) (3) of the Act and Article 23 (1) (4) of the Act, and the above 2. a. Since the conduct of (6) is in violation of Article 3 (2) (1) (3) and Article 23 (1) (4) of the Act, Articles 5 and 24 of the Act are applied to the corrective measures and Article 6 and 24 (2) of the Act are applied to the imposition of fines, respectively, and resolved as ordered.

The KFTC voted as follows.

2017 一月 20th

The λ Commissioner jung jae chan

Vice Chair Hak Hyun Kim

Head Judge Seok Ho Kim

上 Circle Shin Dong Kwon

上 Won Sung Ha Kim

上 Circle Number of active

上 circle rephrase